CHAPTER

75

ENGINE AIR

(LEAP-1B ENGINES)

BOEING

737-7/8/8200/9/10
AIRCRAFT MAINTENANCE MANUAL

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DDG 75-23-02 Left or Right Modulated Turbine Cooling Valve Inoperative - Preparation TASK 75-00-00-040-806-G00			901	PTF ALL
DDG 75-23-02 Left or Right Modulated Turbine Cooling Valve Inoperative - Restoration TASK 75-00-00-440-806-G00			912	PTF ALL
DDG 75-24-01–01 High Pressure Turbine Active Clearance Control (HPTACC) Valve Inoperative - Preparation TASK 75-00-00-040-803-G00			916	PTF ALL
DDG 75-24-01–01 High Pressure Turbine Active Clearance Control (HPTACC) Valve Inoperative - Restoration TASK 75-00-00-440-803-G00			921	PTF ALL
START BLEED/BOOSTER ANTI-ICE VALVE - REMOVAL/INSTALLATION	75-11-01		401	PTF ALL
Start Bleed/Booster Anti-Ice Valve Removal TASK 75-11-01-000-801-G00			401	PTF ALL
Start Bleed/Booster Anti-Ice Valve Installation TASK 75-11-01-400-801-G00			408	PTF ALL
ENGINE COOLING - INSPECTION/CHECK	75-20-00		601	PTF ALL
Engine Cooling Visual Inspection TASK 75-20-00-211-801-G00			601	PTF ALL
MODULATED TURBINE COOLING (MTC) VALVE - MAINTENANCE PRACTICES	75-23-02		201	PTF ALL
Left or Right Modulated Turbine Cooling (MTC) Valve Manual Operation TASK 75-23-02-980-801-G00			201	PTF ALL
MODULATED TURBINE COOLING (MTC) VALVE - REMOVAL/INSTALLATION	75-23-02		401	PTF ALL
Left Modulated Turbine Cooling (MTC) Valve Removal TASK 75-23-02-000-801-G00			401	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124
Left Modulated Turbine Cooling (MTC) Valve Installation TASK 75-23-02-400-801-G00			424	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

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Right Modulated Turbine Cooling (MTC) Valve Removal TASK 75-23-02-000-802-G00			430	PTF ALL
Right Modulated Turbine Cooling (MTC) Valve Installation TASK 75-23-02-400-802-G00			436	PTF ALL
MODULATED TURBINE COOLING ACTUATOR - REMOVAL/INSTALLATION	75-23-03		401	PTF ALL
Left Modulated Turbine Cooling Actuator Removal TASK 75-23-03-000-801-G00			401	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124
Left Modulated Turbine Cooling Actuator Installation TASK 75-23-03-400-801-G00			408	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124
Right Modulated Turbine Cooling Actuator Removal TASK 75-23-03-000-802-G00			411	PTF ALL
Right Modulated Turbine Cooling Actuator Installation TASK 75-23-03-400-802-G00			419	PTF ALL
LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL VALVE (LPTACC) - REMOVAL/INSTALLATION	75-24-01		401	PTF ALL
Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal TASK 75-24-01-000-801-G00			401	PTF ALL
Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation TASK 75-24-01-400-801-G00			407	PTF ALL
LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL MANIFOLDS (LPTACC) - REMOVAL/INSTALLATION	75-24-02		401	PTF ALL
Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Assembly Removal TASK 75-24-02-000-801-G00			401	PTF ALL
Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Assembly Installation TASK 75-24-02-400-801-G00			412	PTF ALL

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Low Pressure Turbine Active Clearance Control (LPTACC) Lower Manifold Assembly Installation TASK 75-24-02-400-802-G00			427	PTF ALL
LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) MANIFOLD BRACKETS - REMOVAL/INSTALLATION	75-24-03		401	PTF ALL
Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Brackets - Removal TASK 75-24-03-000-801-G00			401	PTF ALL
Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Brackets - Installation TASK 75-24-03-400-801-G00			410	PTF ALL
Low Pressure Turbine Active Clearance Control (LPTACC) Lower Manifold Brackets - Removal TASK 75-24-03-000-802-G00			413	PTF ALL
Low Pressure Turbine Active Clearance Control (LPTACC) Lower Manifold Brackets - Installation TASK 75-24-03-400-802-G00			421	PTF ALL
HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL VALVE (HPTACC) - REMOVAL/INSTALLATION	75-24-04		401	PTF ALL
High Pressure Turbine Active Clearance Control (HPTACC) Valve Removal TASK 75-24-04-000-801-G00			401	PTF ALL
High Pressure Turbine Active Clearance Control (HPTACC) Valve Installation TASK 75-24-04-400-801-G00			407	PTF ALL
COMPRESSOR CONTROL - INSPECTION/CHECK	75-30-00		601	PTF ALL
Compressor Control Detailed Inspection TASK 75-30-00-211-801-G00			601	PTF ALL
VARIABLE STATOR VANE SYSTEM - MAINTENANCE PRACTICES	75-31-00		201	PTF ALL
Variable Stator Vane Actuator Manual Operation TASK 75-31-00-910-801-G00			201	PTF ALL

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Left Variable Stator Vane (VSV) Actuator Removal TASK 75-31-02-000-801-G00			401	PTF ALL
Left Variable Stator Vane (VSV) Actuator Installation TASK 75-31-02-400-802-G00			410	PTF ALL
Right Variable Stator Vane (VSV) Actuator Removal TASK 75-31-02-000-802-G00			413	PTF ALL
Right Variable Stator Vane (VSV) Actuator Installation TASK 75-31-02-400-803-G00			422	PTF ALL
VARIABLE BLEED VALVE (VBV) ACTUATOR - REMOVAL/INSTALLATION	75-32-01		401	PTF ALL
Left Variable Bleed Valve Actuator Removal TASK 75-32-01-000-801-G00			401	PTF ALL
Left Variable Bleed Valve Actuator Installation TASK 75-32-01-400-802-G00			406	PTF ALL
Right Variable Bleed Valve Actuator Removal TASK 75-32-01-000-802-G00			408	PTF ALL
Right Variable Bleed Valve Actuator Installation TASK 75-32-01-400-803-G00			418	PTF ALL
TRANSIENT BLEED VALVE - REMOVAL/INSTALLATION	75-33-01		401	PTF ALL
Transient Bleed Valve Removal TASK 75-33-01-000-801-G00			401	PTF ALL
Transient Bleed Valve Installation TASK 75-33-01-400-801-G00			411	PTF ALL
REVERSE BLEED SYSTEM - INSPECTION/CHECK	75-34-00		601	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378
Reverse Bleed System Visual Inspection TASK 75-34-00-200-801-G00			601	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

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Reverse Bleed Check Valve Assembly Removal TASK 75-34-01-000-801-G00			401	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378
Reverse Bleed Check Valve Assembly Installation TASK 75-34-01-400-801-G00			411	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378
REVERSE BLEED BLOWER - REMOVAL/INSTALLATION	75-34-02		401	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378
Reverse Bleed Blower Removal TASK 75-34-02-000-801-G00			401	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378
Reverse Bleed Blower Installation TASK 75-34-02-400-801-G00			406	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378
REVERSE BLEED VALVE - REMOVAL/INSTALLATION	75-34-03		401	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378
Reverse Bleed Valve Removal TASK 75-34-03-000-801-G00			401	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378
Reverse Bleed Valve Installation TASK 75-34-03-400-801-G00			412	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

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ENGINE AIR - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has the maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks prepare the airplane for flight with systems/components that are inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. These are the tasks for the components in the EEC Cooling Systems:

(1) DDG 75-23-02 Left or Right Modulated Turbine Cooling (MTC) Valve Inoperative - Preparation

(2) DDG 75-23-02 Left or Right Modulated Turbine Cooling (MTC) Valve Inoperative - Restoration

(3) DDG 75-24-01–01 High Pressure Turbine Active Clearance Control Valve Inoperative - Preparation

(4) DDG 75-24-01–01 High Pressure Turbine Active Clearance Control Valve Inoperative - Restoration

TASK 75-00-00-040-806-G00

2. DDG 75-23-02 Left or Right Modulated Turbine Cooling Valve Inoperative - Preparation

(Figure 901, Figure 902, and Figure 903)

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with an inoperative left or right Modulated Turbine Cooling (MTC) valve.

NOTE: For Post LEAP-1B Service Bulletin 72-0124, the left side MTC valve and actuator (aft looking forward) are removed.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
36-11-04-000-801	PRSOV Removal (P/B 401)
36-11-04-400-801	PRSOV Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

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Reference	Description	Specification
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A
G51411 [CP8005]	Cable, Safety, Stainless Steel, 0.024 inch (0.61 mm) Diameter	AMS 5689

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Deactivation

SUBTASK 75-00-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-00-00-860-002-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 75-00-00-860-003-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 75-00-00-860-058-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

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(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Left or Right Modulated Turbine Cooling Valve Deactivation

SUBTASK 75-00-00-040-009-G00

(1) Do the steps that follow to lock one of the MTC valves (left or right) for the inoperative MTC system in the open position:

(a) If you want to lock open the left MTC valve [1], do these steps:

1) Make the left MTC valve [1] inoperative (locked position) as follows (Figure 901, Figure 903):

a) Put the left MTC valve [1] in the open position.

<1> Put the left MTC valve [1] and pushrod in a position that permits access for the bolt [2] removal.

b) Cut the safety safety cable, G50065 [CP8006], or cable, G51411 [CP8005], or safety wire, G02345 [CP8001], from the bolt [2].

c) Remove the bolt [2] from the left MTC valve [1] lockout bolt normal position.

d) Lubricate the bolt [2] threads and the lockout position threaded insert on the left MTC valve [1] with Tiodize T8E-H grease, D50355 [CP2961].

e) Put the left MTC valve [1] in the open position and align the hole in the control arm with the lock-out-position threaded insert on the MTC valve body.

f) Swing the MTC valve link to align the link bolt hole with the holes of the control arm and the lock-out-position threaded insert on the MTC valve body.

g) Install the bolt [2] in the left MTC valve [1] lockout position.

h) Tighten the bolt [2] to 25.0 in-lb (2.8 N·m) - 30.0 in-lb (3.4 N·m).

(b) If you want to lock open the right MTC valve [3], do these steps:

1) Remove the Pressure Regulating and Shutoff Valve (PRSOV), if it is necessary (TASK 36-11-04-000-801).

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2) Make the right MTC valve [3] inoperative (locked position) as follows (Figure 902, Figure 903):

a) Put the right MTC valve [3] in the open position.

<1> Put the right MTC valve [3] and pushrod in a position that permits access for the bolt [4] removal.

b) Cut the safety safety cable, G50065 [CP8006], or cable, G51411 [CP8005], or safety wire, G02345 [CP8001], from the bolt [4].

c) Remove the bolt [4] from the right MTC valve [3] lockout bolt normal position.

NOTE: It can be necessary to turn the MTC push-rod by hand CCW (FLA) in order for the bolt to clear the tube and be removed.

d) Lubricate the bolt [4] threads and the lockout position threaded insert on the right MTC valve [3] with the Tiodize T8E-H grease, D50355 [CP2961].

e) Put the right MTC valve [3] in the open position and align the hole in the control arm with the lock-out-position threaded insert on the MTC valve body.

f) Swing the MTC valve link to align the link bolt hole with the holes of the control arm and the lock-out-position threaded insert on the MTC valve body.

g) Install the bolt [4] in the right MTC valve [3] lockout position.

h) Tighten the bolt [4] to 25.0 in-lb (2.8 N·m) - 30.0 in-lb (3.4 N·m).

3) Install the PRSOV, if it is removed (TASK 36-11-04-400-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-00-00-860-059-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

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SUBTASK 75-00-00-860-006-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 75-00-00-860-007-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

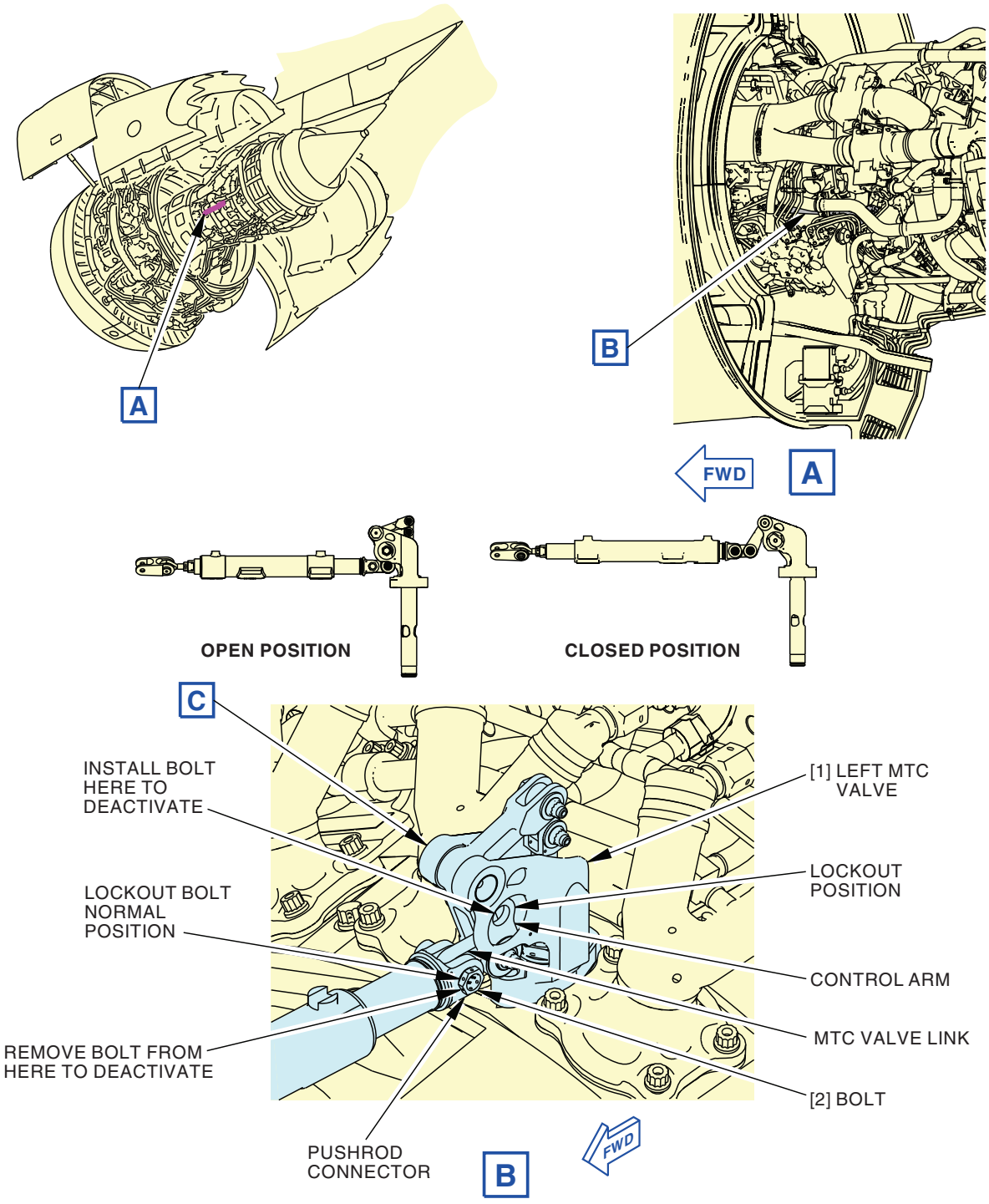
(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-00-00-040-013-G00

(4) If it is necessary, extinguish the MAINT light on the P5-1 panel, do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

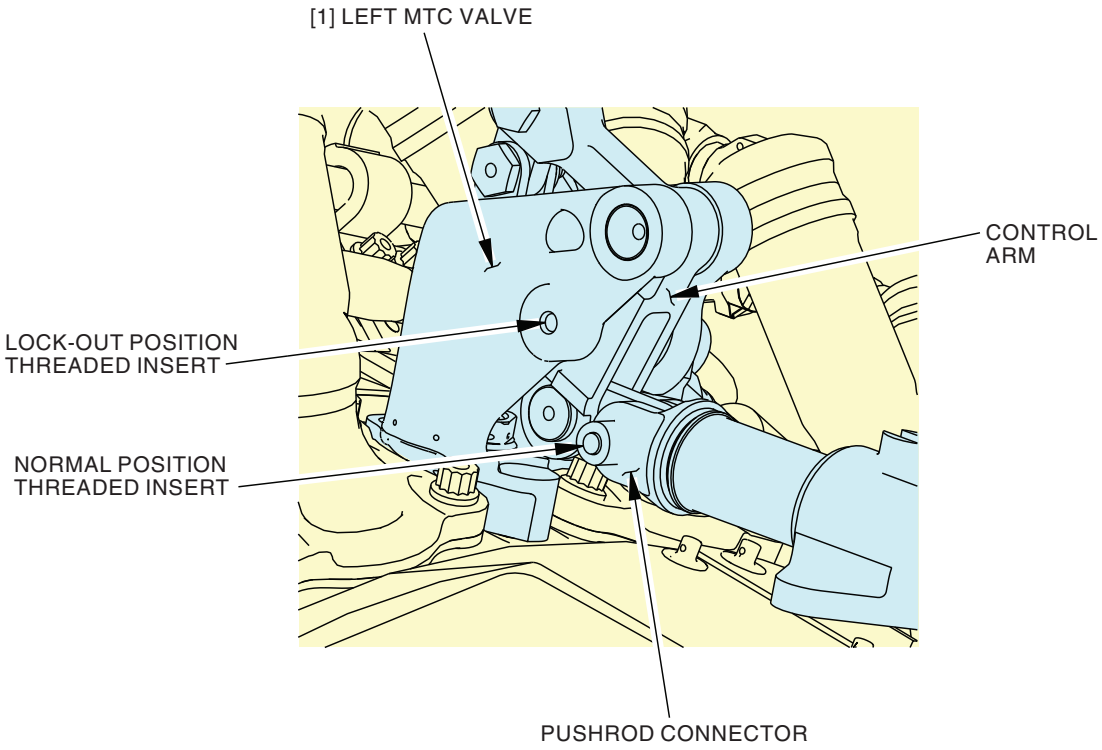
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2455452 S0000571125_V4

Left Modulated Turbine Cooling Valve Inoperative
Figure 901/75-00-00-990-813-G00 (Sheet 1 of 2)

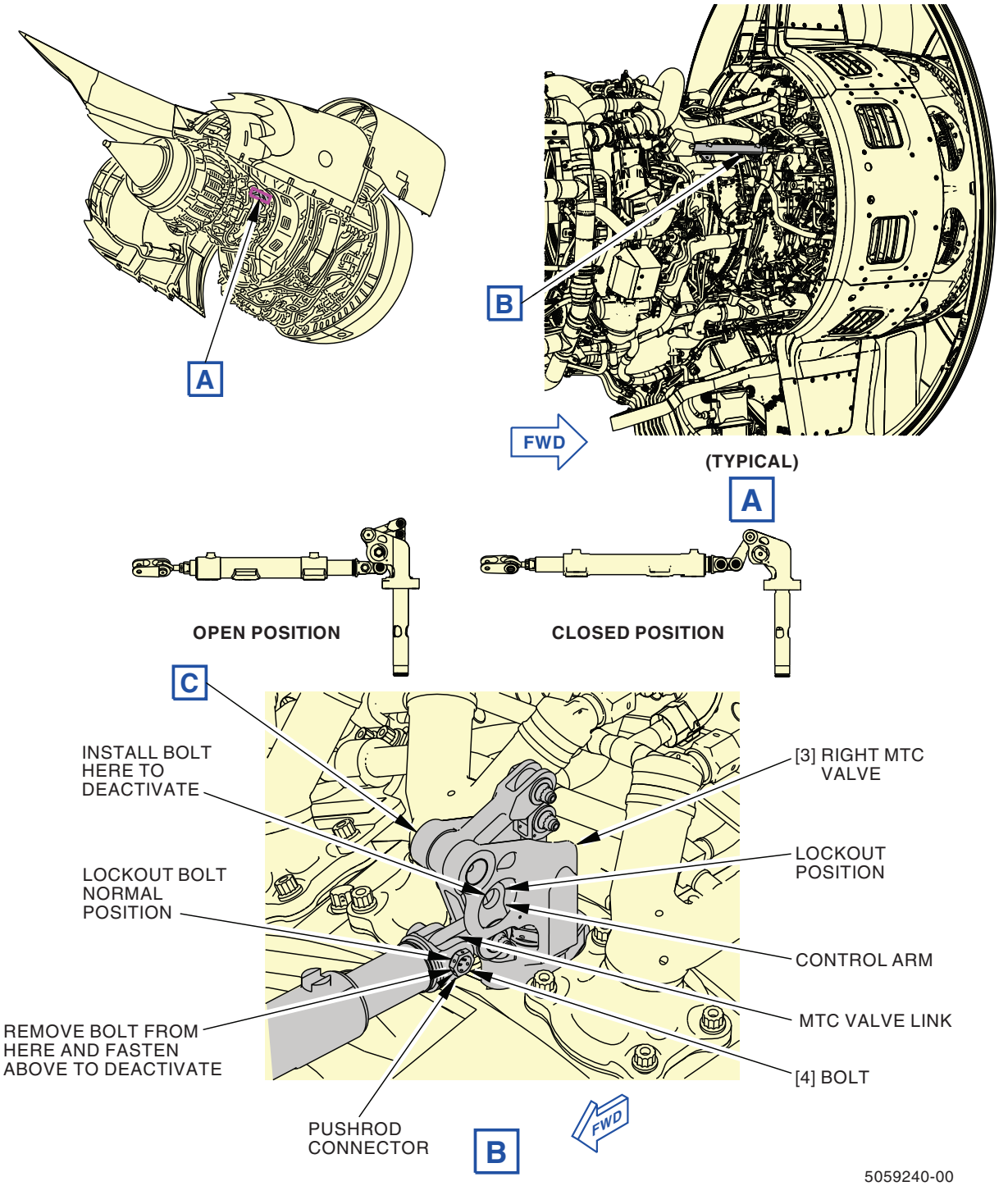
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2499606 S0000585951_V3

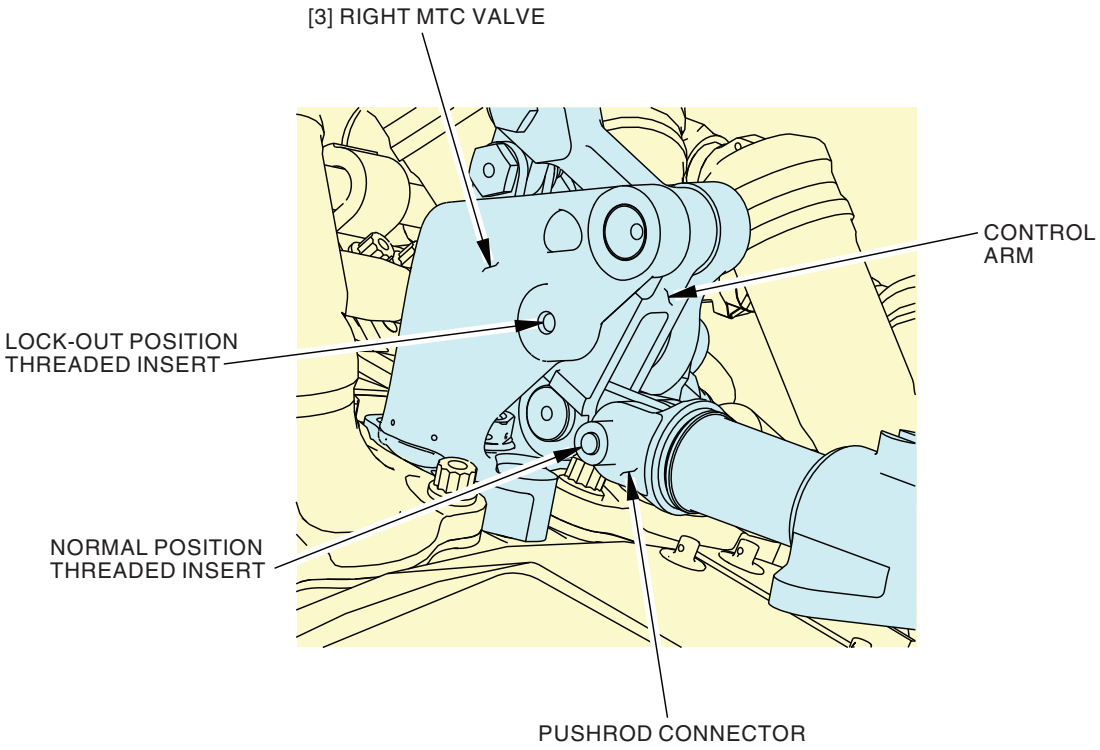
Left Modulated Turbine Cooling Valve Inoperative
Figure 901/75-00-00-990-813-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right Modulated Turbine Cooling Valve Inoperative
Figure 902/75-00-00-990-816-G00 (Sheet 1 of 2)

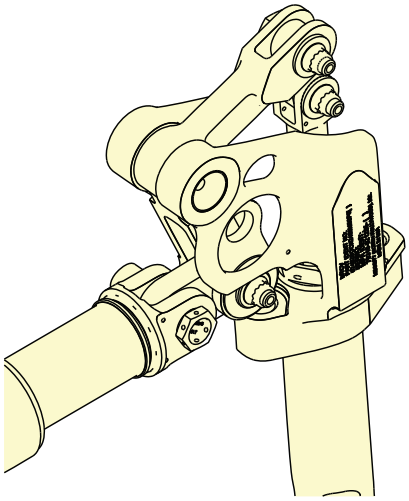
ECCN 9E991 BOEING PROPRIETARY - See title page for details



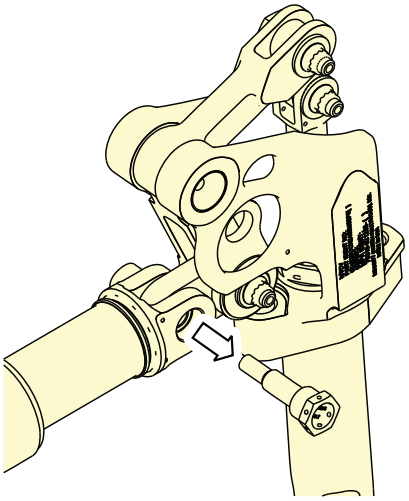
2499611 S0000585953_V3

Right Modulated Turbine Cooling Valve Inoperative
Figure 902/75-00-00-990-816-G00 (Sheet 2 of 2)

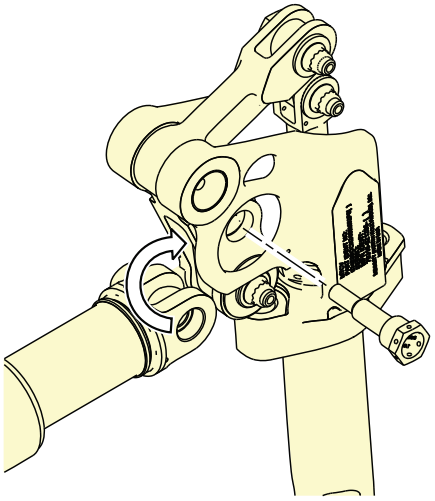
ECCN 9E991 BOEING PROPRIETARY - See title page for details



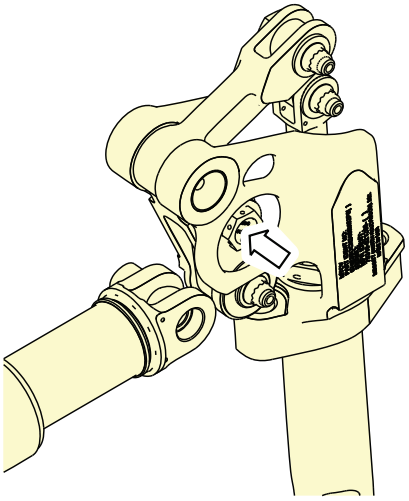
STEP 1: MTC VALVE IN OPEN POSITION WITH THE BOLT INSTALLED IN THE 'NORMAL ' POSITION.



STEP 2: REMOVE THE LOCKOUT BOLT FROM THE 'NORMAL' POSITION.



STEP 3: ALIGN THE HOLE IN CONTROL ARM WITH THE LOCK-OUT-POSITION THREADED INSERT ON THE MTC VALVE BODY. SWING THE MTC VALVE LINK TO ALIGN THE LINK BOLT HOLE WITH THE HOLES OF THE CONTROL ARM AND THE LOCK-OUT POSITION THREADED INSERT ON THE MTC VALVE BODY.

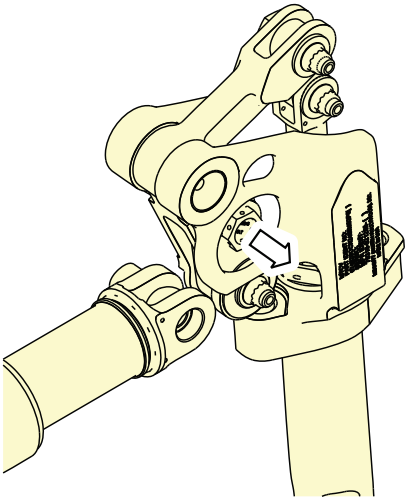


STEP 4: INSTALL THE BOLT IN THE LOCKOUT POSITION AND TIGHTEN THE BOLT TO 25.0 in-lb (2.8 N.m) - 30.0 in-lb (3.4 N.m).

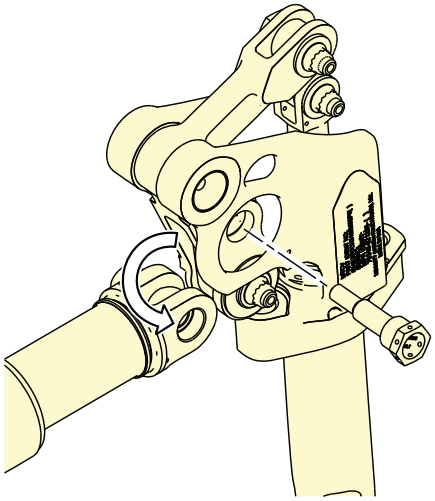
2785593 S0000632329_V1

Left or Right Modulated Turbine Cooling Valve Inoperative: Deactivation
Figure 903/75-00-00-990-817-G00

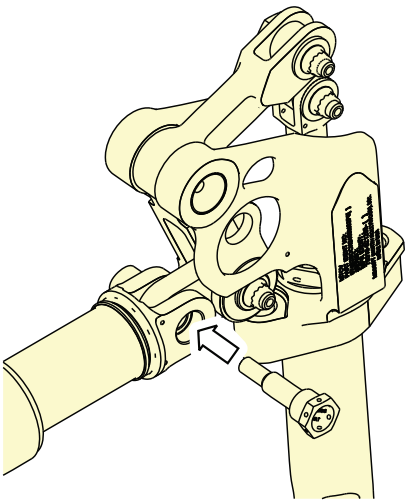
ECCN 9E991 BOEING PROPRIETARY - See title page for details



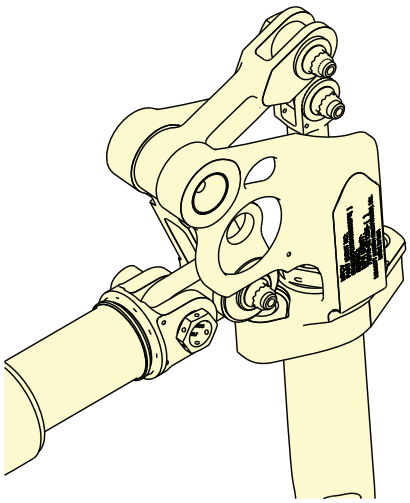
STEP 1: REMOVE THE BOLT FROM THE LOCKOUT POSITION.



STEP 2: SWING THE MTC VALVE LINK TO ALIGN THE LINK BOLT HOLE WITH THE MTC ACTUATOR CLEVIS BOLT HOLE.



STEP 3: INSTALL THE BOLT IN THE LOCKOUT BOLT 'NORMAL' POSITION AND TIGHTEN THE BOLT TO 25.0 in-lb (2.8 N.m) - 30.0 in-lb (3.4 N.m).



STEP 4: MTC VALVE IN OPEN POSITION WITH BOLT INSTALLED IN THE 'NORMAL' POSITION.

2785789 S0000632680_V1

Left or Right Modulated Turbine Cooling Valve Inoperative: Restoration
Figure 904/75-00-00-990-818-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-00-00-440-806-G00

3. DDG 75-23-02 Left or Right Modulated Turbine Cooling Valve Inoperative - Restoration

(Figure 901, Figure 902, and Figure 904)

A. General

(1) This task restores the airplane back to its usual condition after operation with an inoperative left or right Modulated Turbine Cooling (MTC) valve.

NOTE: For Post LEAP-1B Service Bulletin 72-0124, the left side MTC valve and actuator (aft looking forward) are removed.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
36-11-04-000-801	PRSOV Removal (P/B 401)
36-11-04-400-801	PRSOV Installation (P/B 401)
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A
G51411 [CP8005]	Cable, Safety, Stainless Steel, 0.024 inch (0.61 mm) Diameter	AMS 5689

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Activation

SUBTASK 75-00-00-860-017-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-00-00-860-018-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 75-00-00-860-019-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 75-00-00-860-060-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

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1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Left or Right Modulated Turbine Cooling Valve Activation

SUBTASK 75-00-00-440-001-G00

(1) Do the steps that follow to unlock one of the MTC valves (left or right) for the inoperative MTC system:

(a) If you want to unlock the left MTC valve [1], to these steps:

1) Make the left MTC valve [1] operative (not locked position) as follows (Figure 901 and Figure 904):

a) Remove the bolt [2] from the lockout position.

b) Lubricate the threads of the bolt [2] and the threaded insert in the pushrod connector (lockout bolt normal position) with Tiodize T8E-H grease, D50355 [CP2961].

c) Move the control arm of the left MTC valve [1] in order to put the MTC valve link into position between the halves of the pushrod connector and align the bolt holes.

d) Install the bolt [2] in the lockout bolt normal position.

NOTE: Move the left MTC valve [1] to the closed position for access as necessary.

e) Tighten the bolt [2] to 25.0 in-lb (2.8 N·m) - 30.0 in-lb (3.4 N·m).

f) Install safety safety cable, G50065 [CP8006], or cable, G51411 [CP8005], or safety wire, G02345 [CP8001], to the lockout bolt [2].

(b) If you want to unlock the right MTC valve [3], to these steps:

1) Remove the Pressure Regulating and Shutoff Valve (PRSOV), if it is necessary (TASK 36-11-04-000-801).

2) Make the right MTC valve [3] operative (not locked position) as follows (Figure 902, Figure 904):

a) Remove the bolt [4] from the right MTC valve [3] lockout position.

b) Lubricate the threads of the bolt [4] and the threaded insert in the pushrod connector (lockout bolt normal position) with Tiodize T8E-H grease, D50355 [CP2961].

c) Move the control arm of the right MTC valve [3] in order to put the MTC valve link into position between the halves of the pushrod connector and align the bolt holes.

d) Install the bolt [4] in the right MTC valve [3] lockout bolt normal position.

NOTE: It can be necessary to turn the MTC push-rod by hand CCW (FLA) in order for the bolt to clear the tube and be installed.

e) Move the right MTC valve [3] to the closed position for access as necessary.

f) Tighten the bolt [4] to 25.0 in-lb (2.8 N·m) - 30.0 in-lb (3.4 N·m).

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g) Install safety safety cable, G50065 [CP8006], or cable, G51411 [CP8005], or safety wire, G02345 [CP8001], to the lockout bolt [4].

3) Install the PRSOV, if it is removed (TASK 36-11-04-400-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-00-00-860-061-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-00-00-860-022-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 75-00-00-860-023-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-00-00-710-002-G00

(4) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.

SUBTASK 75-00-00-720-001-G00

(5) Uncheck the applicable MAINT light Status Message(s) on the MAINT LIGHT menu (TASK 31-65-00-750-801).

END OF TASK

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TASK 75-00-00-040-803-G00

4. DDG 75-24-01–01 High Pressure Turbine Active Clearance Control (HPTACC) Valve Inoperative - Preparation

(Figure 905)

A. General

(1) This task provides the instructions on how to prepare the airplane for flight with an inoperative High Pressure Turbine Active Clearance Control (HPTACC) valve.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Deactivation

SUBTASK 75-00-00-860-010-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 75-00-00-860-011-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 75-00-00-860-012-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 75-00-00-860-013-G00

(4) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. High Pressure Turbine Active Clearance Control Valve Deactivation

SUBTASK 75-00-00-040-003-G00

(1) Deactivate the inoperative HPTACC valve [1] in the locked position as follows:

NOTE: The HPTACC valve must be in the closed position.

(a) Rotate the valve linkage [6] clockwise to move the valve into the closed position.

(b) Loosen the two lockout bolts [2] on the HPTACC valve [1].

NOTE: The two lockout bolts must be loose enough so the tab on the unlock plate can pass under the tabs on the face plate.

(c) Lift the face plate [4] and push the unlock plate [3] down until you see "LOCK" on the lock plate [5].

(d) Tighten the lockout bolts [2] to 80.0 in-lb (9.04 N·m) - 85.0 in-lb (9.60 N·m).

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-00-00-860-014-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-00-00-440-012-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 75-00-00-860-016-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

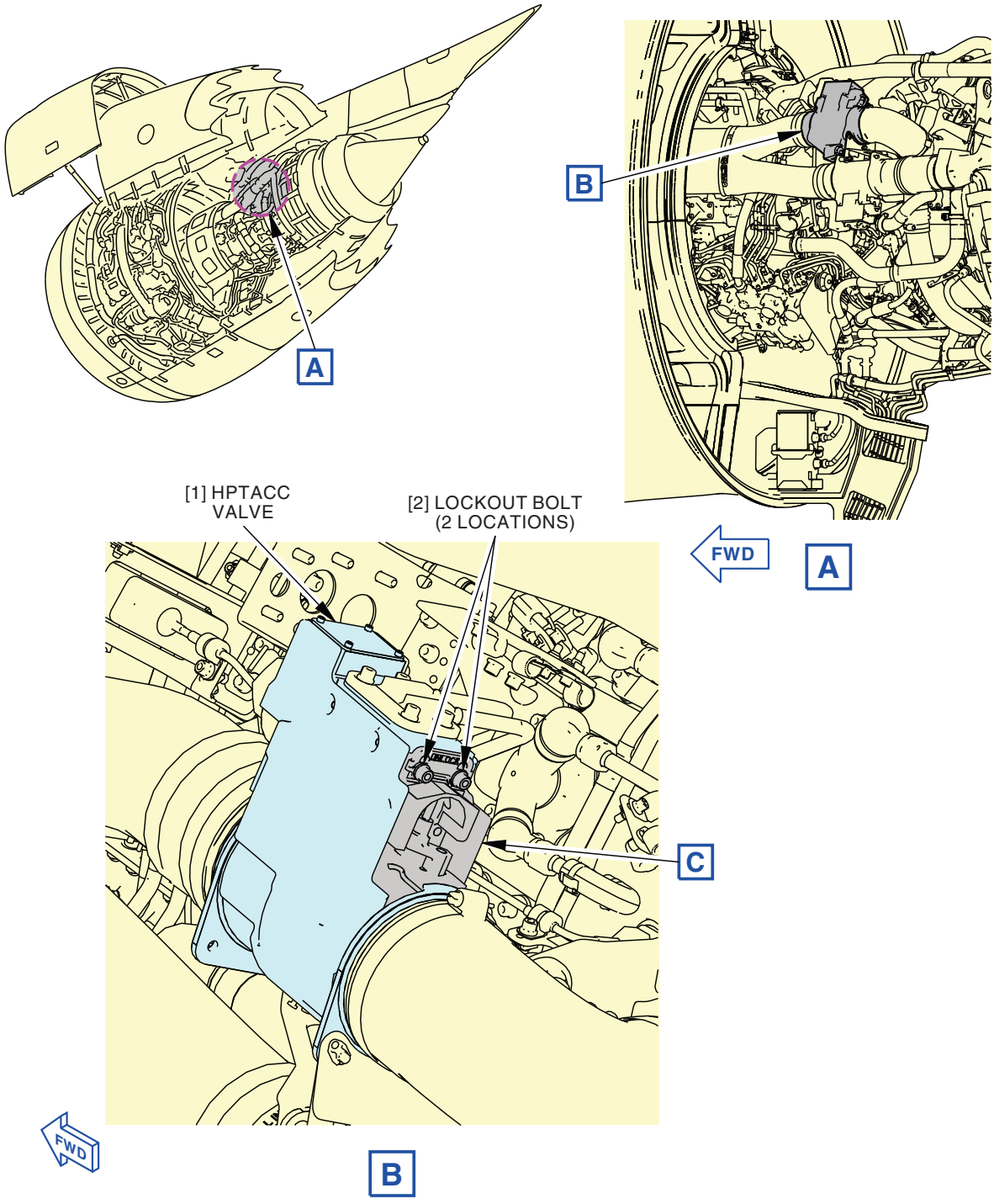
(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-00-00-040-012-G00

(4) If it is necessary, extinguish the MAINT light on the P5-1 panel, do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

END OF TASK

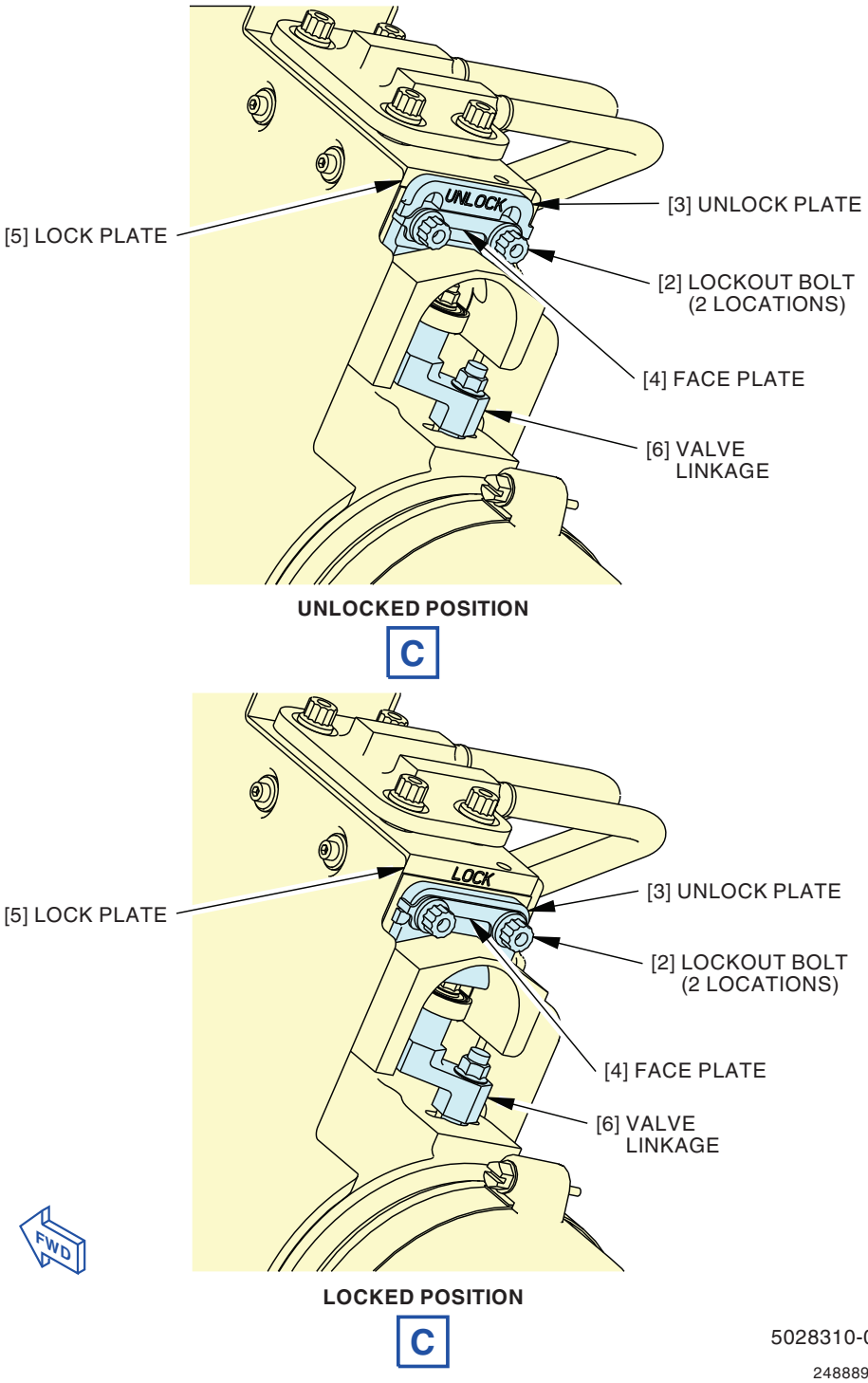
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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High Pressure Turbine Active Clearance Control Valve Inoperative
Figure 905/75-00-00-990-803-G00 (Sheet 1 of 2)

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High Pressure Turbine Active Clearance Control Valve Inoperative
Figure 905/75-00-00-990-803-G00 (Sheet 2 of 2)

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TASK 75-00-00-440-803-G00

5. DDG 75-24-01–01 High Pressure Turbine Active Clearance Control (HPTACC) Valve Inoperative - Restoration

(Figure 905)

A. General

(1) This task provides the instructions on how to put the airplane back to its usual condition after operation with an inoperative High Pressure Turbine Active Clearance Control (HPTACC) valve.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Activation

SUBTASK 75-00-00-860-026-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-00-00-860-027-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 75-00-00-860-028-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 75-00-00-860-029-G00

(4) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. High Pressure Turbine Active Clearance Control Valve Activation

SUBTASK 75-00-00-440-003-G00

(1) Put the HPTACC valve [1] to the operative (unlocked position) as follows:

(a) Loosen the two lockout bolts [2] on the HPTACC valve [1].

NOTE: The two lockout bolts must be loose enough so the tab on the unlock plate can pass under the tabs on the face plate.

(b) Lift the face plate [4] and pull the unlock plate [3] up until you see "UNLOCK" on the unlock plate [3].

(c) Tighten the lockout bolts [2] to 80.0 in-lb (9.04 N·m) - 85.0 in-lb (9.60 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-00-00-860-030-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-00-00-860-031-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 75-00-00-860-032-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 75-00-00-740-001-G00

(4) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00

SUBTASK 75-00-00-740-003-G00

(5) Uncheck the applicable MAINT light Status Message(s) on the MAINT LIGHT menu (TASK 31-65-00-750-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

START BLEED/BOOSTER ANTI-ICE VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Start bleed/booster anti-ice valve removal

(2) Start bleed/ booster anti-ice valve installation.

TASK 75-11-01-000-801-G00

2. Start Bleed/Booster Anti-Ice Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the start bleed and booster anti-ice valve.

(2) The start bleed and booster anti-ice valve is referred to as the start bleed/BAI valve.

(3) The start bleed/BAI valve is installed at the 5:00 o'clock position on the engine core.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-11-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

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1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-11-01-010-001-G00

(2) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Start Bleed/Booster Anti-Ice Valve Removal

SUBTASK 75-11-01-020-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [4] from the start bleed/booster anti-ice valve [1] as follows:

(a) Disconnect the electrical connector [4] from the start bleed/booster anti-ice valve [1] (TASK 70-00-01-910-803-G00).

(b) Install protective covers on the electrical connector [4] and the electrical receptacles on the start bleed/booster anti-ice valve [1].

SUBTASK 75-11-01-020-002-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Disconnect the SBV-BAI drain tube [5] from the start bleed/booster anti-ice valve [1] as follows:

(a) Put a fuel resistant container (5 gal), STD-201 below the SBV-BAI drain tube [5].

(b) Disconnect the SBV-BAI drain tube [5] from the start bleed/booster anti-ice valve [1].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

SUBTASK 75-11-01-020-003-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Disconnect the start-TBV manifolds [6] (2 locations) from the start bleed/booster anti-ice valve [1] as follows:

(a) Cut and remove the safety wire or safety cable from the B-nuts on the start-TBV manifolds [6].

(b) Put a fuel resistant container (5 gal), STD-201, below the start-TBV manifolds [6].

(c) Disconnect the B-nuts on the start-TBV manifolds [6] (2 locations) from the start bleed/booster anti-ice valve [1].

(d) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

SUBTASK 75-11-01-020-009-G00

(4) Loosen but do not remove the two bolts [34] on the saddle clamp [35].

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SUBTASK 75-11-01-010-003-G00

(5) Remove the two bolts [3] that attach the start bleed/booster anti-ice valve [1] to the SBV bracket [2].

SUBTASK 75-11-01-020-004-G00

(6) Remove the coupling [9] that attaches the BAI air tube [7] to the start bleed/booster anti-ice valve [1].

SUBTASK 75-11-01-020-006-G00

(7) Remove the coupling [11] that attaches the start bleed air tube [12] to the start bleed/booster anti-ice valve [1].

SUBTASK 75-11-01-020-007-G00

(8) Remove and discard the seal [10].

SUBTASK 75-11-01-420-001-G00

(9) Remove the coupling [14] that attaches the start bleed leg air duct [15] to the start bleed/booster anti-ice valve [1].

SUBTASK 75-11-01-010-004-G00

(10) Remove the start bleed/booster anti-ice valve [1] from the engine.

SUBTASK 75-11-01-020-013-G00

(11) Remove and discard the seal [8] and the seal [13].

SUBTASK 75-11-01-860-002-G00

(12) Install protective covers on these components:

(a) The start bleed/booster anti-ice valve [1]

(b) The SBV-BAI drain tube [5]

(c) The start-TBV manifolds [6]

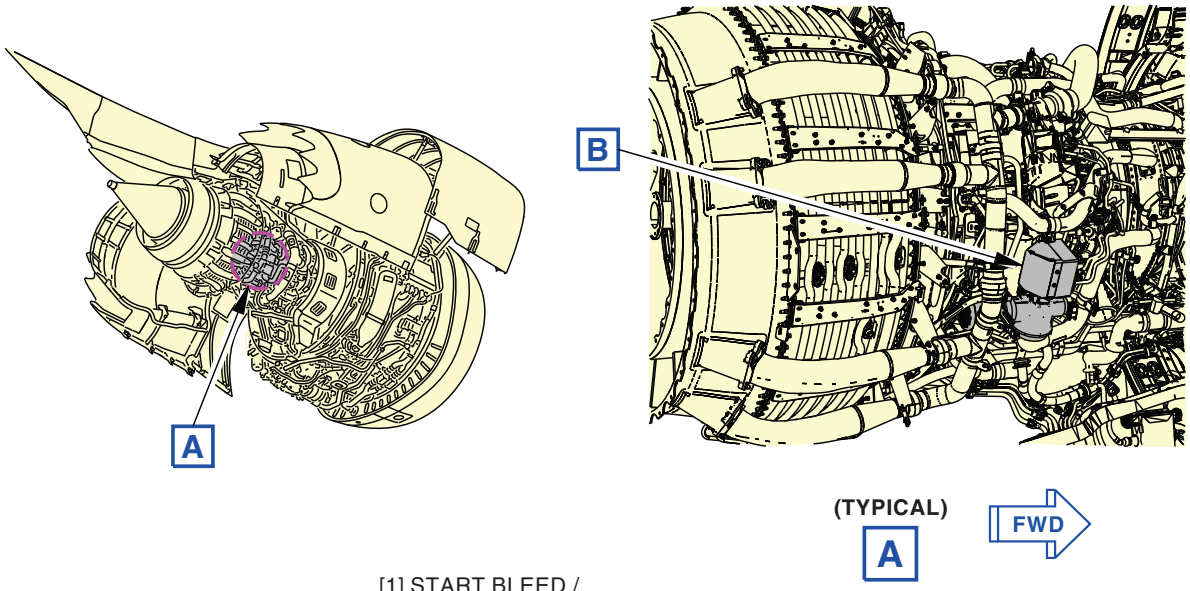
(d) The BAI air tube [7]

(e) The start bleed air tube [12]

(f) The start bleed leg air duct [15].

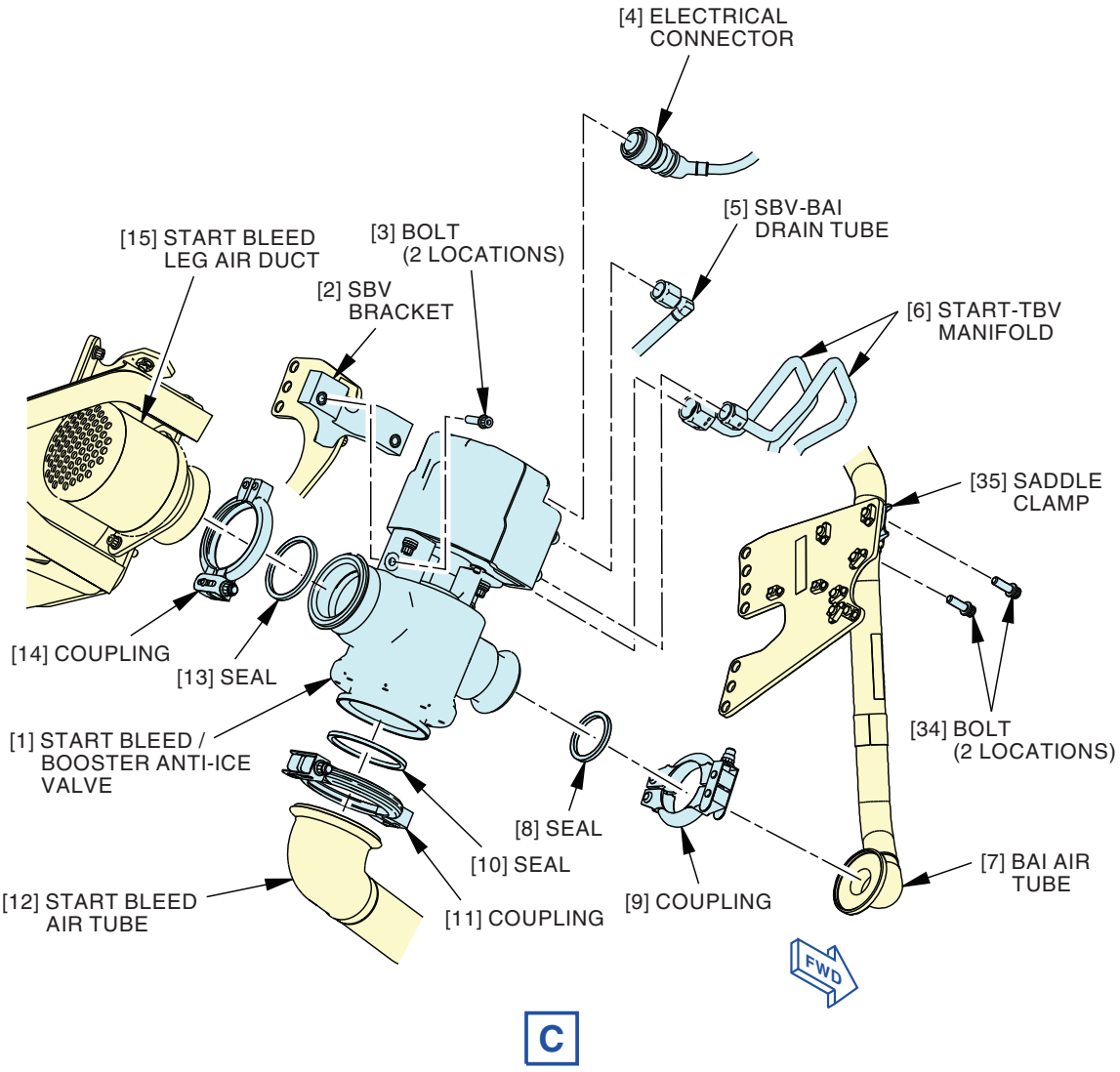
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Start Bleed/ Booster Anti-Ice Valve Installation
Figure 401/75-11-01-990-801-G00 (Sheet 1 of 3)

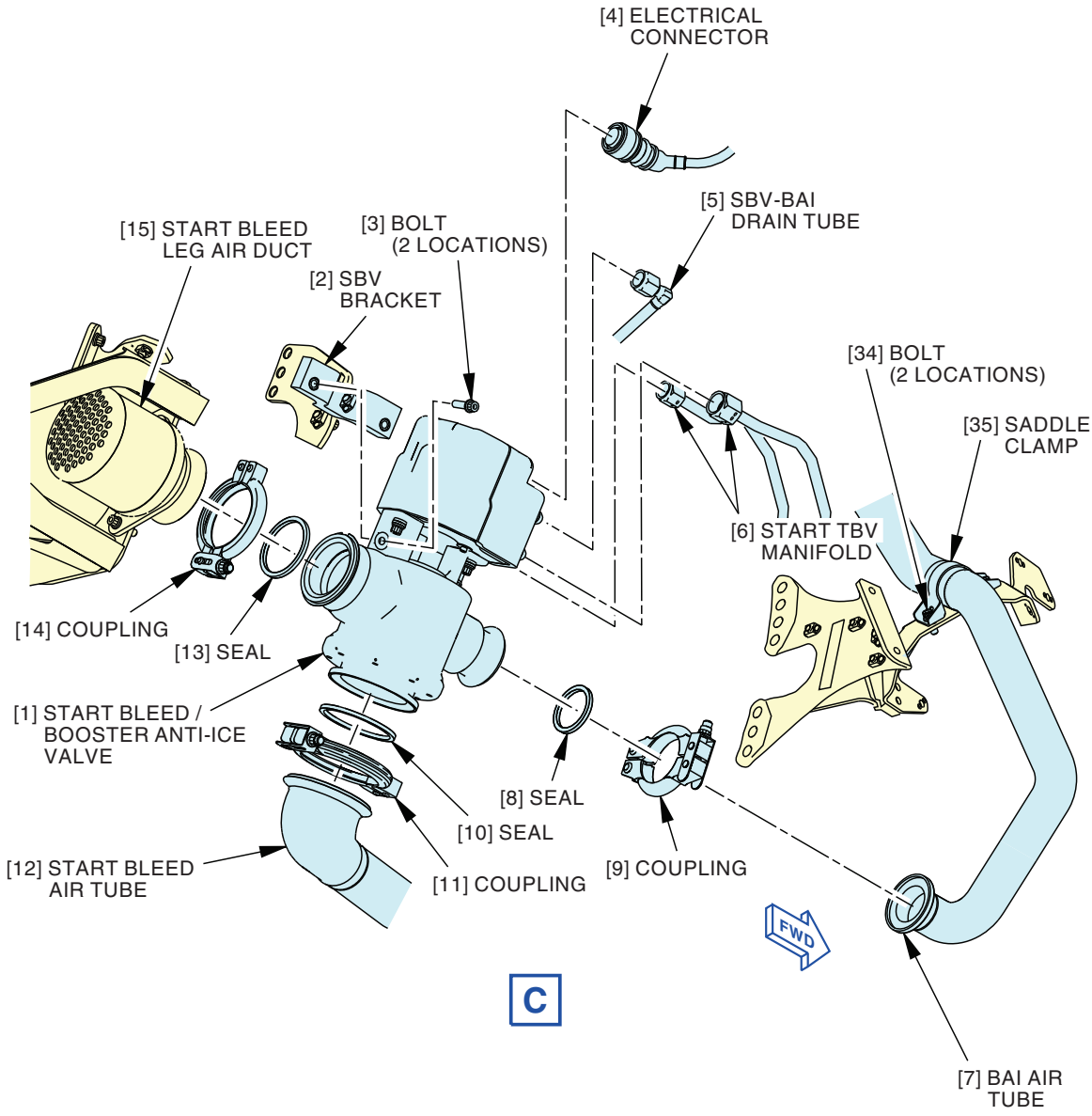
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Start Bleed/ Booster Anti-Ice Valve Installation
Figure 401/75-11-01-990-801-G00 (Sheet 2 of 3)

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Start Bleed/ Booster Anti-Ice Valve Installation
Figure 401/75-11-01-990-801-G00 (Sheet 3 of 3)

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TASK 75-11-01-400-801-G00

3. Start Bleed/Booster Anti-Ice Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the start bleed and booster anti-ice valve.

(2) The start bleed/booster anti-ice valve is referred to as the start bleed/BAI valve.

(3) You must do the tests that are listed in the power plant test reference table after you install the start bleed/BAI valve.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Start bleed/booster anti-ice valve	75-11-01-01-060	PTF ALL
8	Seal	75-11-00-03-230	PTF ALL
10	Seal	75-23-00-02-100	PTF ALL
13	Seal	75-11-01-01-040	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

H. Start Bleed/ Booster Anti-Ice Valve Installation

SUBTASK 75-11-01-020-008-G00

(1) Remove the protective covers from these components:

(a) The start bleed/booster anti-ice valve [1]

(b) The SBV-BAI drain tube [5]

(c) The start-TBV manifolds [6]

(d) The BAI air tube [7]

(e) The start bleed air tube [12]

(f) The start bleed leg air duct [15].

SUBTASK 75-11-01-420-016-G00

(2) Install the new seal [13], the new seal [10], and the new seal [8] into the ports of the start bleed/booster anti-ice valve [1].

SUBTASK 75-11-01-420-002-G00

(3) Put the start bleed/booster anti-ice valve [1] in position on the SBV bracket [2].

SUBTASK 75-11-01-420-006-G00

(4) Install the coupling [14] that attaches the start bleed leg air duct [15] to the start bleed/booster anti-ice valve [1].

(a) Tighten the coupling [14] as needed to align the start bleed/booster anti-ice valve [1] mounting holes with holes in the SBV bracket [2].

SUBTASK 75-11-01-420-003-G00

(5) Apply engine oil, D00599 [CP2442] on the threads and friction surfaces of the three bolts [3].

SUBTASK 75-11-01-420-004-G00

(6) Install the two bolts [3] that attach the start bleed/booster anti-ice valve [1] to the SBV bracket [2].

(a) Tighten the three bolts [3] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 75-11-01-420-008-G00

(7) Install the coupling [11] that attaches the start bleed air tube [12] to the start bleed/booster anti-ice valve [1].

(a) Tighten the coupling [11] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 75-11-01-410-001-G00

(8) Install the coupling [9] that attaches the BAI air tube [7] to the start bleed/booster anti-ice valve [1].

(a) Tighten the coupling [9] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 75-11-01-420-013-G00

(9) Tighten the coupling [14] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 75-11-01-420-017-G00

(10) Tighten the two bolts [34] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

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SUBTASK 75-11-01-410-002-G00

(11) Connect the start-TBV manifolds [6] to the start bleed/booster anti-ice valve [1].

(a) Tighten the larger start-TBV manifold [6] B-nut to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).

(b) Tighten the smaller start-TBV manifold [6] B-nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

SUBTASK 75-11-01-410-003-G00

(12) Connect the SBV-BAI drain tube [5] to the start bleed/booster anti-ice valve [1].

(a) Tighten the SBV-BAI drain tube [5] to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

SUBTASK 75-11-01-420-010-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(13) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [4] to the start bleed/booster anti-ice valve [1] as follows:

(a) Remove the protective covers from the start bleed/booster anti-ice valve [1] and the electrical connector [4].

(b) Connect the electrical connector [4] to the start bleed/booster anti-ice valve [1] (TASK 70-00-01-910-803-G00).

1) Tighten the electrical connector [4].

SUBTASK 75-11-01-420-018-G00

(14) Install a safety wire, G02345 [CP8001] or safety safety cable, G50065 [CP8006] on the start-TBV manifold [6] B-nuts (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-11-01-410-004-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

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1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-11-01-860-003-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Start Bleed/ Booster Anti-Ice Valve Installation Test

SUBTASK 75-11-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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ENGINE COOLING - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Engine Cooling Visual Inspection.

TASK 75-20-00-211-801-G00

2. Engine Cooling Visual Inspection

A. General

(1) This task gives the instructions to do visual inspection of the engine cooling system.

(2) You do an inspection of these components in this task:

(a) The Low Pressure Turbine Active Clearance Control (LPTACC) valve

(b) The High Pressure Turbine Active Clearance Control (HPTACC) valve

(c) The high pressure turbine and low pressure turbine cooling system air and fuel tubes

(d) The left and right Modulated Turbine Cooling (MTC) valves and actuators

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
70-00-01-300-801-G00	Correction of Leaks (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-00-05-200-801-G00	Tube and Hose Inspection Criteria and Damage Limits (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-22-01-000-801-G00	Electronic Engine Control (EEC) Electrical Harness (HJ3A) Removal (P/B 401)
73-22-01-400-801-G00	Electronic Engine Control (EEC) Electrical Harness (HJ3A) Installation (P/B 401)
74-21-01-400-801-G00	Left Ignition Lead Installation (P/B 401)
74-21-01-400-802-G00	Right Ignition Lead Installation (P/B 401)
75-23-02-000-801-G00	Left Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)
75-23-02-000-802-G00	Right Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)
75-23-02-400-801-G00	Left Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
75-23-02-400-802-G00	Right Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
75-23-03-000-801-G00	Left Modulated Turbine Cooling Actuator Removal (P/B 401)
75-23-03-000-802-G00	Right Modulated Turbine Cooling Actuator Removal (P/B 401)
75-23-03-400-801-G00	Left Modulated Turbine Cooling Actuator Installation (P/B 401)
75-23-03-400-802-G00	Right Modulated Turbine Cooling Actuator Installation (P/B 401)
75-24-01-000-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
75-24-01-400-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation (P/B 401)
75-24-04-000-801-G00	High Pressure Turbine Active Clearance Control (HPTACC) Valve Removal (P/B 401)
75-24-04-400-801-G00	High Pressure Turbine Active Clearance Control (HPTACC) Valve Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 75-20-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-20-00-860-005-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 75-20-00-040-001-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-20-00-860-002-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Engine Cooling Visual Inspection

SUBTASK 75-20-00-211-001-G00

(1) Examine the fuel tubes and manifolds for the MTC, HPTACC and LPTACC as follows (TASK 70-00-05-200-801-G00):

(a) Examine the areas that you can see for damage or contamination.

SUBTASK 75-20-00-211-002-G00

(2) Examine the air tubes and manifolds for the MTC, HPTACC and LPTACC and the High Pressure Turbine (HPT)/Low Pressure Turbine (LPT) cooling manifolds as follows (TASK 70-00-05-200-801-G00):

(a) Examine the areas that you can see for damage or contamination.

SUBTASK 75-20-00-210-001-G00

(3) Examine the Core Compartment Cooling (CCC) air duct as follows (TASK 70-00-05-200-801-G00):

(a) Look for signs of air leaks at the connections of the forward flexible duct and the igniter lead cooling duct (TASK 70-00-01-300-801-G00).

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1) If you see signs of air leaks, tighten the hose clamps on the CCC air manifold (TASK 74-21-01-400-801-G00 and TASK 74-21-01-400-802-G00).

(b) Look for cracks in the CCC air manifold.

1) If you see a crack, replace the CCC air manifold.

SUBTASK 75-20-00-210-002-G00

(4) Examine the LPTACC valve as follows:

(a) Look for signs of fuel leaks.

1) It is permitted to have a valve that has a fuel stain, if you do not find drops of fuel.

NOTE: A valve shaft that is wet is permitted, if there are no drops of fuel.

2) A fuel leak that makes drops of fuel is not permitted.

3) A fuel leak from the valve housing or shaft is not permitted.

4) If you find signs of a fuel leak, do these steps:

a) Replace the gasket seal between the Active Clearance Control (ACC) fuel manifold the LPTACC valve (TASK 75-24-01-000-801-G00 and TASK 75-24-01-400-801-G00).

b) Make sure that the installation of the fuel tubes that connect to the ACC fuel manifold is correct and tight.

<1> If the installation of the fuel tubes are not correct or are loose, re-install and tighten the fuel tube B-nuts (TASK 75-24-01-400-801-G00).

c) If the fuel leak continues, replace the LPTACC valve (TASK 75-24-01-000-801-G00 and TASK 75-24-01-400-801-G00).

(b) Look for cracks and dents in the LPTACC valve.

1) If you see a crack or a dent, replace the LPTACC valve (TASK 75-24-01-000-801-G00 and TASK 75-24-01-400-801-G00).

(c) Look for signs of air leaks at the connections of the flexible duct and the hose clamps on the ends of the LPTACC valve.

1) If you see signs of an air leak, do these steps.

a) Identify the air leakage (TASK 70-00-01-300-801-G00).

b) Install and tighten the hose clamps on the LPTACC valve (TASK 75-24-01-400-801-G00).

(d) Examine the electrical receptacles and electrical connectors for the LPTACC valve.

1) If the electrical connector is loose, tighten the electrical connector (TASK 70-00-01-910-803-G00).

a) If an electrical connector has damage that prevents correct installation, replace the wiring harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

2) If there is damage to the electrical receptacle that prevents the correct installation of the electrical connector, do this step:

a) Replace the LPTACC valve (TASK 75-24-01-000-801-G00 and TASK 75-24-01-400-801-G00).

3) Small damage to the LPTACC valve electrical receptacle threads is permitted.

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a) Repair the damage until you can tighten the connector and the connector can hold torque.

SUBTASK 75-20-00-210-003-G00

(5) Examine the HPTACC valve as follows:

(a) Look for signs of fuel leaks.

1) It is permitted to have a valve that has a fuel stain if you do not find drops of fuel.

NOTE: A valve shaft that is wet is permitted, if there are no drops of fuel.

2) A fuel leak that makes drops of fuel is not permitted.

3) A fuel leak from the valve housing or shaft is not permitted.

4) If you find signs of a fuel leak, do these steps:

a) Replace the gasket seal between the ACC fuel manifold the HPTACC valve (TASK 75-24-04-000-801-G00 and TASK 75-24-04-400-801-G00).

b) Make sure that the installation of the fuel tubes that connect to the ACC fuel manifold is correct and tight.

c) If the installation of the fuel tubes are not correct or are loose, re-install and tighten the fuel tube B-nuts (TASK 75-24-04-400-801-G00).

d) If the fuel leak continues, replace the HPTACC valve (TASK 75-24-04-000-801-G00 and TASK 75-24-04-400-801-G00).

(b) Look for cracks and dents in the HPTACC valve.

1) If you see a crack or dent, replace the HPTACC valve (TASK 75-24-04-000-801-G00 and TASK 75-24-04-400-801-G00).

(c) Look for signs of air leaks at the connections of the flexible duct and the hose clamps on the ends of the HPTACC valve.

1) Identify the air leakage (TASK 70-00-01-300-801-G00).

2) If you see signs of an air leak, do these steps.

a) Re-install and tighten the hose clamps on the HPTACC valve (TASK 75-24-04-400-801-G00).

(d) Examine the electrical receptacle and connector for the HPTACC valve.

1) If the electrical connector is loose, tighten the electrical connector (TASK 70-00-01-910-803-G00).

a) If an electrical connector has damage that prevents its correct installation, replace the wiring harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

2) If there is damage to the electrical receptacle that prevents the correct installation of the electrical connector, do this step:

a) Replace the HPTACC valve (TASK 75-24-04-000-801-G00 and TASK 75-24-04-400-801-G00).

3) Small damage to the electrical receptacle threads is permitted.

a) Repair the damage to the connector until you can tighten the connector and the connector can hold torque.

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SUBTASK 75-20-00-210-004-G00

(6) Examine the left and right MTC valve and actuator as follows:

NOTE: For Post LEAP-1B Service Bulletin 72-0124, the left side MTC valve and actuator (aft looking forward) are removed.

(a) Look for signs of fuel leaks at the MTC actuator.

1) It is permitted to have an actuator that has a fuel stain, if you do not find drops of fuel.

NOTE: An actuator shaft that is wet is permitted, if there are no drops of fuel.

2) A fuel leak that makes drops of fuel is not permitted.

3) A fuel leak from the actuator housing or shaft is not permitted.

4) If you find signs of a fuel leak, do these steps:

a) Replace the gasket seal between the MTC fuel manifold the MTC actuator (TASK 75-23-03-000-801-G00 or TASK 75-23-03-000-802-G00 and TASK 75-23-03-400-801-G00 or TASK 75-23-03-400-802-G00).

b) Make sure that the installation of the fuel tubes connecting to the MTC fuel manifold is correct and tight.

<1> If the installation of the fuel tubes are not correct or are loose, re-install and tighten the fuel tube B-nuts (TASK 75-23-03-400-801-G00 or TASK 75-23-03-400-802-G00).

c) If the fuel leak continues, replace the MTC actuator (TASK 75-23-03-000-801-G00 or TASK 75-23-03-000-802-G00 and TASK 75-23-03-400-801-G00 or TASK 75-23-03-400-802-G00).

(b) If the MTC actuator is loose, tighten the bolts and nuts (TASK 75-23-03-400-801-G00 or TASK 75-23-03-400-802-G00).

(c) If there are cracks and dents on the MTC actuator, scores on the piston rod, or if the piston rod is bent, replace the MTC actuator (TASK 75-23-03-000-801-G00 or TASK 75-23-03-000-802-G00 and TASK 75-23-03-400-801-G00 or TASK 75-23-03-400-802-G00).

(d) Examine the electrical receptacle and electrical connectors for the MTC actuator.

1) If the electrical connector is loose, tighten the electrical connector (TASK 70-00-01-910-803-G00).

a) If an electrical connector has damage that prevents its correct installation, replace the wiring harness (TASK 73-22-01-000-801-G00 and TASK 73-22-01-400-801-G00).

2) If there is damage to the electrical receptacle that prevents the correct installation of the electrical connector, do this step:

a) Replace the MTC actuator (TASK 75-23-03-000-801-G00 or TASK 75-23-03-000-802-G00 and TASK 75-23-03-400-801-G00 or TASK 75-23-03-400-802-G00).

3) Small damage to the MTC actuator electrical receptacle threads is permitted.

a) Repair the damage until you can tighten the connector and the connector can hold torque.

(e) If the MTC valve is loose, tighten the bolts (TASK 75-23-02-000-801-G00 or TASK 75-23-02-000-802-G00).

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(f) Examine the MTC valve for signs of air leaks.

1) If you see signs of an air leak at the combustor case, replace the seals on the MTC valve at the combustor case.

2) If the leak continues, replace the MTC valve (TASK 75-23-02-000-801-G00 or TASK 75-23-02-000-802-G00 and TASK 75-23-02-400-801-G00 or TASK 75-23-02-400-802-G00).

(g) Look for cracks, dents, and damage to MTC valve linkage.

1) If you see a crack, dent, or damage to linkage, replace the MTC valve (TASK 75-23-02-000-801-G00 or TASK 75-23-02-000-802-G00 and TASK 75-23-02-400-801-G00 or TASK 75-23-02-400-802-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-20-00-860-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-20-00-440-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

SUBTASK 75-20-00-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MODULATED TURBINE COOLING (MTC) VALVE - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) A manual operation of the left or right MTC valve.

TASK 75-23-02-980-801-G00

2. Left or Right Modulated Turbine Cooling (MTC) Valve Manual Operation

(Figure 201, Figure 202)

A. General

(1) This task gives the instructions to manually operate the left or right Modulated Turbine Cooling (MTC) valve.

(2) The modulated turbine cooling valve is referred to as the MTC valve.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124

NOTE: The left side MTC valve and actuator (aft looking forward) are removed.

PTF ALL

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
36-11-04-000-801	PRSOV Removal (P/B 401)
36-11-04-400-801	PRSOV Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
75-00-00-040-806-G00	DDG 75-23-02 Left or Right Modulated Turbine Cooling Valve Inoperative - Preparation (P/B 901)
75-23-02-000-801-G00	Left Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)
75-23-02-000-802-G00	Right Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)
75-23-02-400-801-G00	Left Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
75-23-02-400-802-G00	Right Modulated Turbine Cooling (MTC) Valve Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Consumable Materials

Reference	Description	Specification
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A
G51411 [CP8005]	Cable, Safety, Stainless Steel, 0.024 inch (0.61 mm) Diameter	AMS 5689

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Manual Operation

SUBTASK 75-23-02-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-23-02-860-006-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 75-23-02-040-001-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 75-23-02-860-007-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Manual Operation of the Left or Right Modulated Turbine Cooling (MTC) Valve

SUBTASK 75-23-02-980-001-G00

(1) Do these steps to manually operate the left or right MTC valves:

(a) Manually actuate the left MTC valve [1] as follows (Figure 201):

NOTE: The manual operation of the left MTC valve will determine if the left MTC valve is serviceable.

1) Do a check of the bolt [2] of the left MTC valve [1].

a) If the bolt [2] is installed in the bolt normal position, continue to manually operate the left MTC valve [1].

b) If the bolt [2] is installed in the lockout position, do the steps that follow:

<1> Cut the safety cable or safety wire from the bolt [2].

<2> Move the left MTC valve [1] actuating arm to align the MTC valve link with the holes in the MTC valve pushrod connector.

<3> Lubricate the bolt [2] threads and threads of the left MTC actuator clevis thread insert with Tiodize T8E-H grease, D50355 [CP2961].

<4> Install the bolt [2] in the bolt normal position threaded insert on the left MTC valve [1].

<5> Tighten the bolt [2] to 25 in-lb (2.8 N·m) - 30 in-lb (3.4 N·m).

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<6> Install safety safety cable, G50065 [CP8006], or cable, G51411 [CP8005], or safety wire, G02345 [CP8001], to the bolt [2].

<7> Remove the bolt and nut that attach the left MTC valve [1] and clevis to the left MTC actuator (TASK 75-23-02-000-801-G00).

2) Move the operating lever by hand to manually operate the left MTC valve [1] through its full range.

a) If the left MTC valve [1] fails, do the manual actuation check:

<1> Lock the left MTC valve [1] out (TASK 75-00-00-040-806-G00), or

<2> Replace the left MTC valve [1] (TASK 75-23-02-000-801-G00 and TASK 75-23-02-400-801-G00).

3) Install and tighten the nut and bolt that attach the left MTC valve [1] and clevis to the left MTC actuator (TASK 75-23-02-400-801-G00).

(b) Manually actuate the right MTC valve [3] as follows (Figure 202):

NOTE: The manual operation of the right MTC valve will determine if the right MTC valve is serviceable.

1) If it is necessary, remove the Pressure Regulating and Shutoff Valve (PRSOV) (TASK 36-11-04-000-801).

2) Do a check of the bolt [4] of the right MTC valve [3].

a) If the bolt [4] is installed in the bolt normal position, continue to manually operate the right MTC valve [3].

b) If the bolt [4] is installed in the lockout position, do the steps that follow:

<1> Cut the safety cable or safety wire from the bolt [4].

<2> Move the right MTC valve [3] actuating arm to align the MTC valve link with the holes in the MTC valve pushrod connector.

<3> Lubricate the bolt [4] threads and threads of the right MTC actuator clevis thread insert with Tiodize T8E-H grease, D50355 [CP2961].

<4> Install the bolt [4] in the bolt normal position threaded insert on the right MTC valve [3].

<5> Tighten the bolt [4] to 25 in-lb (2.8 N·m) - 30 in-lb (3.4 N·m).

<6> Install safety safety cable, G50065 [CP8006], or cable, G51411 [CP8005], or safety wire, G02345 [CP8001], to the bolt [4].

<7> Remove the bolt and nut that attach the right MTC valve [3] and clevis to the right MTC actuator (TASK 75-23-02-000-802-G00).

3) Move the operating lever by hand to manually operate the right MTC valve [3] through its full range.

a) If the right MTC valve [3] fails, do the manual actuation check:

<1> Lock the right MTC valve [3] out (TASK 75-00-00-040-806-G00), or

<2> Replace the right MTC valve [3] (TASK 75-23-02-000-802-G00 and TASK 75-23-02-400-802-G00).

4) Install and tighten the nut and bolt that attach the right MTC valve [3] and clevis to the right MTC actuator (TASK 75-23-02-400-802-G00).

5) Install the PRSOV, if it is removed (TASK 36-11-04-400-801).

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I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-23-02-860-008-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-23-02-860-009-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 75-23-02-860-010-G00

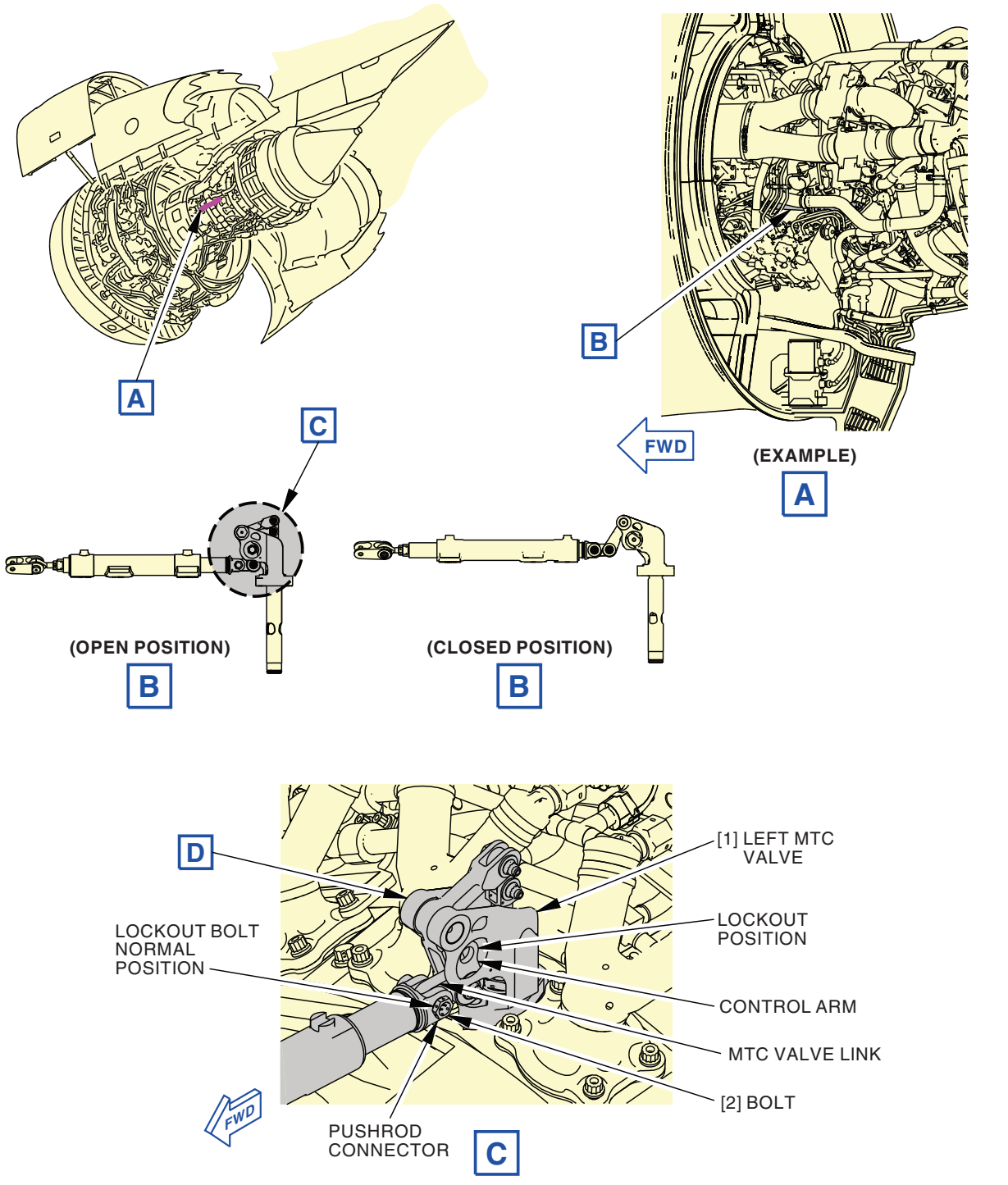
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

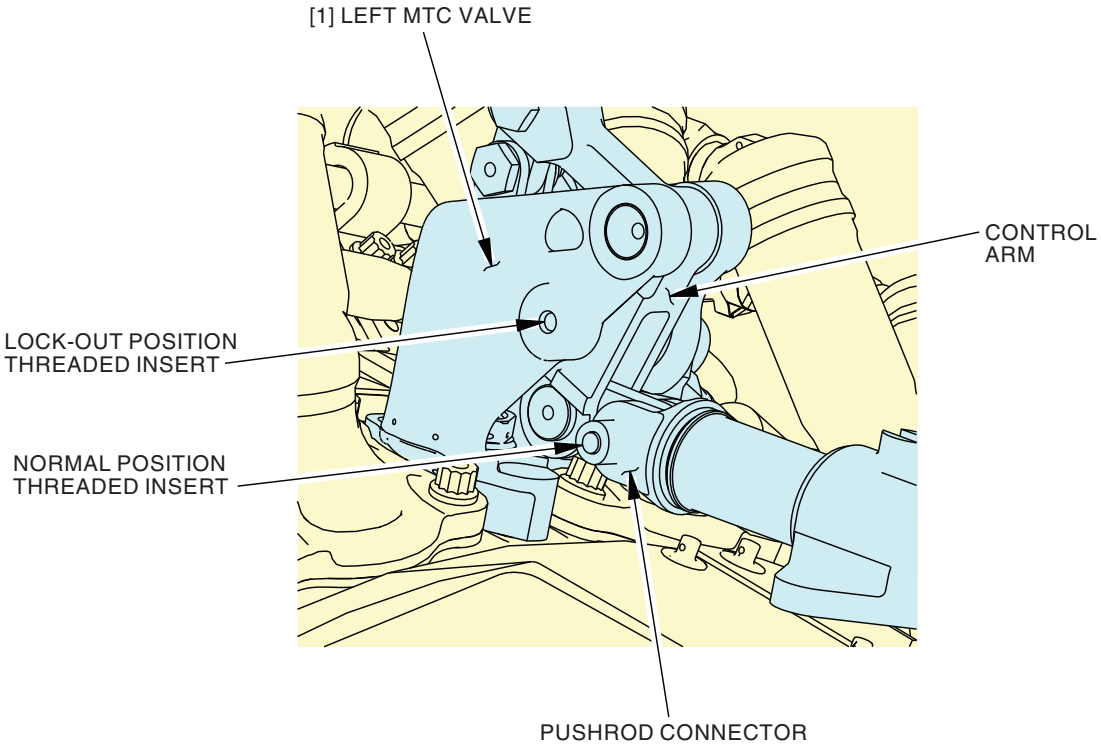
——— END OF TASK ———

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Left Modulated Turbine Cooling Valve
Figure 201/75-23-02-990-810-G00 (Sheet 1 of 2)

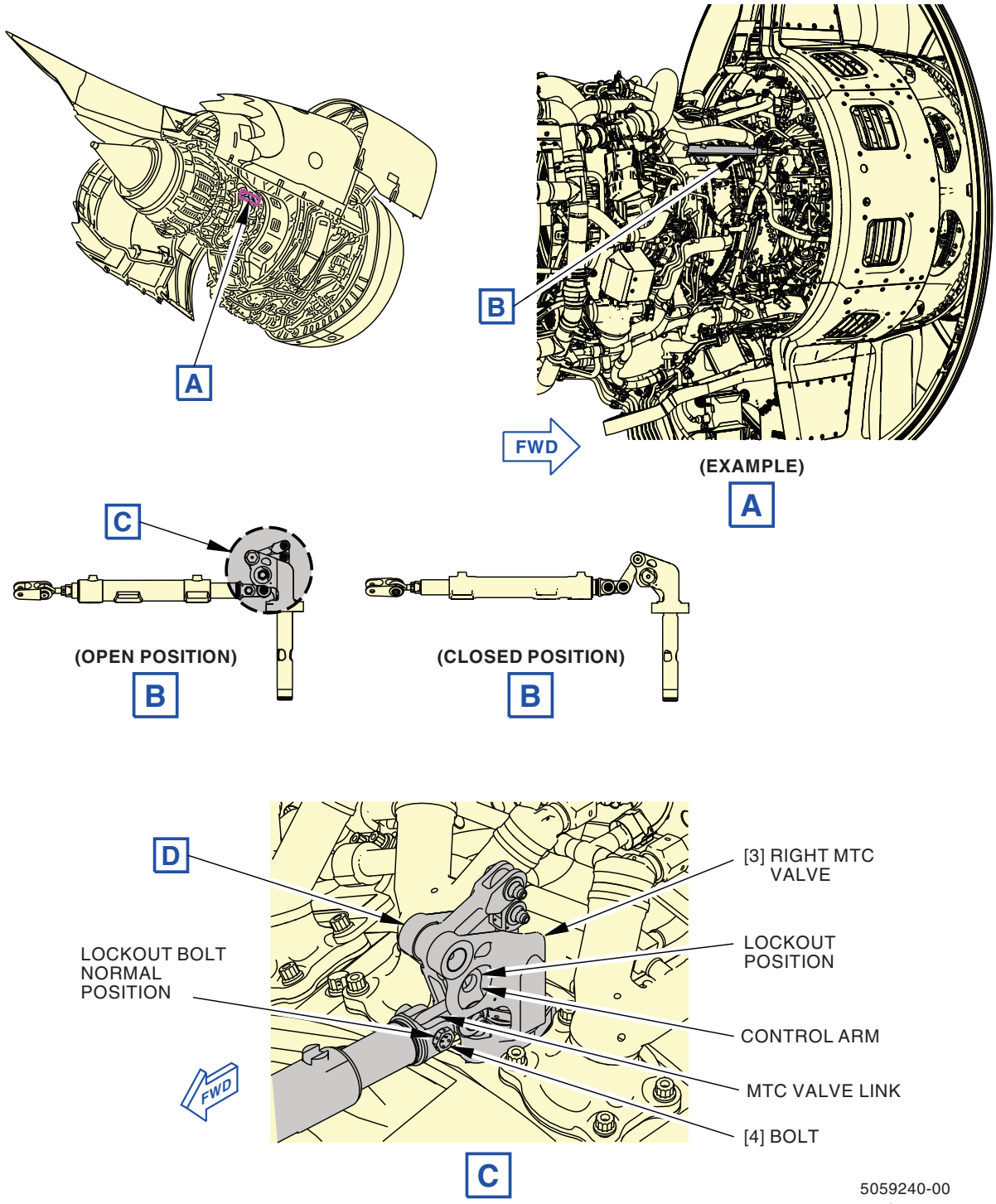
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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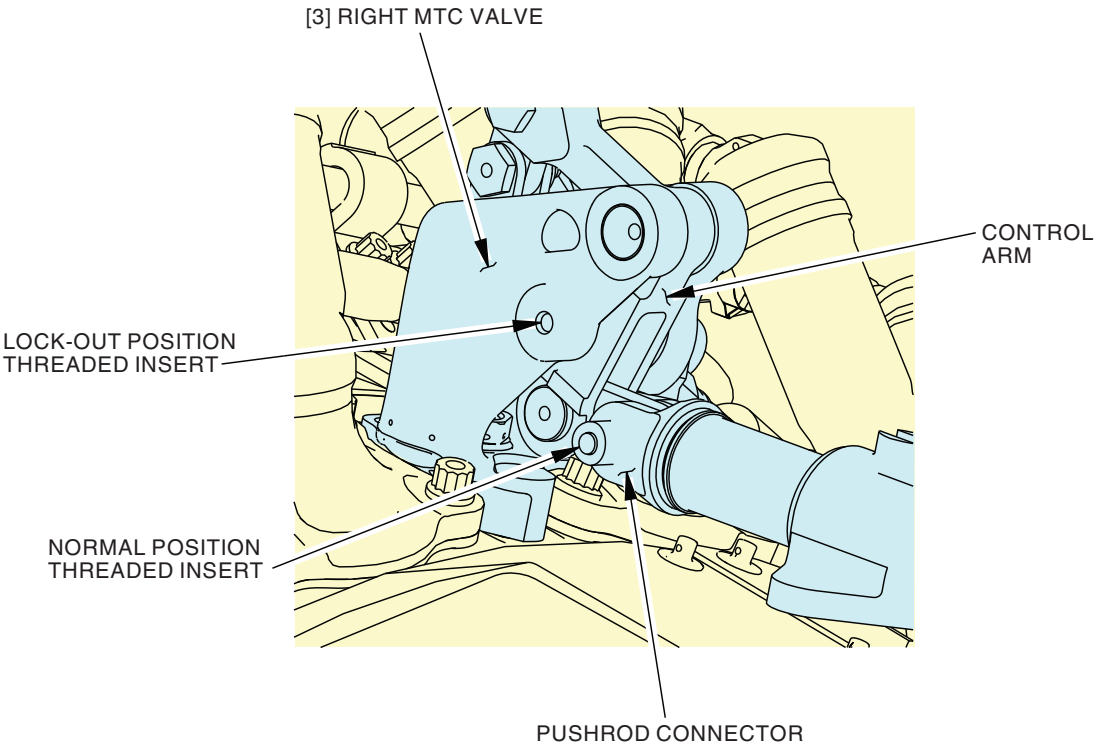
Left Modulated Turbine Cooling Valve
Figure 201/75-23-02-990-810-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right Modulated Turbine Cooling Valve
Figure 202/75-23-02-990-811-G00 (Sheet 1 of 2)

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3058365 S0000821876_V1

Right Modulated Turbine Cooling Valve
Figure 202/75-23-02-990-811-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MODULATED TURBINE COOLING (MTC) VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

(1) Left Modulated Turbine Cooling (MTC) Valve Removal

(2) Left Modulated Turbine Cooling (MTC) Valve Installation

PTF ALL

(3) Right Modulated Turbine Cooling (MTC) Valve Removal

(4) Right Modulated Turbine Cooling (MTC) Valve Installation.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

TASK 75-23-02-000-801-G00

2. Left Modulated Turbine Cooling (MTC) Valve Removal

(Figure 401, Figure 402, Figure 403, and Figure 404)

A. General

(1) This task gives the instructions to remove the left Modulated Turbine Cooling (MTC) valve.

(2) The left modulated turbine cooling valve is referred to as the left MTC valve.

(3) The left MTC valve is installed at the 8:00 o'clock position on the engine core.

(4) To get access to the left MTC valve open the left thrust reverser..

(5) The MTC valve and valve body must be maintained as a set.

B. References

Reference	Title
36-11-01-000-801	Engine Pneumatic Duct Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(Continued)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

G. Prepare for the Removal

SUBTASK 75-23-02-860-001-G00

(1) Do these steps to make sure that the fuel control valve and spar valve stay in the closed position:

(a) Make sure that the L (R) START LEVER switch is in the CUTOFF position.

1) Install a DO NOT OPERATE tag, STD-858, on the L (R) START LEVER switch.

(b) Make sure that the applicable SPAR VALVE CLOSED and ENG VALVE CLOSED lights, on the P5 forward overhead panel, are on dimly.

SUBTASK 75-23-02-865-001-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-23-02-010-001-G00

(3) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

H. Left Modulated Turbine Cooling (MTC) Valve Removal

SUBTASK 75-23-02-010-002-G00

(1) If it is necessary, remove the applicable ECS duct [1A] on the left side of the engine (TASK 36-11-01-000-801).

PTF ALL; ALL 737 MAX, PRE LEAP-1B SERVICE BULLETIN 72-0050 AND PRE LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 75-23-02-010-005-G00

(2) Loosen the start and TBV manifold [91] as follows (Figure 402):

(a) Put a fuel resistant container (5 gal), STD-201, below the start and TBV manifold [91] at the SCU [39].

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Remove the four bolts [135] that attach the start and TBV manifold [91] and the gasket seal [134] to the SCU [39].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Remove the bolt [122] and bolt [121] that attach the loop clamp [123], loop clamp [120], and start and TBV manifold [91] to the bracket [109] and bracket [108].

(e) Remove the bolt [112], bolt [114], and bolt [115] that attach the loop clamp [111], loop clamp [113], loop clamp [116], and start and TBV manifold [91] to the bracket [118] and bracket [117].

(f) Remove the two bolts [119] that attach the start and TBV manifold [91] to the bracket [110].

(g) Remove the bolt [129], bolt [130], bolt [128], and bolt [127] that attach the loop clamp [126], loop clamp [131], loop clamp [132], loop clamp [125], and start and TBV manifold [91] to the bracket [124] and bracket [133].

(h) Remove and discard the gasket seal [134].

(i) Put the protective covers on the SCU [39] and start and TBV manifold [91].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 75-23-02-020-001-G00

(3) Loosen the start and TBV manifold [91] as follows (Figure 402):

(a) Put a fuel resistant container (5 gal), STD-201, below the start and TBV manifold [91] at the SCU [39].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

> **WARNING** DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Remove the four bolts [135] that attach the start and TBV manifold [91] and gasket seal [134] to the SCU [39].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Remove the two bolts [140] that attach the loop clamps [137] and start and TBV manifold [91] to the bracket [138] and bracket [139].

(e) Remove the two bolts [136] that attach the loop clamps [137] and start and TBV manifold [91] to the bracket [138].

(f) Remove the two bolts [119] that attach the clamp [142] and start and TBV manifold [91] to the bracket [141].

(g) Remove the two bolts [143] that attach the loop clamp [113], loop clamp [116], and start and TBV manifold [91] to the bracket [144].

(h) Remove the bolt [112] that attaches the loop clamp [111] and start and TBV manifold [91] to the bracket [144].

(i) Remove the bolt [129], bolt [130], bolt [128], and bolt [127] that attach the loop clamp [126], loop clamp [131], loop clamp [132], loop clamp [125], and start and TBV manifold [91] to the bracket [124] and bracket [133].

(j) Remove and discard the gasket seal [134].

(k) Put the protective covers on the SCU [39] and the start and TBV manifold [91].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 75-23-02-020-002-G00

(4) Loosen the start and TBV manifold [91] as follows (Figure 402):

(a) Put a fuel resistant container (5 gal), STD-201, below the start and TBV manifold [91] at the SCU [39].

> **WARNING** DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Remove the four bolts [135] that attach the start and TBV manifold [91] and the gasket seal [134] to the SCU [39].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Remove the two bolts [121] and two bolts [122] that attach the loop clamps [120] and start and TBV manifold [91] to the bracket [109]

(e) Remove the three bolts [112] and bolt [114] that attach the loop clamps [111]and start and TBV manifold [91] to the bracket [118].

(f) Remove the three bolts [150] and bolt [119] that attach the loop clamps [145] and start and TBV manifold [91] to the bracket [110].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(g) Remove the two bolts [127] that attach the rubber spacer [147] and start and TBV manifold [91] to the bracket [124].

(h) Remove and discard the gasket seal [134].

(i) Put the protective covers on the SCU [39] and start and TBV manifold [91].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

SUBTASK 75-23-02-010-007-G00

(5) Remove the LPT ACC drain tube [171] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the LPT ACC drain tube [171].

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Disconnect the LPT ACC drain tube [171] from the ACC valve drain tube [174].

(c) Disconnect the LPT ACC drain tube [171] from the ACC fuel manifold [48].

(d) Remove the bolt [175] that attaches the loop clamp [172] and LPT ACC drain tube [171] to the bracket [173].

(e) Remove the LPT ACC drain tube [171] from the engine.

(f) Put the protective covers on the LPT ACC drain tube [171], ACC fuel manifold [48], and ACC valve drain tube [174].

SUBTASK 75-23-02-010-008-G00

(6) Remove the left MTC valve [197] as follows:

NOTE: The MTC valve and valve body must be kept as a set.

(a) Remove the bolt [207] that attaches the fuel nozzle (position 13) [208] and bracket [206] to the combustion case [191].

(b) Disconnect the secondary fuel manifold hose [210] B-nut from the SCU [39].

(c) Remove the bolt [211] that attaches the loop clamp [212] and secondary fuel manifold hose [210] to the bracket [209].

(d) Remove the bolt [200] and nut [201] that attach the clevis [199] to the left MTC actuator [202].

(e) Put the left MTC valve [197] in the open position.

(f) Apply penetrating fluid, D50186 [CP2691], to the bolt [194], bolt [195], and threads that extend through the combustion case [191] inserts.

NOTE: The step to apply penetrating fluid, D50186 [CP2691], is necessary to decrease the risk of damage to the threaded inserts.

1) Give penetrating fluid, D50186 [CP2691], time to soak into threads.

(g) Remove the bolt [195] that attaches the left MTC valve [197] to the combustion case [191].

(h) Remove the bolt [194] that attaches the left MTC valve [197] to the combustion case [191].

(i) Remove the two bolts [196] that attach the left MTC valve [197] to the bracket [192].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(j) Remove the two bolts [198] that attach the left MTC valve [197] to the bracket [203].

CAUTION DURING THE REMOVAL OR INSTALLATION, MAKE SURE THAT THE VALVE STAYS PERPENDICULAR TO THE CASE. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(k) Remove the left MTC valve [197] from the engine by rotating the left MTC valve body between the start and TBV manifold [91] and the SCU [39].

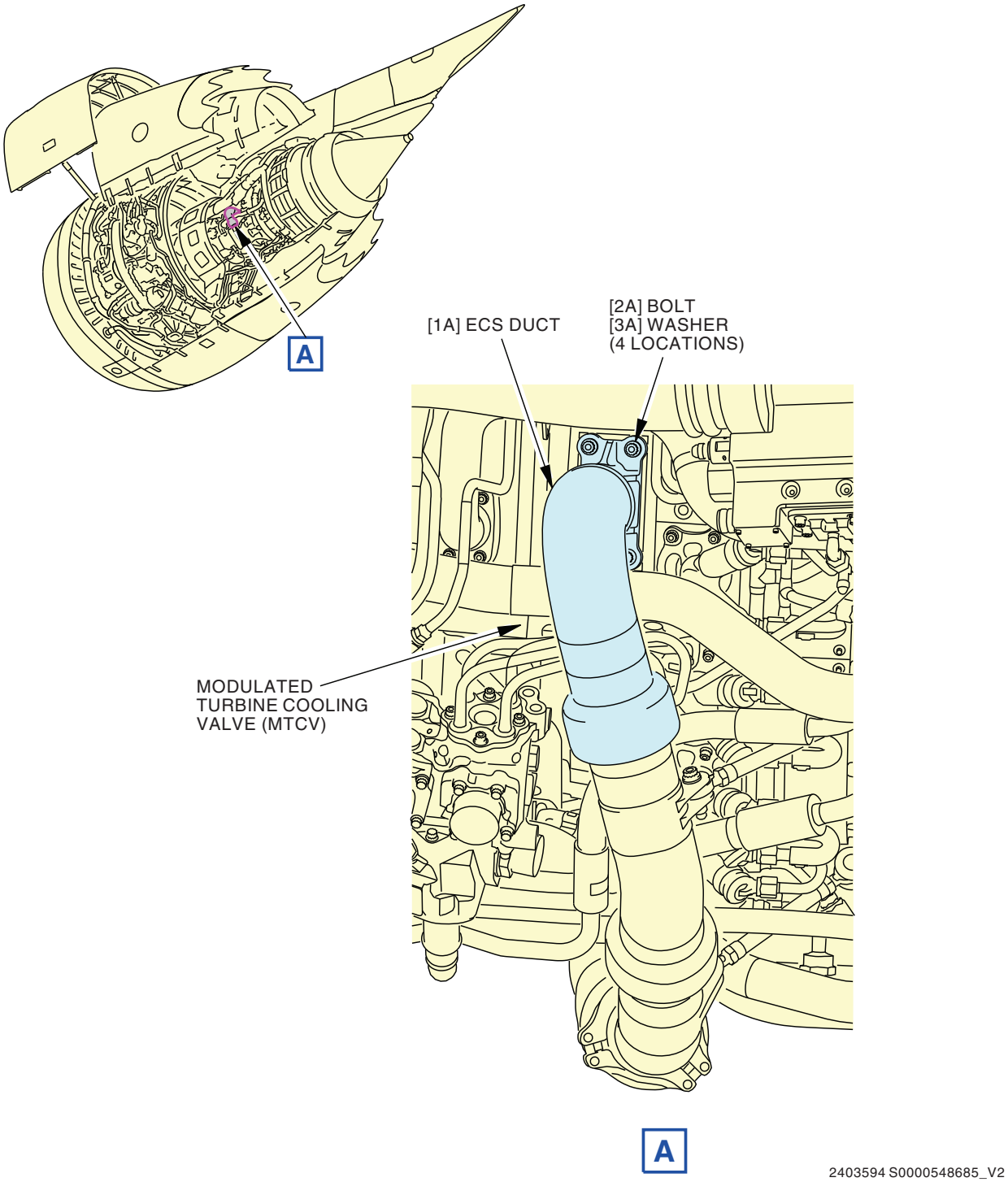
(l) Remove and discard the internal pressure seal [193].

(m) Remove the piston rings [233].

(n) Put the protective covers on the open port of the combustion case [191].

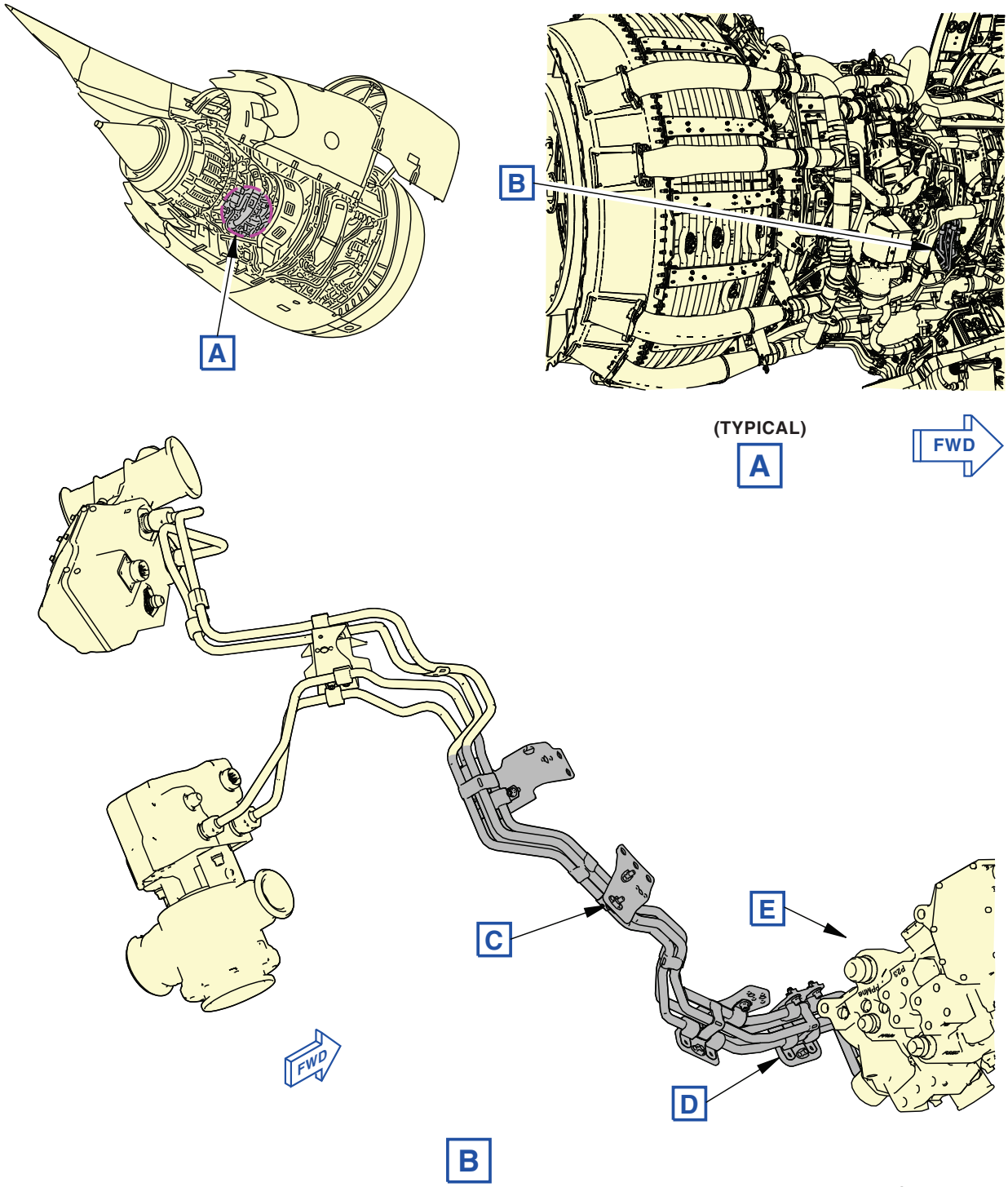
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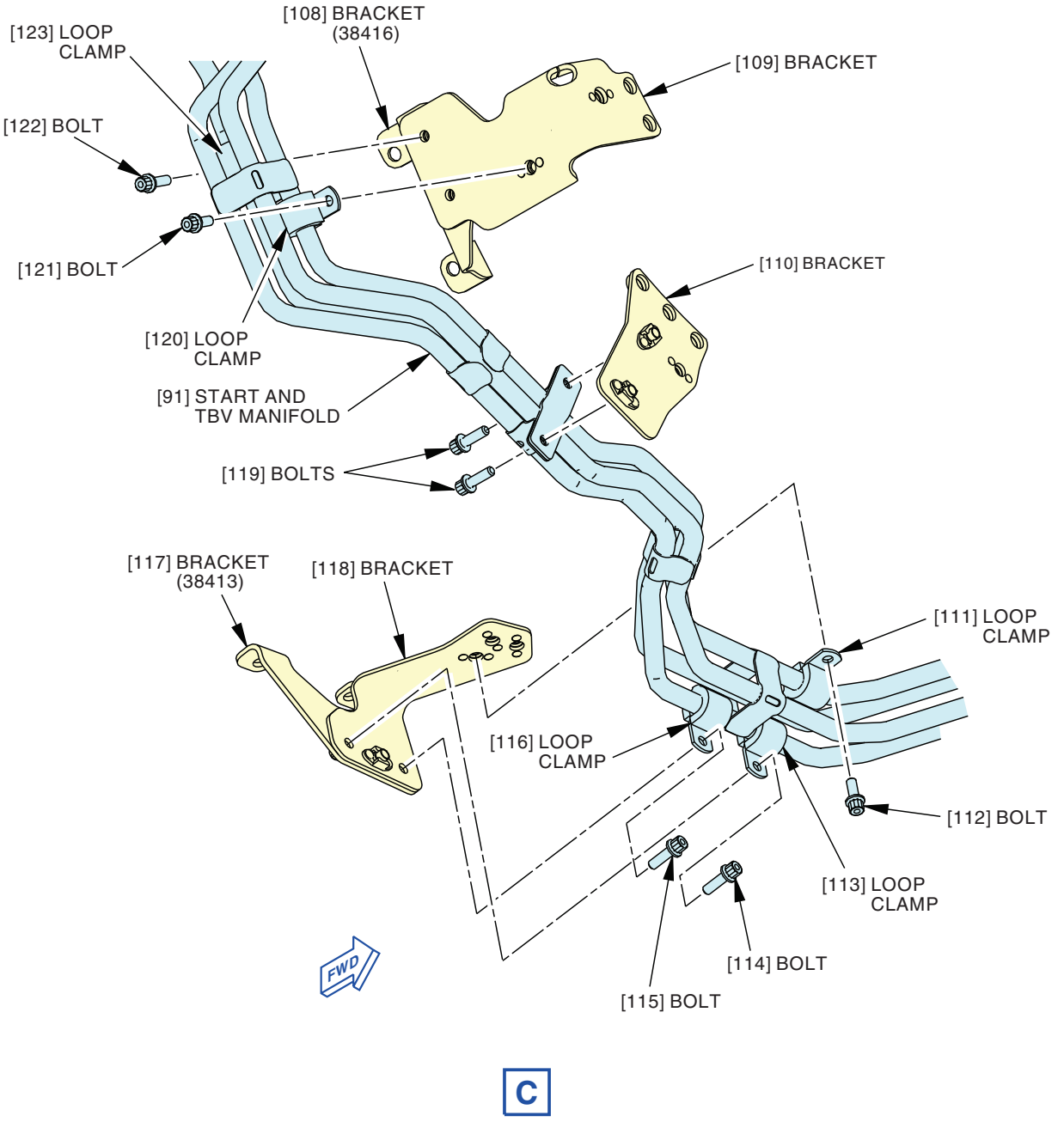
ECS Duct Installation
Figure 401/75-23-02-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



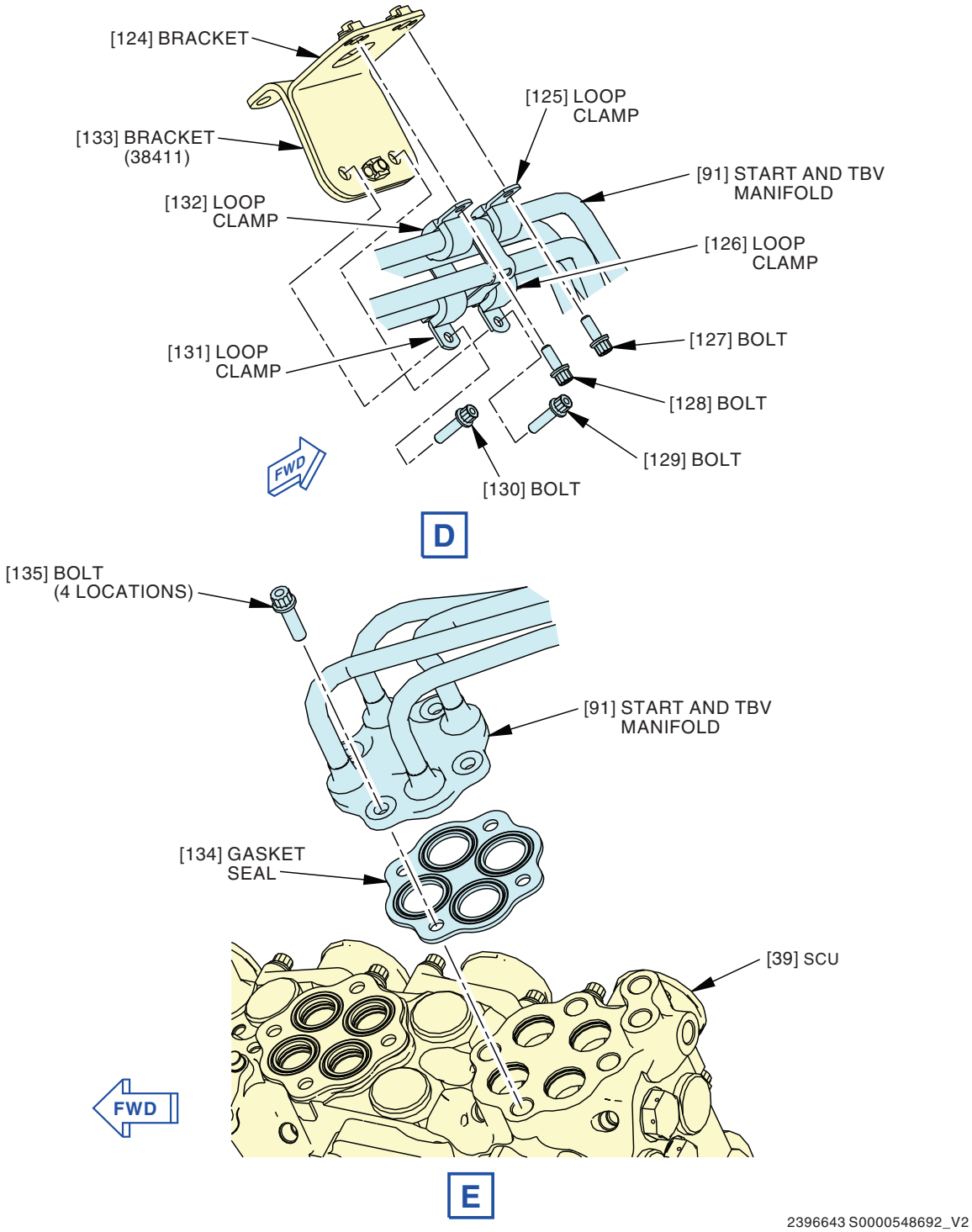
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 1 of 11)

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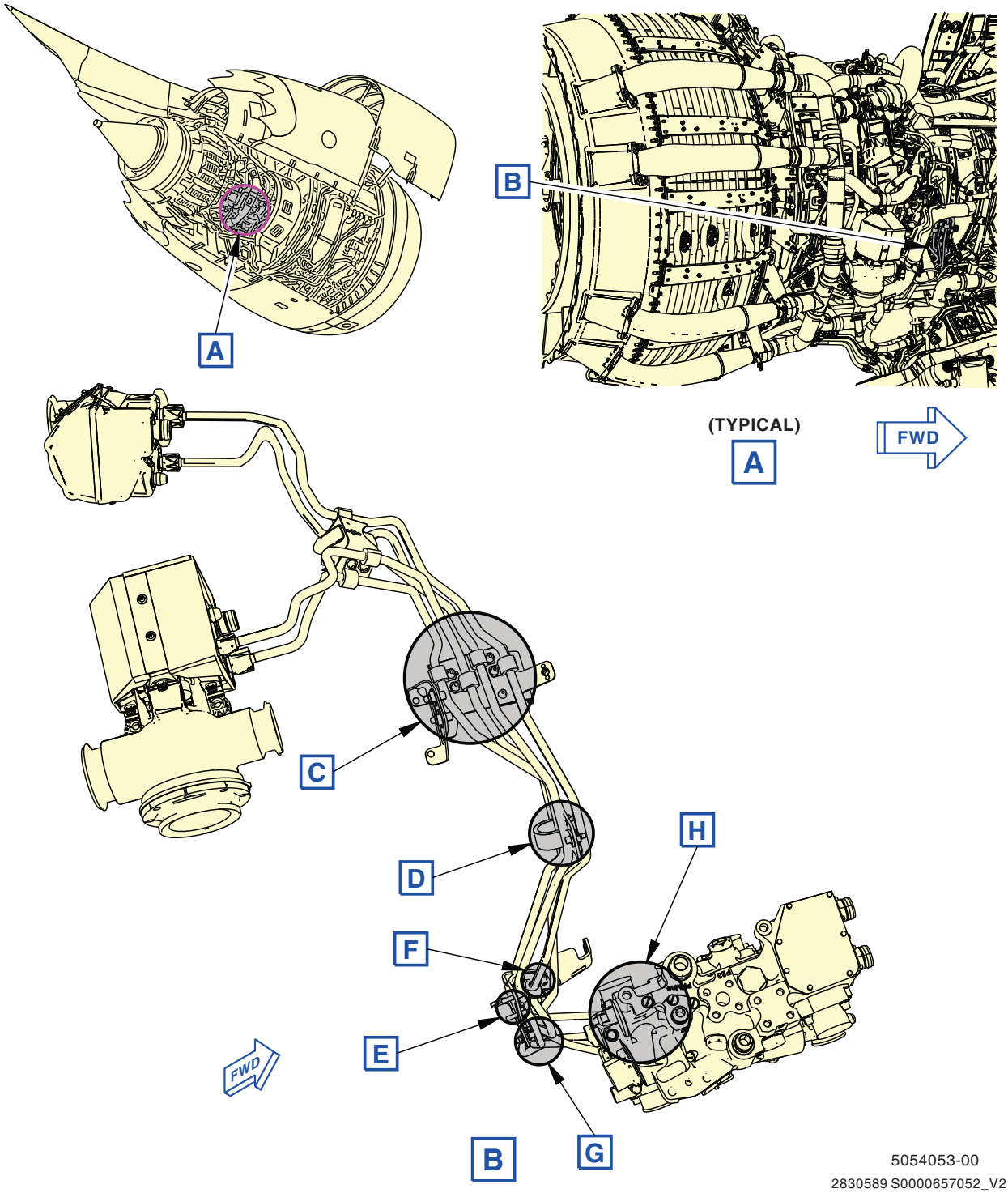
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 2 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



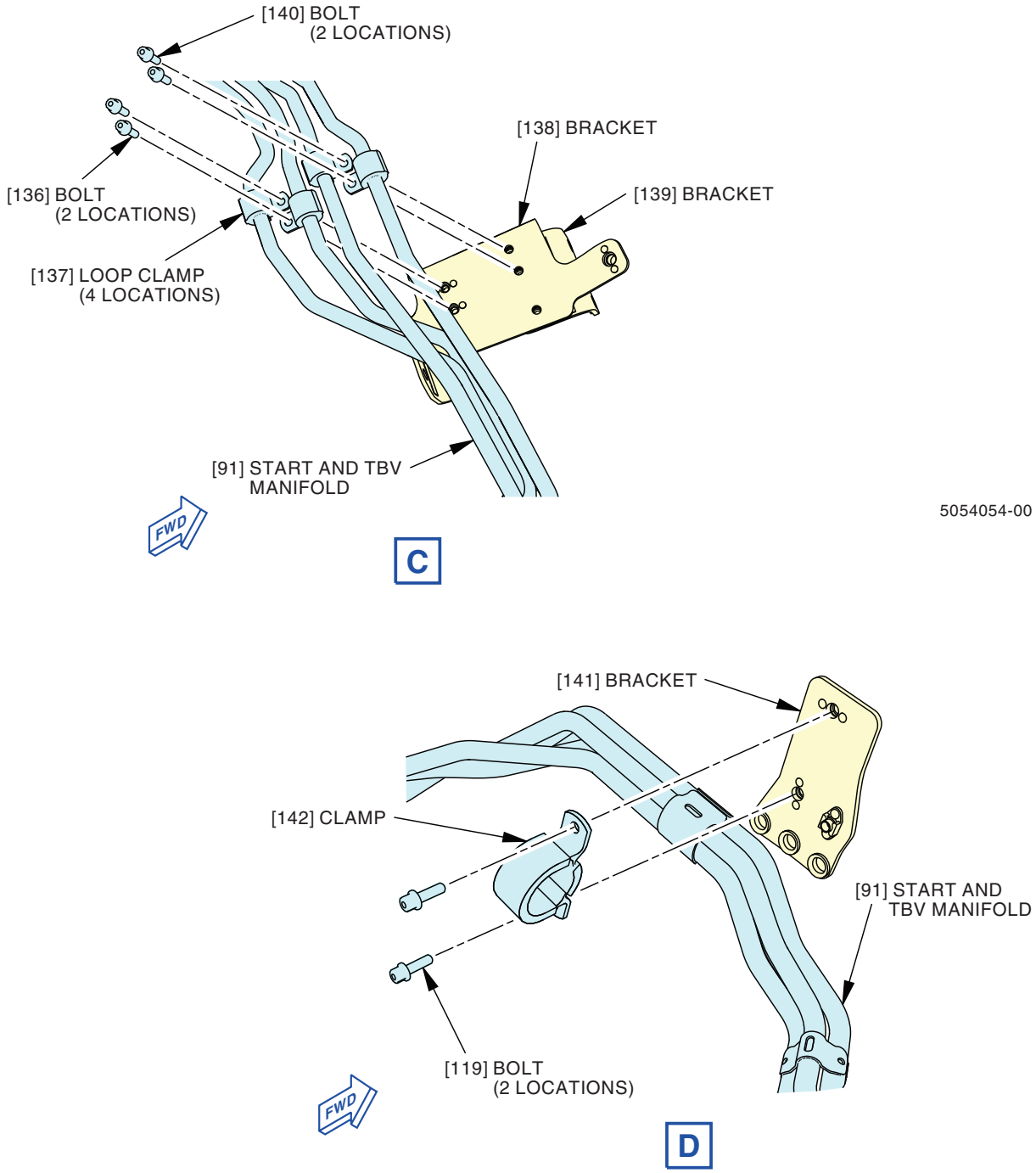
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 3 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



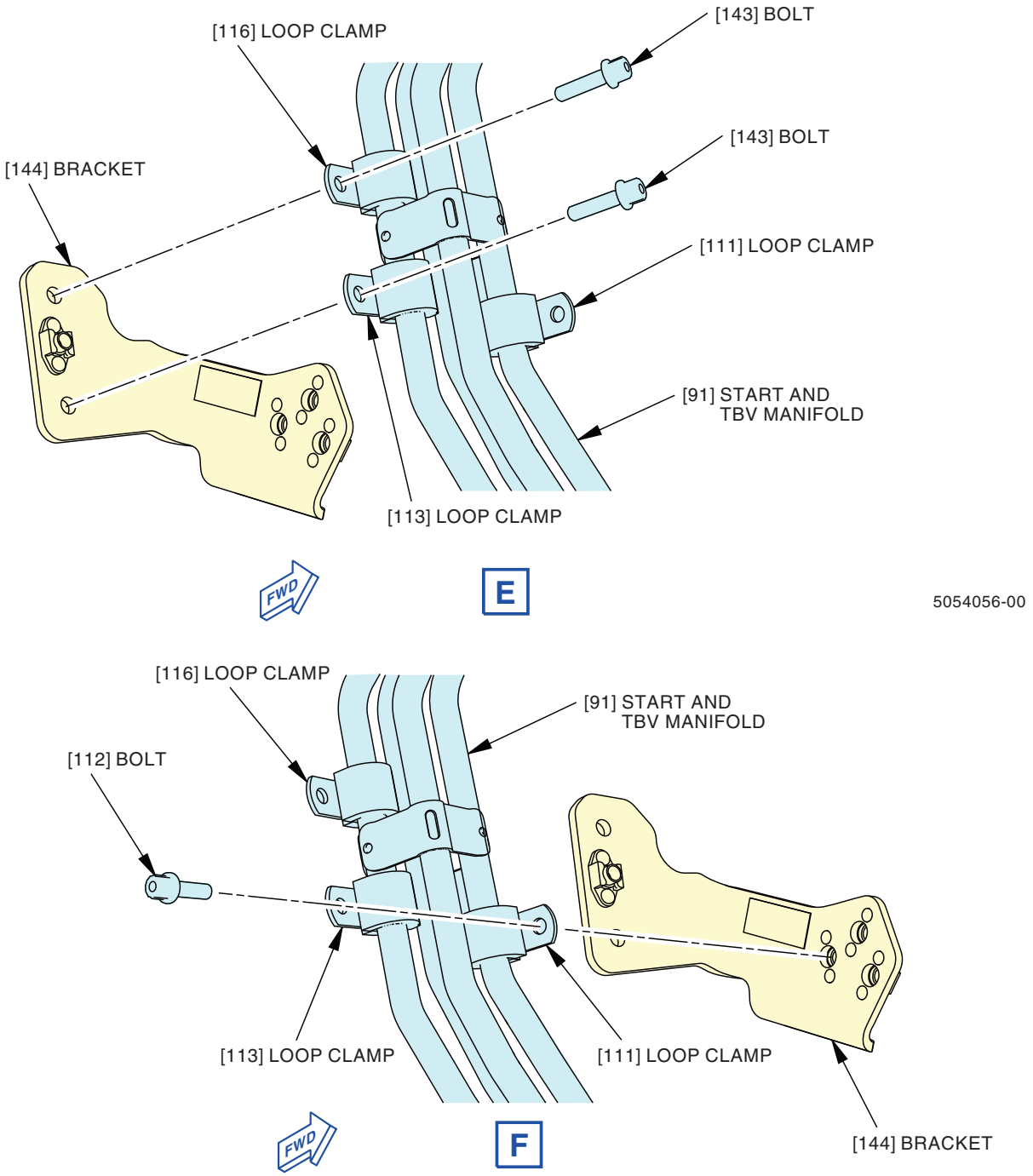
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 4 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



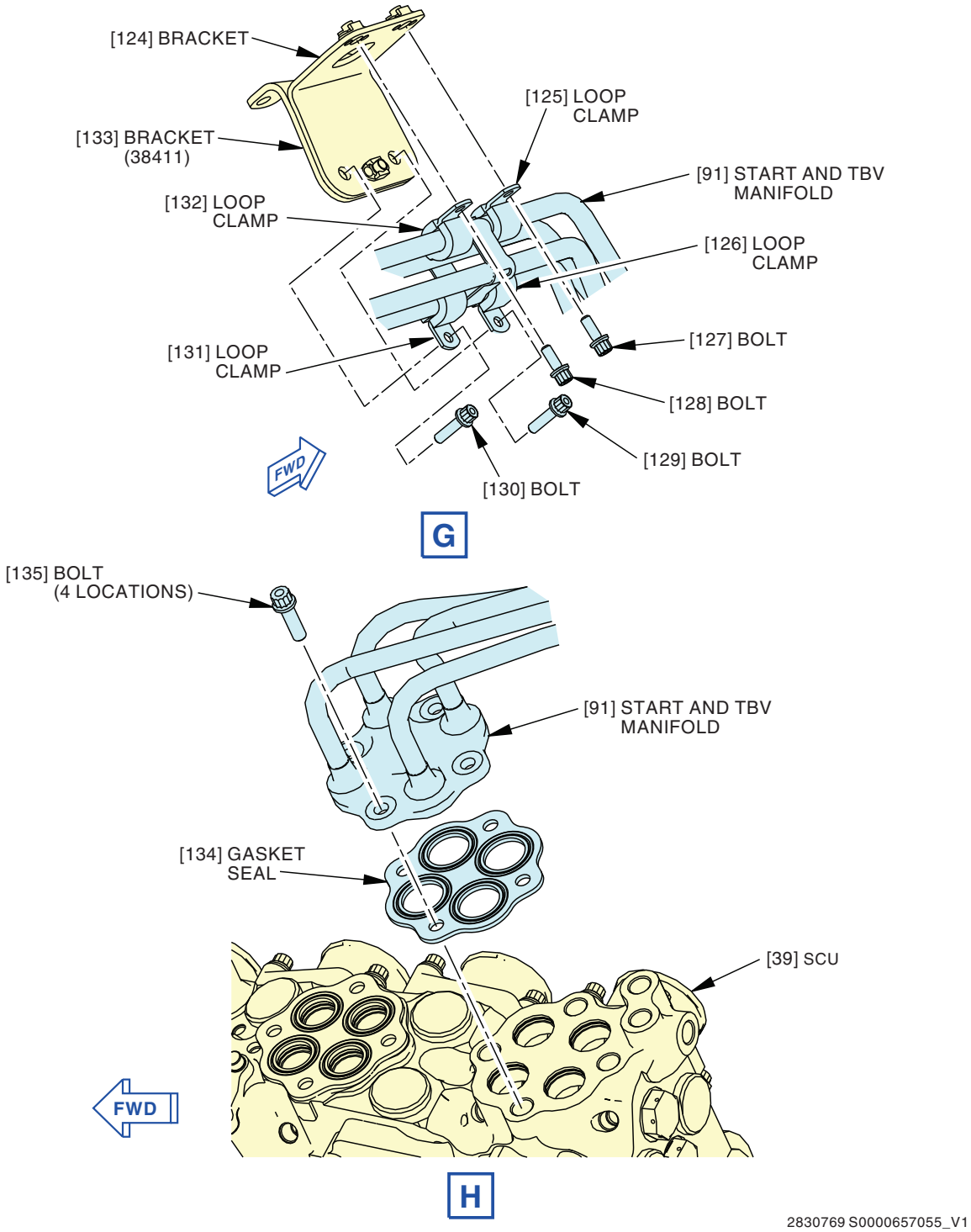
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 5 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



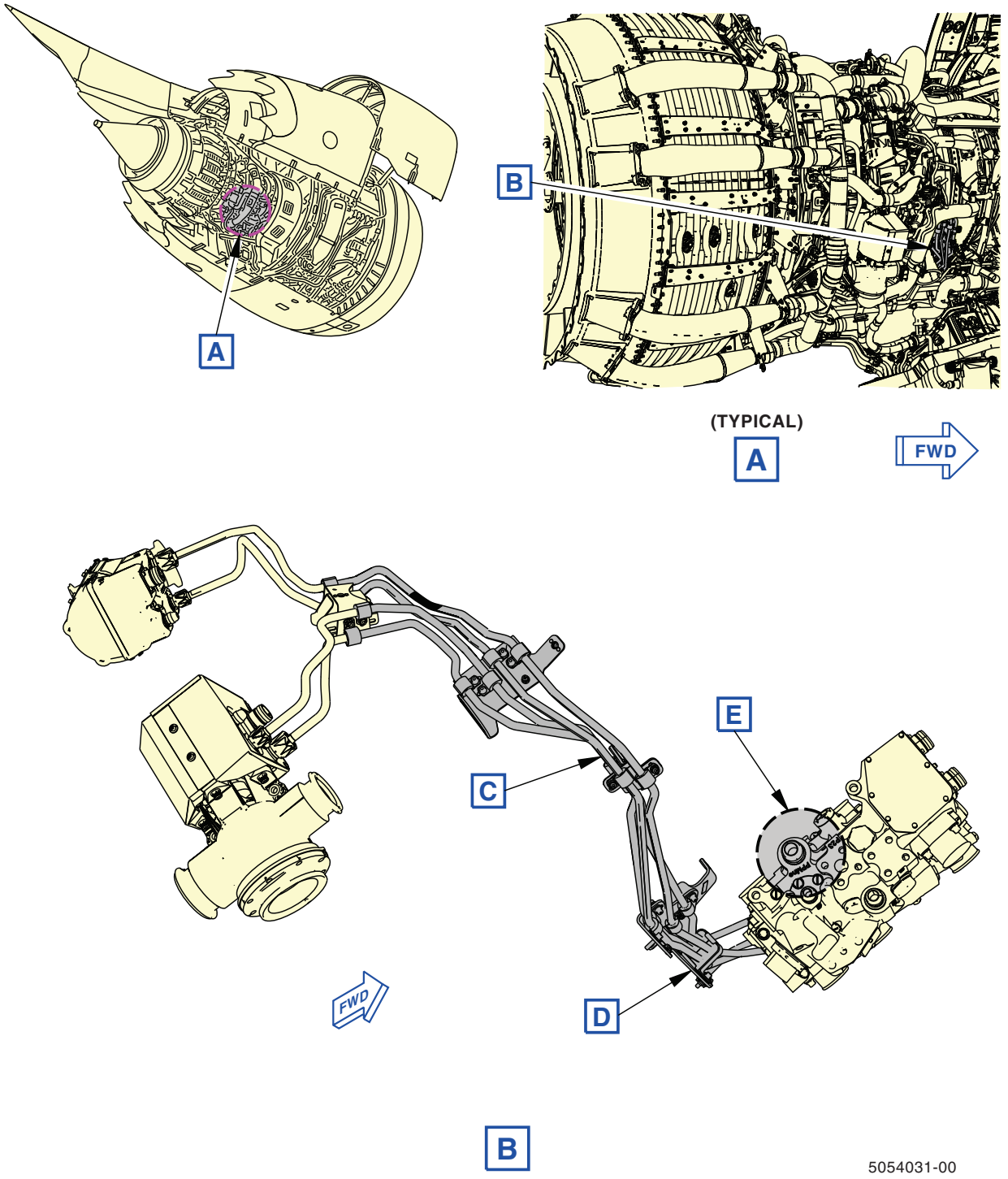
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 6 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



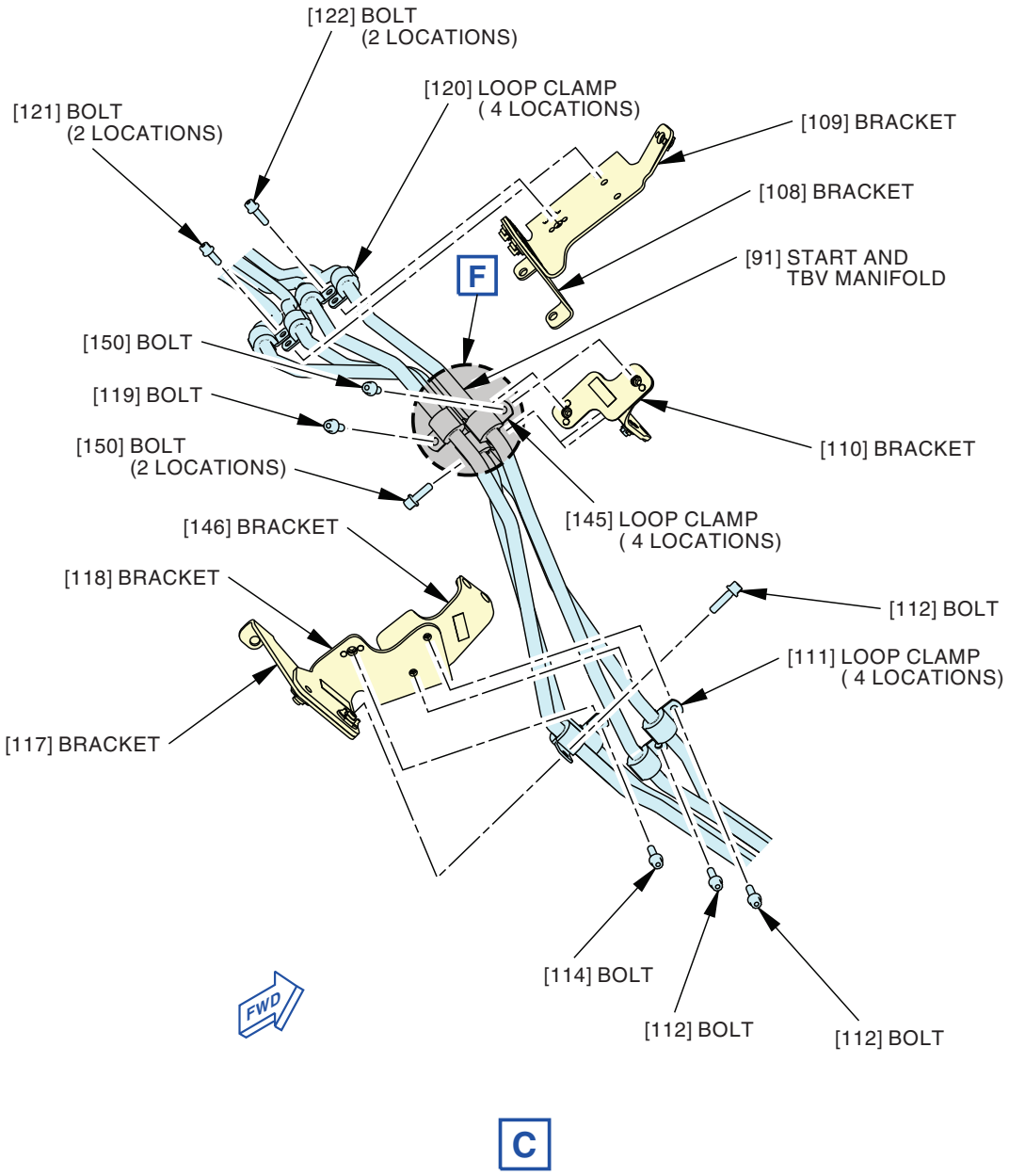
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 7 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 8 of 11)

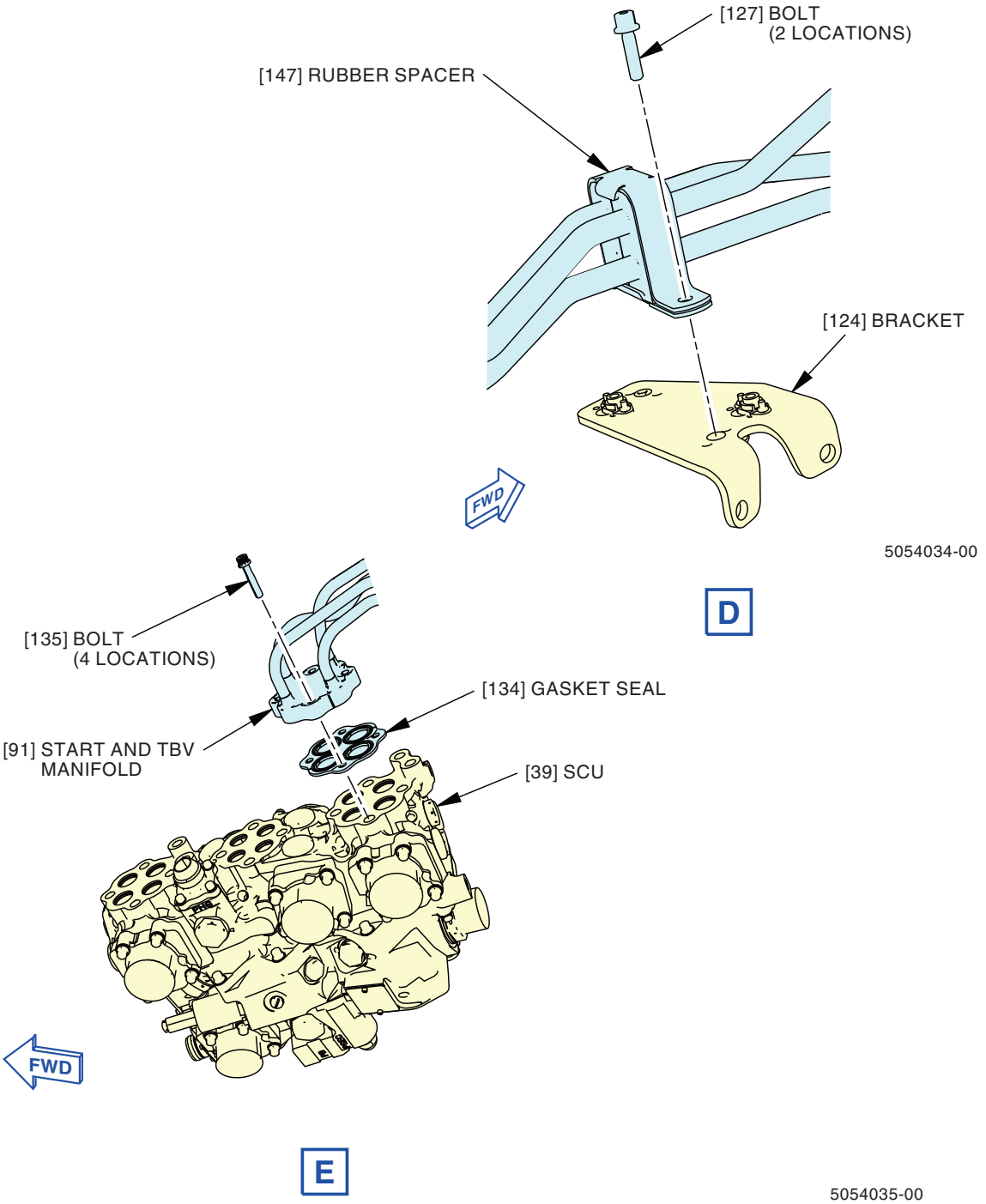
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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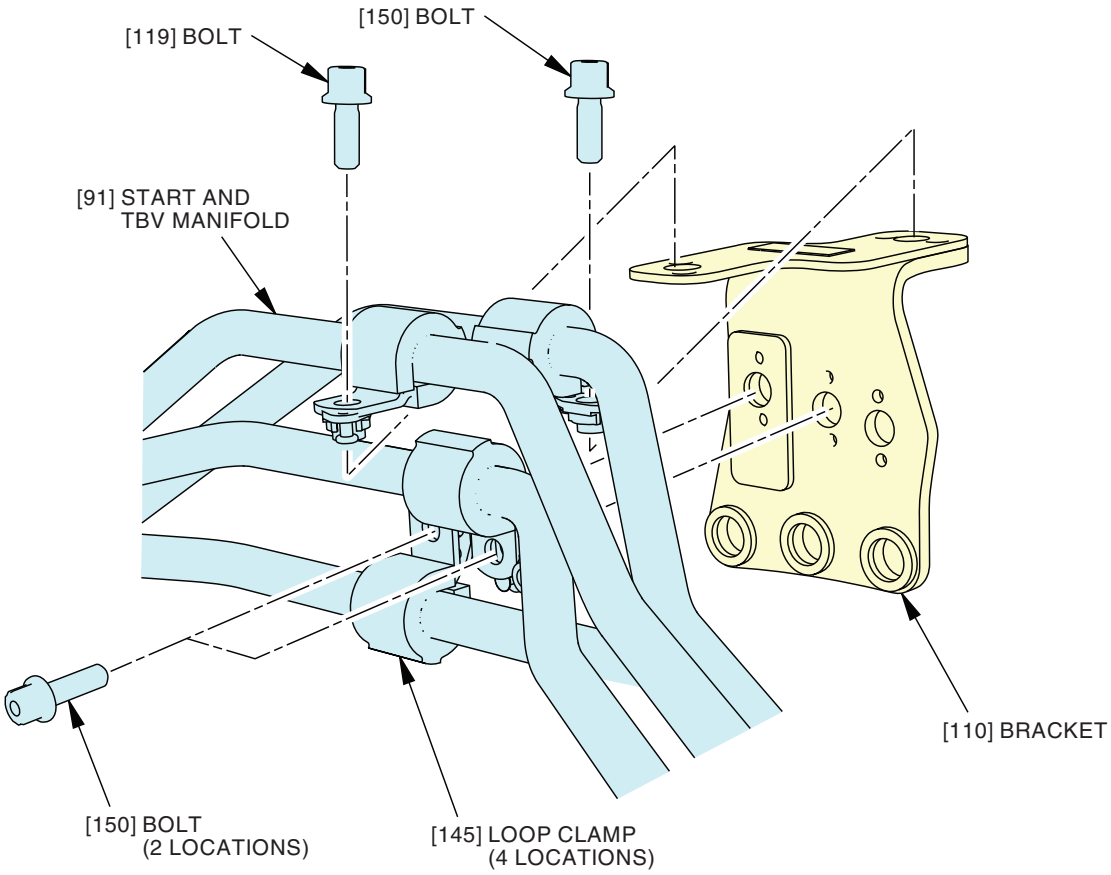
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 9 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 10 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

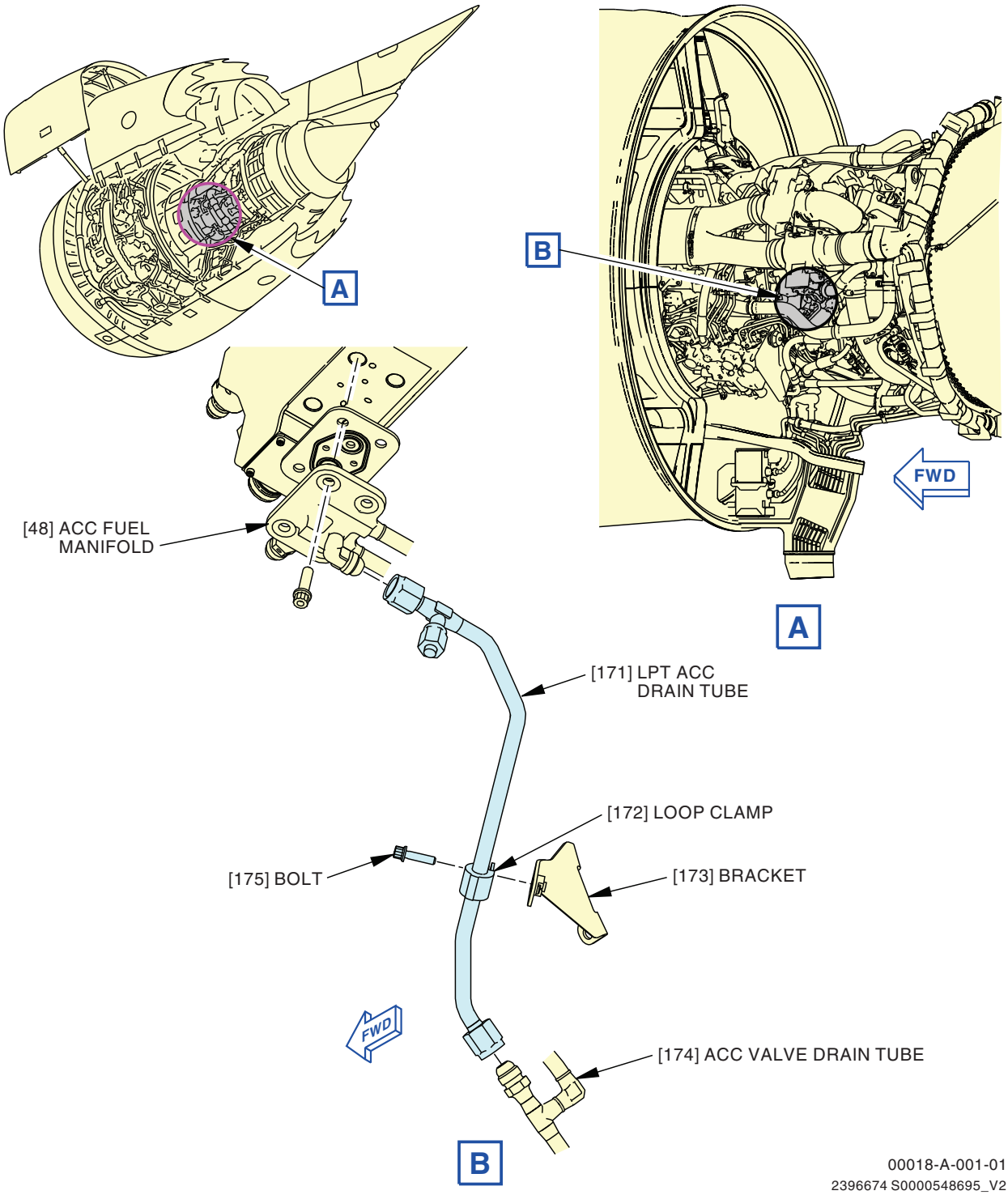


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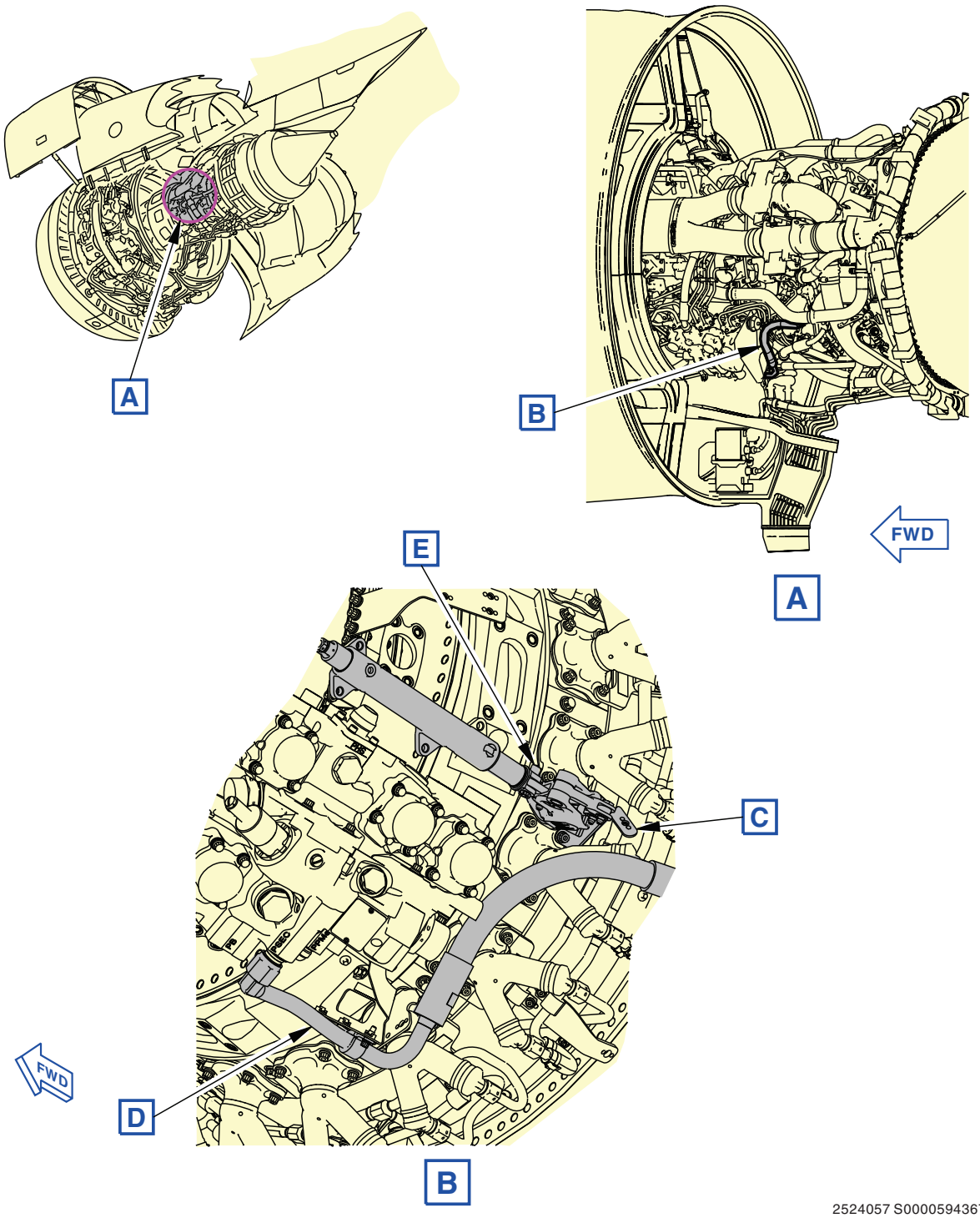
Start and TBV Manifold Installation
Figure 402/75-23-02-990-805-G00 (Sheet 11 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPT ACC Drain Tube Installation
Figure 403/75-23-02-990-807-G00

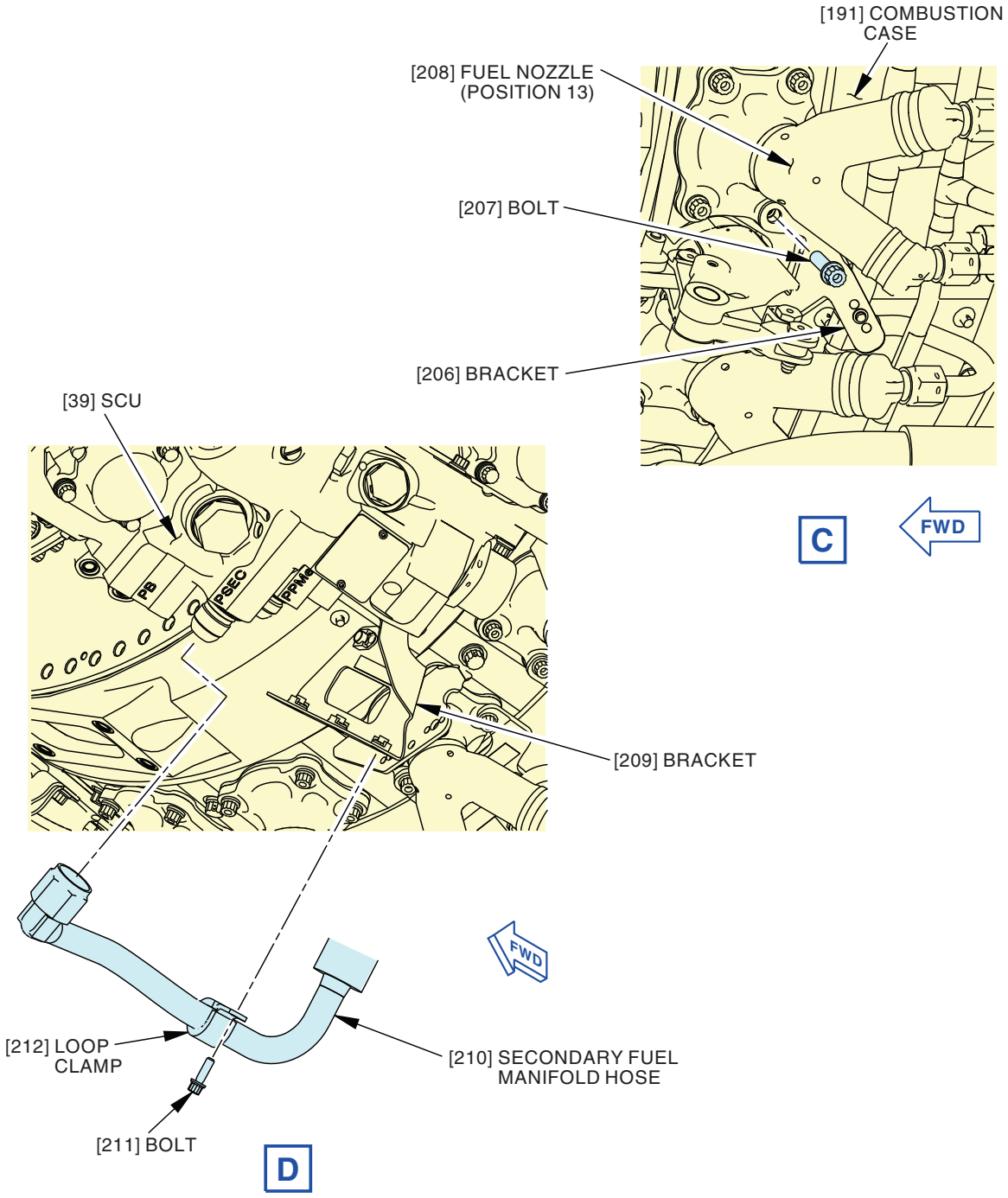
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Left Modulated Turbine Cooling (MTC) Valve Installation
Figure 404/75-23-02-990-801-G00 (Sheet 1 of 4)

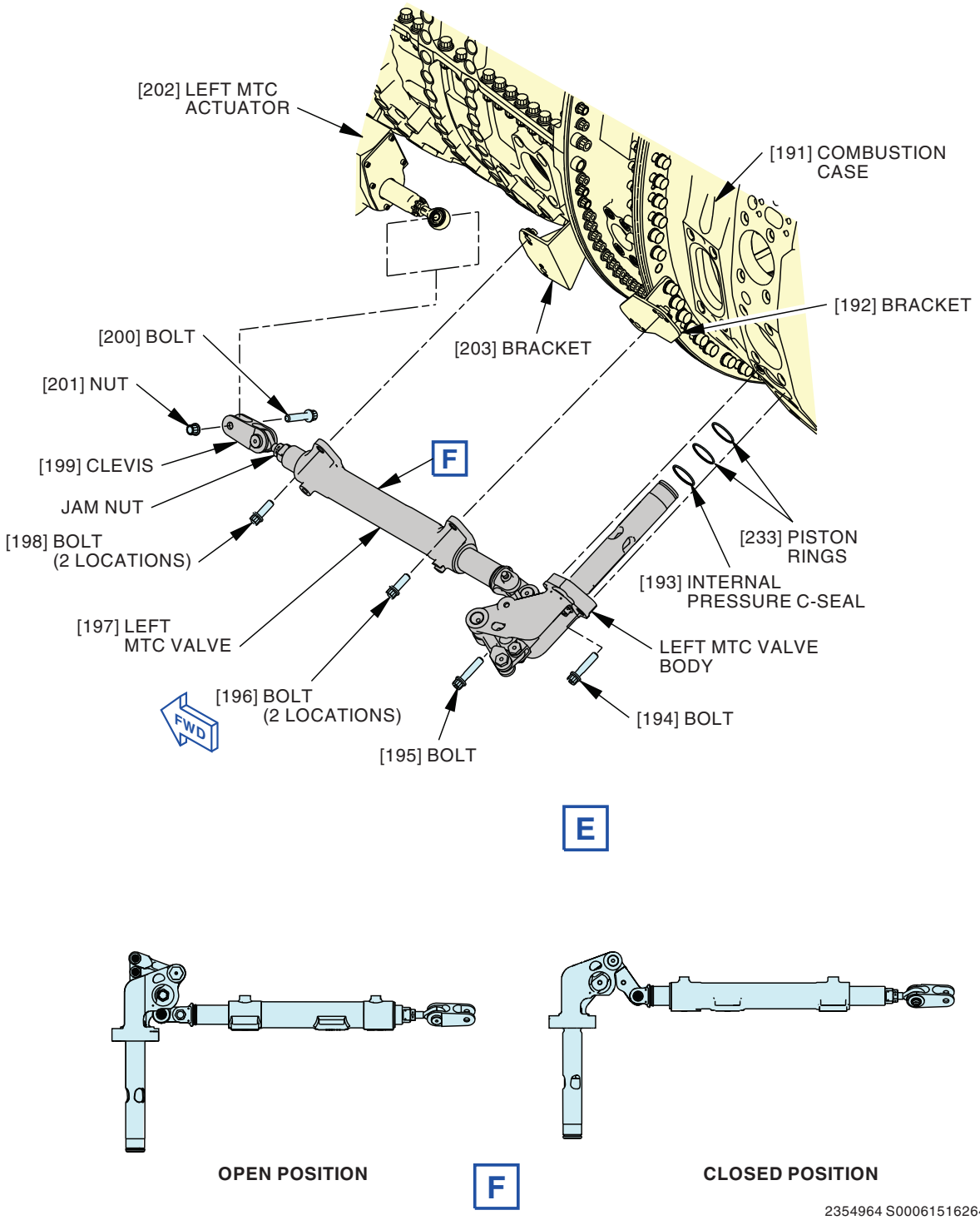
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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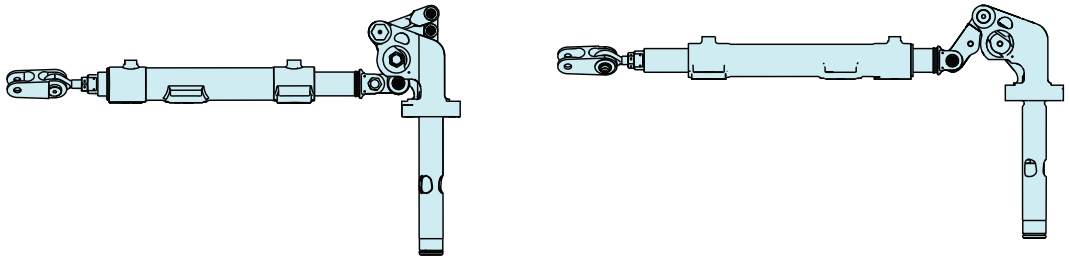
Left Modulated Turbine Cooling (MTC) Valve Installation
Figure 404/75-23-02-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left Modulated Turbine Cooling (MTC) Valve Installation
Figure 404/75-23-02-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(OPEN POSITION)

C

(CLOSED POSITION)

C

00020-A-001-01
2396781 S0000548680_V2

Left Modulated Turbine Cooling (MTC) Valve Installation
Figure 404/75-23-02-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-23-02-400-801-G00

3. Left Modulated Turbine Cooling (MTC) Valve Installation

(Figure 401, Figure 402, Figure 403, and Figure 404)

A. General

(1) This task gives the instructions to install the left modulated turbine cooling valve.

(2) The left modulated turbine cooling valve is referred to as the left MTC valve.

(3) You must do the tests that are listed in the power plant test reference table after you install the left MTC valve.

(4) The MTC valve and valve body must be maintained as a set.

B. References

Reference	Title
36-11-01-400-801	Engine Pneumatic Duct Installation (P/B 401)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A50553 [CP2397]	Adhesive - Neoprene High Performance Rubber and Gasket, 3M EC-1300L	
D00599 [CP2442]	Oil - Engine	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2
G51435 [CP2635]	Wax - Stick	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
134	Gasket seal	73-11-00-20-140	PTF ALL
197	Left MTC valve	75-23-02-01-010	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

G. Left Modulated Turbine Cooling (MTC) Valve Installation

SUBTASK 75-23-02-430-001-G00

(1) Install the left MTC valve [197] as follows:

NOTE: The MTC valve and the MTC valve body must be kept as a set.

(a) Remove the protective cover from the open port on the combustion case [191].

(b) Install the piston rings [233] on the left MTC valve [197].

1) Apply 3M EC-1300L adhesive, A50553 [CP2397], in the ring grooves at the end of the left MTC valve [197].

2) Let the adhesive become tacky for 15-20 minutes before you install the piston rings [233].

3) Install and compress the piston rings [233] into the ring grooves.

NOTE: It is possible to cure with heat not more than 250°F (121°C).

a) Make sure that the end gaps of the piston rings [233] are 180 degrees apart.

(c) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the bolt [194] and bolt [195] and engine oil, D00599 [CP2442], on the threads of the two bolts [196], two bolts [198], and bolt [200].

NOTE: You can use engine oil, D00599 [CP2442], as an alternative for the bolt [194] and bolt [195].

(d) Put the new internal pressure seal [193] on the left MTC valve [197].

1) Apply stick wax, G51435 [CP2635], on the new internal pressure seal [193].

CAUTION

DURING THE REMOVAL OR INSTALLATION, MAKE SURE THAT THE VALVE STAYS PERPENDICULAR TO THE CASE. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(e) Put the left MTC valve [197] in position on the bracket [192] and bracket [203].

1) Put the body of the left MTC valve [197] between the start and TBV manifold [91] and the SCU [39] mount and into the combustion case [191].

(f) Install the two bolts [198] that attach the left MTC valve [197] to the bracket [203].

(g) Install the two bolts [196] that attach the left MTC valve [197] to the bracket [192].

(h) Tighten the bolts [198] and bolts [196] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

(i) Put the clevis [199] in a position clear of the left MTC actuator [202] rod end before you extend the actuator.

(j) Apply 45 psig (310 kPa) MAX pressure to extend the left MTC actuator [202] rod.

NOTE: If it is necessary, reduce the pressure and rotate the left MTC actuator [202] rod end to align with the clevis [199]. Apply the 45 psig (310 kPa) MAX pressure again.

(k) If the holes in the clevis [199] do not align with the hole of the rod end, do the steps that follow:

1) Remove the safety wire.

2) Loosen the jam nut on the left MTC valve [197] rod end.

3) Loosen the jam nut on the left MTC valve [197].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Adjust the rod end to get alignment.

(l) Adjust the rod end in ½ turn increments to get the approximate 45 degree orientation to the mounting bracket [203] centerline.

(m) Tighten the jam nut to 85 in-lb (9.6 N·m) - 95 in-lb (10.7 N·m).

(n) Tightly attach the rod end jam nut.

(o) If the hole in the clevis still does not align with the hole in the rod end, do the steps that follow:

1) Remove the red plastic shipping bolt from the left MTC valve [197].

2) Move the push rod on the left MTC valve [197] until the holes align.

(p) Install the bolt [200] and nut [201] that attach the clevis [199] to the left MTC actuator [202].

(q) Tighten the nut [201] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

(r) Put the left MTC valve [197] in the open position.

(s) Install the bolt [195] that attaches the left MTC valve [197] to the combustion case [191].

(t) Install the bolt [194] that attaches the left MTC valve [197] to the combustion case [191].

(u) Tighten the bolt [194] and bolt [195] as follows:

1) If Tiodize T8E-H grease, D50355 [CP2961], is used, tighten the bolt [194] and bolt [195] to 130.0 in-lb (14.7 N·m).

2) If oil, D00599 [CP2442], is used, tighten the bolt [194] and bolt [195] to 205.0 in-lb (23.2 N·m).

(v) Cycle the left MTC actuator [202] to full retracted and full extended positions several times.

1) Make sure that the linkage has smooth, unrestricted motion of the push rod and the left MTC valve [197].

(w) Remove the pressure from the left MTC actuator [202].

(x) Connect the secondary fuel manifold hose [210] B-nut to the SCU [39].

1) Tighten the secondary fuel manifold hose [210] B-nut to 662.4 in-lb (74.8 N·m) - 777.6 in-lb (87.9 N·m).

(y) Install the bolt [211] that attaches the loop clamp [212] and secondary fuel manifold hose [210] to the bracket [209].

1) Tighten the bolt [211] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

(z) Install the bolt [207] that attaches the fuel nozzle (position 13) [208] and bracket [206] to the combustion case [191].

1) Tighten the bolt [207] to 189.0 in-lb (21.4 N·m) - 221.0 in-lb (25.0 N·m).

SUBTASK 75-23-02-430-002-G00

(2) Install the LPT ACC drain tube [171] as follows:

(a) Remove the protective covers from the LPT ACC drain tube [171], ACC fuel manifold [48], and ACC valve drain tube [174].

(b) Apply engine oil, D00599 [CP2442] on the threads of the bolt [175].

(c) Put the LPT ACC drain tube [171] in position between the ACC fuel manifold [48] and the ACC valve drain tube [174].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(d) Connect the LPT ACC drain tube [171] B-nut to the ACC fuel manifold [48].

1) Tighten the LPT ACC drain tube [171] B-nut at the ACC fuel manifold [48] to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

(e) Connect the LPT ACC drain tube [171] to the ACC valve drain tube [174].

1) Tighten the LPT ACC drain tube [171] B-nut at the ACC valve drain tube [174] to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

(f) Install the bolt [175] that attaches the loop clamp [172] and LPT ACC drain tube [171] to the bracket [173].

(g) Tighten the bolt [175] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

PTF ALL; ALL 737 MAX, PRE LEAP-1B SERVICE BULLETIN 72-0050 AND PRE LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 75-23-02-430-004-G00

(3) Install the hardware to the start and TBV manifold [91] as follows (Figure 402):

(a) Remove the protective covers on the SCU [39] and the start and TBV manifold [91].

(b) Apply engine oil, D00599 [CP2442] on the threads of the bolt [112], bolt [114], bolt [115], two bolts [119], and bolt [121].

(c) Apply engine oil, D00599 [CP2442] on the threads of the bolt [122], bolt [127], bolt [128], bolt [129], bolt [130], and four bolts [135].

(d) Install a new gasket seal [134] on the SCU [39] port.

(e) Install the four bolts [135] that attach the start and TBV manifold [91] to the SCU [39].

1) Tighten the four bolts [135] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(f) Install the bolt [129], bolt [130], bolt [128], and bolt [127] that attach the loop clamp [126], loop clamp [131], loop clamp [132], loop clamp [125], and start and TBV manifold [91] to the bracket [124] and bracket [133].

(g) Install the bolt [112], bolt [114], and bolt [115] that attach the loop clamp [111], loop clamp [113], loop clamp [116], and start and TBV manifold [91] to the bracket [118] and bracket [117].

(h) Install the bolt [122] and bolt [121] that attach the loop clamp [123], loop clamp [120], and start and TBV manifold [91] to the bracket [109] and bracket [108].

(i) Tighten the bolt [114] and bolt [115] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

(j) Tighten the bolt [121], bolt [128], bolt [129], and bolt [130] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

(k) Tighten the bolt [112], bolt [122], and bolt [127] to 51.0 in-lb (5.8 N·m) - 59.0 in-lb (6.7 N·m).

(l) Install the two bolts [119] that attach the start and TBV manifold [91] to the bracket [110].

1) Tighten the bolts [119] to 51.0 in-lb (5.8 N·m) - 59.0 in-lb (6.7 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 75-23-02-420-001-G00

(4) Install the hardware to the start and TBV manifold [91] as follows (Figure 402):

(a) Remove the protective covers on the SCU [39] and the start and TBV manifold [91].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(b) Apply engine oil, D00599 [CP2442] on the threads of the bolt [112], two bolts [143], two bolts [119], two bolts [140], and two bolts [136].

(c) Apply engine oil, D00599 [CP2442] on the threads of the bolt [127], bolt [128], bolt [129], bolt [130], and four bolts [135].

(d) Install a new gasket seal [134] on the SCU [39] port.

(e) Install the four bolts [135] that attach the start and TBV manifold [91] to the SCU [39].

(f) Install the bolt [129], bolt [130], bolt [128], and bolt [127] that attach the loop clamp [126], loop clamp [131], loop clamp [132], loop clamp [125], and start and TBV manifold [91] to the bracket [124] and bracket [133].

(g) Install the bolt [112] and two bolts [143] that attach the loop clamp [111], loop clamp [113], loop clamp [116], and start and TBV manifold [91] to the bracket [144].

(h) Install the two bolts [140] that attach the loop clamps [137] and start and TBV manifold [91] to the bracket [138] and bracket [139].

(i) Install the two bolts [136] that attach the loop clamps [137] and start and TBV manifold [91] to the bracket [138].

(j) Install the two bolts [119] that attach the clamp [142] and start and TBV manifold [91] to the bracket [141].

(k) Tighten the four bolts [135] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(l) Tighten the two bolts [143] and two bolts [136] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

(m) Tighten the bolt [127], bolt [128], bolt [129], and bolt [130] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

(n) Tighten the two bolts [140], bolt [112], and two bolts [119] to 50.0 in-lb (5.6 N·m) - 60.0 in-lb (6.8 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 75-23-02-420-002-G00

(5) Install the hardware to the start and TBV manifold [91] as follows:

(a) Remove the protective covers on the SCU [39] and the start and TBV manifold [91].

(b) Apply engine oil, D00599 [CP2442] on the threads of the three bolts [112], bolt [114], three bolts [150], bolt [119], two bolts [122] and two bolts [121].

(c) Apply engine oil, D00599 [CP2442] on the threads of the two bolts [127] and four bolts [135].

(d) Install a new gasket seal [134] on the SCU [39] port.

(e) Install the four bolts [135] that attach the start and TBV manifold [91] to the SCU [39].

1) Tighten the four bolts [135] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(f) Install the two bolts [127] that attach the rubber spacer [147] and start and TBV manifold [91] to the bracket [124].

(g) Install the three bolts [112] and bolt [114] that attach the loop clamps [111] and start and TBV manifold [91] to the bracket [118].

(h) Install the two bolts [121] and two bolts [122] that attach the loop clamps [120] and start and TBV manifold [91] to the bracket [109].

(i) Tighten the two bolts [121] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(j) Tighten the bolt [114] and two bolts [122] to 50.0 in-lb (5.6 N·m) - 60.0 in-lb (6.8 N·m).

(k) Install the three bolts [150] and bolt [119] that attach the start and TBV manifold [91] to the bracket [110].

1) Tighten the three bolts [150] to 50.0 in-lb (5.6 N·m) - 60.0 in-lb (6.8 N·m).

2) Tighten the bolt [119] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

(l) Tighten the three bolts [112] and two bolts [127] to 51.0 in-lb (5.8 N·m) - 59.0 in-lb (6.7 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

SUBTASK 75-23-02-410-003-G00

(6) Install the applicable ECS duct [1A] on the left side of the engine (TASK 36-11-01-400-801).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-23-02-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-23-02-865-002-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

I. Left Modulated Turbine Cooling (MTC) Valve Installation Test

SUBTASK 75-23-02-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

——— END OF TASK ———

TASK 75-23-02-000-802-G00

4. Right Modulated Turbine Cooling (MTC) Valve Removal

(Figure 405 and Figure 406)

A. General

(1) This task gives the instructions to remove the right modulated turbine cooling valve.

(2) The right modulated turbine cooling valve is referred to as the right MTC valve.

(3) The right MTC valve is installed at the 2:00 o'clock position on the engine core.

(4) You must open the right thrust reverser to get access to the right MTC valve.

(5) The MTC valve and valve body must be kept as a set.

B. References

Reference	Title
36-11-01-000-801	Engine Pneumatic Duct Removal (P/B 401)
36-11-04-000-801	PRSOV Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Removal

SUBTASK 75-23-02-860-002-G00

(1) Do these steps to make sure that the fuel control valve and spar valve stay in the closed position:

(a) Make sure that the L (R) START LEVER switch is in the CUTOFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Install a DO NOT OPERATE tag, STD-858, on the L (R) START LEVER switch.

(b) Make sure that the applicable SPAR VALVE CLOSED and ENG VALVE CLOSED lights, on the P5 forward overhead panel, are on dimly.

SUBTASK 75-23-02-860-003-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-23-02-010-009-G00

(3) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Right Modulated Turbine Cooling (MTC) Valve Removal

SUBTASK 75-23-02-010-010-G00

(1) Remove the upper and lower ECS ducts from the right side of the engine (TASK 36-11-01-000-801).

(a) Remove the couplings [12A] from the ECS ducts.

SUBTASK 75-23-02-010-011-G00

(2) Remove the PRSOV valve [11A] from the engine (TASK 36-11-04-000-801).

SUBTASK 75-23-02-010-012-G00

(3) Remove the right MTC valve [228] as follows:

NOTE: The MTC valve and valve body must be maintained as a set.

(a) Remove the bolt [227] and nut [225] that attach the clevis [226] to the right MTC actuator [222].

(b) Apply penetrating fluid, D50186 [CP2691], to the bolts [229], bolts [230] and threads that extend through the combustion case inserts.

1) Let the penetrant time to soak into threads.

NOTE: It is necessary to apply penetrating fluid, D50186 [CP2691], to decrease the risk of damage to the threaded inserts.

(c) Put the right MTC valve [228] in the open position.

(d) Remove the bolt [229] that attaches the right MTC valve [228] to the combustion case [191].

(e) Remove the bolt [230] that attaches the right MTC valve [228] to the combustion case [191].

(f) Remove the two bolts [224] that attach the right MTC valve [228] to the bracket [232].

(g) Remove the two bolts [223] that attach the right MTC valve [228] to the bracket [221].

CAUTION

DURING THE REMOVAL OR INSTALLATION, MAKE SURE THAT THE VALVE STAYS PERPENDICULAR TO THE CASE. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(h) Remove the right MTC valve [228] from the engine.

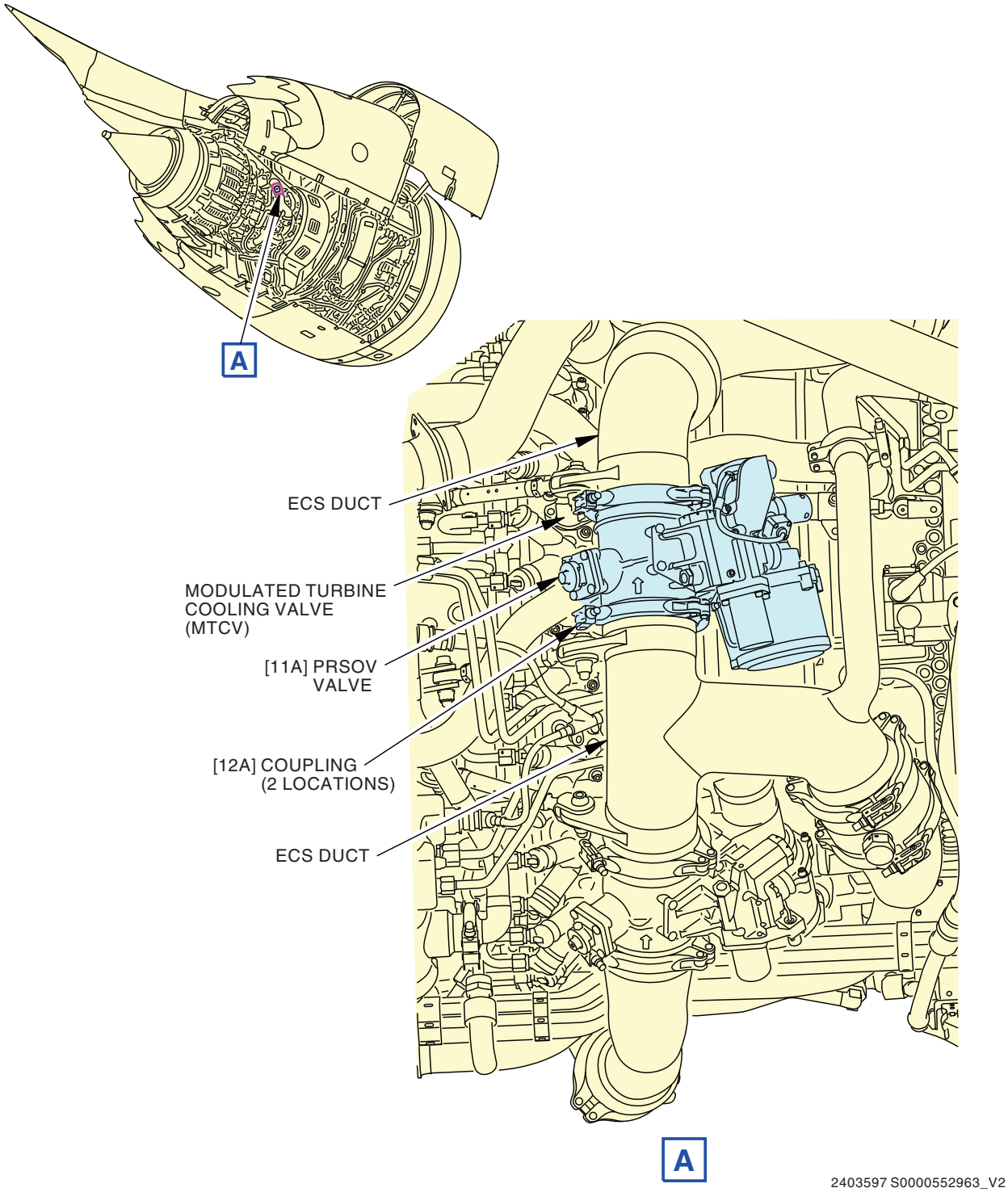
(i) Remove and discard the internal pressure seal [231].

(j) Remove the piston rings [233].

(k) Put a protective cover on the open port on the combustion case [191].

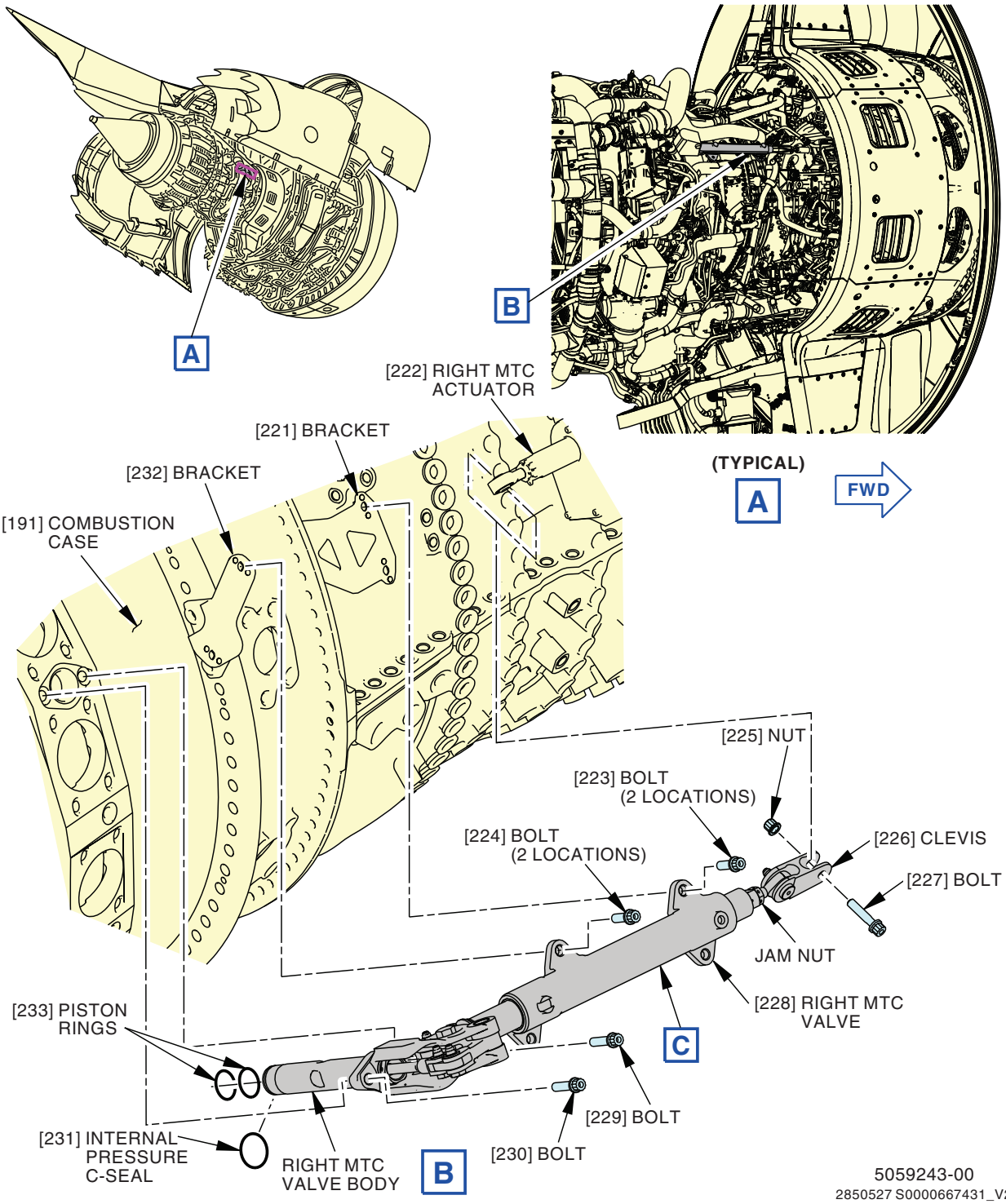
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



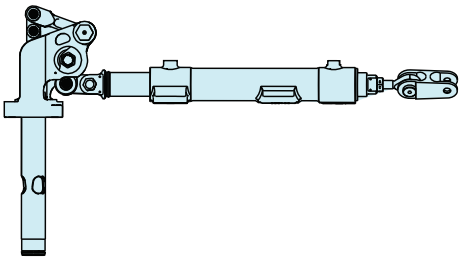
ECS Duct Installation
Figure 405/75-23-02-990-809-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



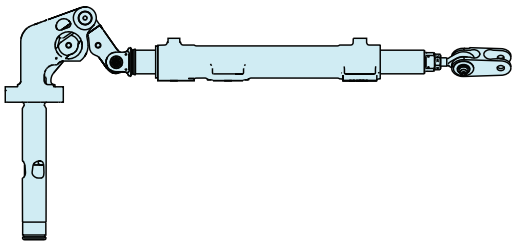
Right Modulated Turbine Cooling (MTC) Valve Installation
Figure 406/75-23-02-990-808-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(OPEN POSITION)

C



(CLOSED POSITION)

C

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Right Modulated Turbine Cooling (MTC) Valve Installation
Figure 406/75-23-02-990-808-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-23-02-400-802-G00

5. Right Modulated Turbine Cooling (MTC) Valve Installation

(Figure 405 and Figure 406)

A. General

(1) This task gives the instructions to install the right modulated turbine cooling valve.

(2) The right modulated turbine cooling valve is referred to as the right MTC valve.

(3) You must do the tests that are listed in the power plant test reference table after you install the right MTC valve.

(4) The MTC valve and valve body must be kept as a set.

B. References

Reference	Title
36-11-01-400-801	Engine Pneumatic Duct Installation (P/B 401)
36-11-04-400-801	PRSOV Installation (P/B 401)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A50173	Adhesive - 3M Super 77 Multipurpose Spray	
A50553 [CP2397]	Adhesive - Neoprene High Performance Rubber and Gasket, 3M EC-1300L	
D00599 [CP2442]	Oil - Engine	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2
G51435 [CP2635]	Wax - Stick	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
228	Right MTC valve	75-23-02-02-010	PTF ALL
231	Internal pressure seal	75-23-02-02-060	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Right Modulated Turbine Cooling (MTC) Valve Installation

SUBTASK 75-23-02-410-005-G00

(1) Install the right MTC valve [228] as follows:

NOTE: The MTC valve and valve body must be kept as a set.

(a) Remove a protective cover from the MTC port on the combustion case [191].

(b) Install the piston rings [233] on the right MTC valve [228].

1) Apply Super 77 spray adhesive, A50173, or 3M EC-1300L adhesive, A50553 [CP2397], in the ring grooves at the end of the valve probe.

2) Let the adhesive become tacky for 15-20 minutes before you install the piston rings [233].

3) Install and compress the piston rings [233] into the ring grooves.

NOTE: It is possible to cure with heat not more than 250°F (121°C).

a) Make sure that the piston rings [233] end gaps are 180 degrees apart.

(c) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the bolt [229] and bolt [230], and engine oil, D00599 [CP2442], on the threads of the two bolts [223], two bolts [224], bolt [227].

NOTE: You can use engine oil, D00599 [CP2442], as an alternative for bolt [229] and bolt [230].

(d) Apply stick wax, G51435 [CP2635], on the new internal pressure seal [231].

(e) Put the internal pressure seal [231] on the right MTC valve [228].

CAUTION

DURING THE REMOVAL OR INSTALLATION, MAKE SURE THAT THE VALVE STAYS PERPENDICULAR TO THE CASE. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(f) Put the right MTC valve [228] in position on the bracket [232] and bracket [221].

(g) Install the two bolts [223] that attach the right MTC valve [228] to the bracket [221].

(h) Install the two bolts [224] that attach the right MTC valve [228] to the bracket [232].

(i) Tighten the bolts [223] and bolts [224] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

(j) Put the clevis [226] in a position clear of the right MTC actuator [222] rod end before you extend the actuator.

(k) Apply 45 psig (310 kPa) MAX pressure to extend the right MTC actuator [222] rod.

NOTE: If it is necessary, reduce the pressure and rotate the right MTC actuator rod end to align with the clevis. Apply the 45 psig (310 kPa) MAX pressure again.

(l) If the holes in the clevis [226] do not align with the hole of the rod end, do these steps:

1) Remove the safety wire.

2) Loosen the jam nut on the right MTC valve [228] rod end.

3) Loosen the jam nut on the right MTC valve [228].

4) Adjust the rod end to get alignment.

(m) Adjust the rod end in ½ turn increments to get approximate 45 degree orientation to the mounting bracket [221] centerline.

(n) Tighten the jam nut to 85 in-lb (9.6 N·m) - 95 in-lb (10.7 N·m).

(o) Tightly attach the rod end jam nut.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(p) If the hole in the clevis still does not align with the hole in the rod end, do the steps that follow:

1) Remove the red plastic shipping bolt from the right MTC valve [228].

2) Move the push rod on the right MTC valve [228] until the holes align.

(q) Install the bolt [227] and nut [225] that attach the clevis [226] to the right MTC actuator [222].

(r) Tighten the nut [225] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

(s) Put the right MTC valve [228] in the open position.

(t) Install the bolt [229] that attaches the right MTC valve [228] to the combustion case [191].

(u) Install the bolt [230] that attaches the right MTC valve [228] to the combustion case [191].

(v) Tighten the bolt [230] and bolt [229] as follows:

1) If Tiodize T8E-H grease, D50355 [CP2961], is used, tighten the bolt [229] and bolt [230] to 130.0 in-lb (14.7 N·m).

2) If engine oil, D00599 [CP2442], is used, tighten the bolt [229] and bolt [230] to 205.0 in-lb (23.2 N·m).

(w) Cycle the right MTC actuator [222] to the full retracted and full extended positions several times.

1) Make sure that the linkage has smooth, unrestricted motion of the push rod and the right MTC valve [228].

(x) Remove the pressure from the right MTC actuator [222].

SUBTASK 75-23-02-410-006-G00

(2) Install the upper and lower ECS ducts on the right side of the engine (TASK 36-11-01-400-801).

SUBTASK 75-23-02-410-007-G00

(3) Install the PRSOV valve [11A] (TASK 36-11-04-400-801).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-23-02-410-008-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-23-02-860-004-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

I. Right Modulated Turbine Cooling (MTC) Valve Installation Test

SUBTASK 75-23-02-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MODULATED TURBINE COOLING ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has four tasks:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

(1) Left Modulated Turbine Cooling (MTC) Actuator Removal

(2) Left Modulated Turbine Cooling (MTC) Actuator Installation

PTF ALL

(3) Right Modulated Turbine Cooling (MTC) Actuator Removal

(4) Right Modulated Turbine Cooling (MTC) Actuator Installation.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

TASK 75-23-03-000-801-G00

2. Left Modulated Turbine Cooling Actuator Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the left Modulated Turbine Cooling (MTC) actuator.

(2) The left MTC actuator is at the 8:00 o'clock position on the engine compressor case.

(3) You must open the left thrust reverser to get access to the left MTC actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

E. Prepare for the Removal

SUBTASK 75-23-03-860-001-G00

(1) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

SUBTASK 75-23-03-860-002-G00

(2) Do these steps to make sure that the fuel control valve and spar valve stay in the closed position:

(a) Make sure that the L (R) START LEVER switch is in the CUTOFF position.

1) Install a DO NOT OPERATE tag, STD-858, on the L (R) START LEVER switch.

(b) Make sure that the applicable SPAR VALVE CLOSED and ENG VALVE CLOSED lights, on the P5 forward overhead panel, are on dimly.

F. Left Modulated Turbine Cooling Actuator Removal

SUBTASK 75-23-03-010-001-G00

(1) Remove the segment shroud 4 (TASK 72-23-04-000-801-G00).

SUBTASK 75-23-03-010-003-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR IS CLEAN BEFORE YOU DISCONNECT IT. DIRTY ELECTRICAL CONNECTORS CAN CAUSE INDICATION PROBLEMS.

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(CAUTION PRECEDES)

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO LOOSEN THE ELECTRICAL CONNECTOR. DO NOT USE PLIERS THAT HAVE METAL JAWS. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [3] and the electrical connector [4] from the left MTC actuator [1] and the left VSV actuator [2] (TASK 70-00-01-910-803-G00).

(a) Install the protective covers on the electrical receptacle on the left MTC actuator [1], the left VSV actuator [2], the electrical connector [3], and the electrical connector [4].

SUBTASK 75-23-03-610-001-G00

(3) Drain the MTC VSV drain manifold [6] as follows:

(a) Put a fuel resistant container (5 gal), STD-201 below the drain plug [17].

(b) Remove the drain plug [17] from the MTC VSV drain manifold [6].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Install the drain plug [17] on the MTC VSV drain manifold [6].

1) Tighten the drain plug [17] to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

(e) Move the fuel resistant container (5 gal), STD-201 away.

SUBTASK 75-23-03-610-002-G00

(4) Drain the MTC fuel manifold [7] as follows:

(a) Put a fuel resistant container (5 gal), STD-201 below the MTC fuel manifold [7].

(b) Disconnect the fuel tube [8] from the MTC fuel manifold [7].

1) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(c) Disconnect the MTC fuel tube [9] from the MTC fuel manifold [7].

1) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Disconnect the MTC VSV drain manifold [6] from the MTC fuel manifold [7].

1) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(e) Move the fuel resistant container (5 gal), STD-201 away.

SUBTASK 75-23-03-020-001-G00

(5) Disconnect the fuel tube [8] from the MTC fuel manifold [7].

SUBTASK 75-23-03-020-002-G00

(6) Disconnect the MTC fuel tube [9] from the MTC fuel manifold [7].

SUBTASK 75-23-03-020-003-G00

(7) Disconnect the MTC VSV drain manifold [6] from the MTC fuel manifold [7].

SUBTASK 75-23-03-010-004-G00

(8) Remove the left MTC actuator [1] as follows:

(a) Remove the four bolts [10] that attach the MTC fuel manifold [7] and the fuel tube bracket [11] to the left MTC actuator [1].

(b) Remove the MTC fuel manifold [7] from the left MTC actuator [1].

(c) Remove and discard the gasket seal [5] from the left MTC actuator [1].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(d) Remove the bolt [12] and the nut [14] that attach the left MTC actuator [1] rod from the MTC cooling valve link [13].

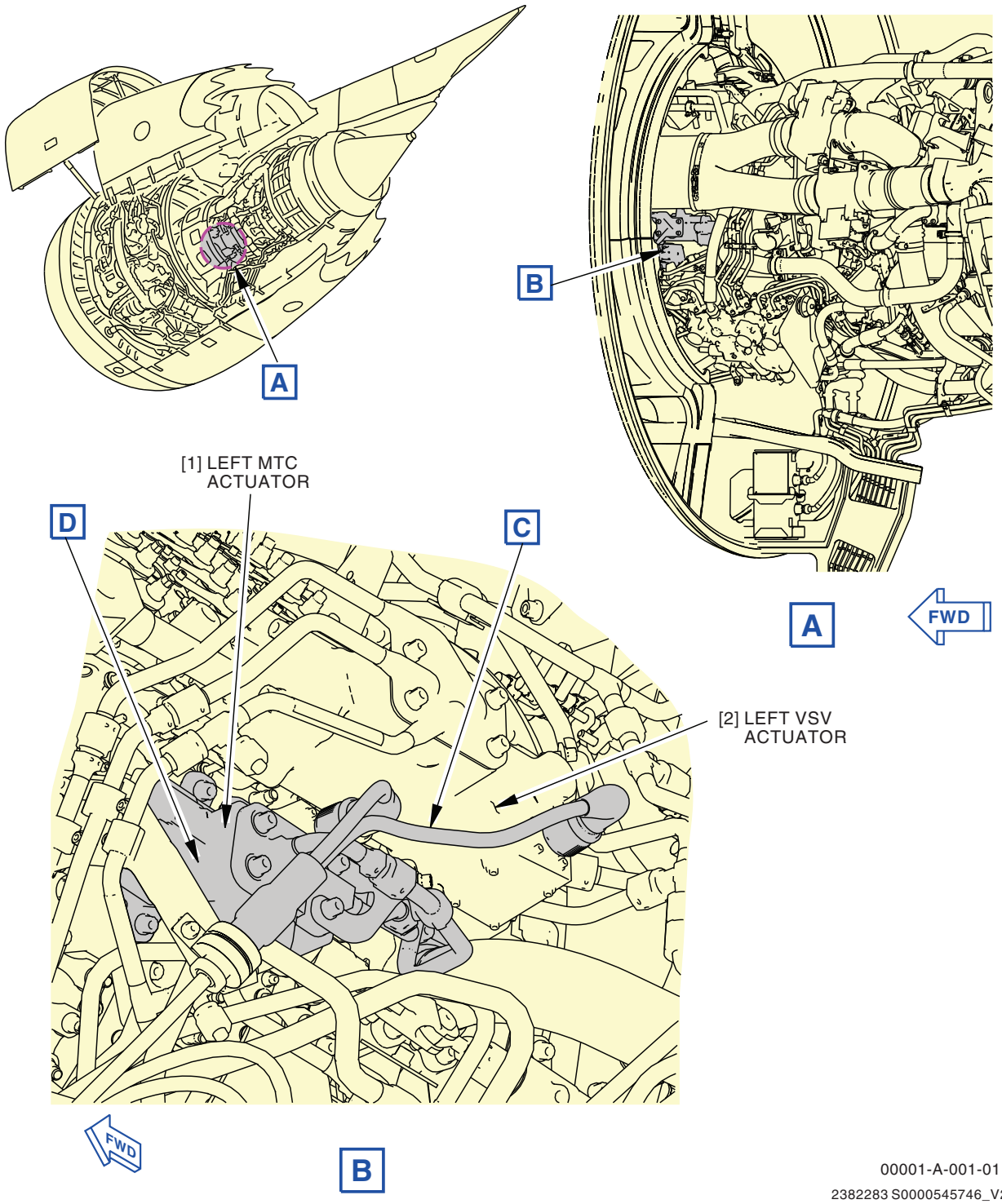
(e) Remove the four bolts [15] that attach the left MTC actuator [1] to the MTC bracket [16].

(f) Remove the left MTC actuator [1] from the engine.

(g) Install the protective covers on the left MTC actuator [1] open ports, the MTC fuel manifold [7], the fuel tube [8], the MTC fuel tube [9], and the MTC VSV drain manifold [6].

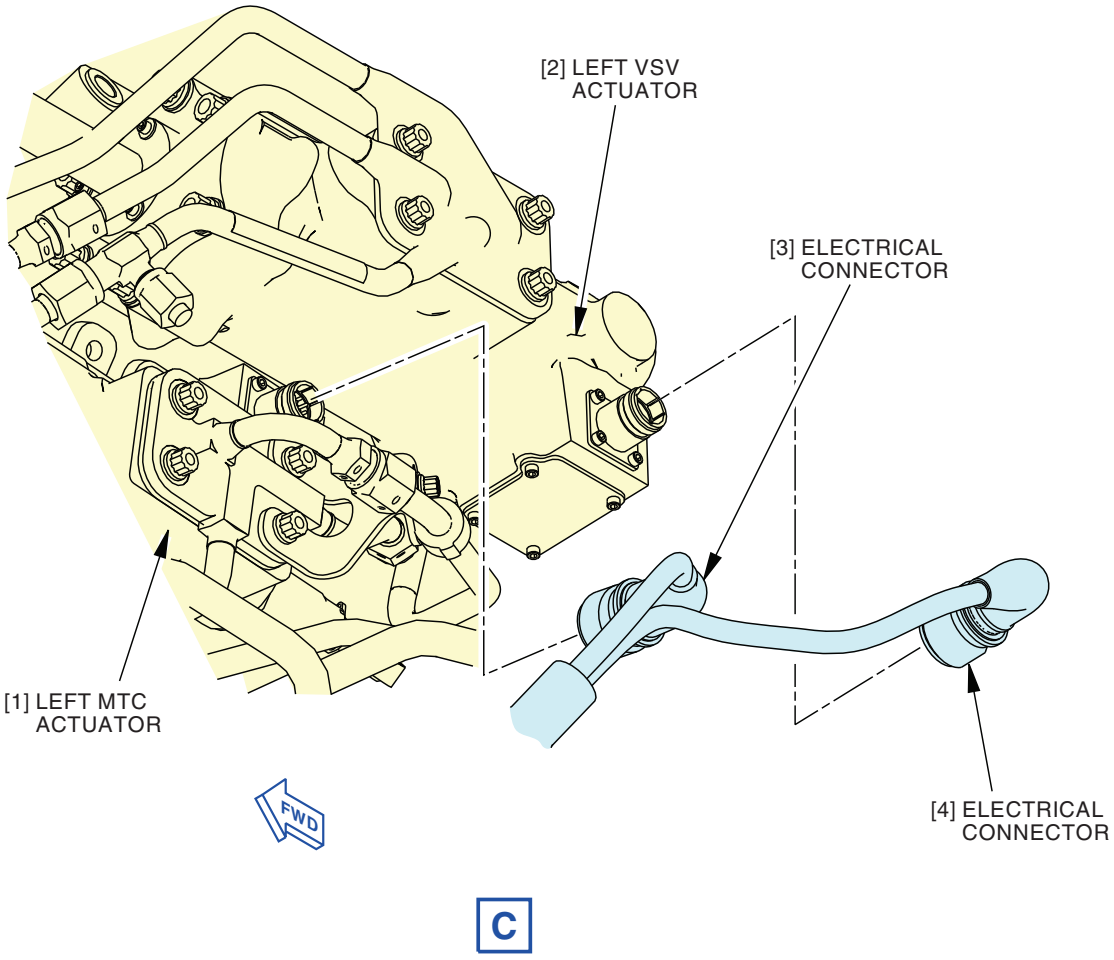
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left MTC Actuator Installation
Figure 401/75-23-03-990-801-G00 (Sheet 1 of 3)

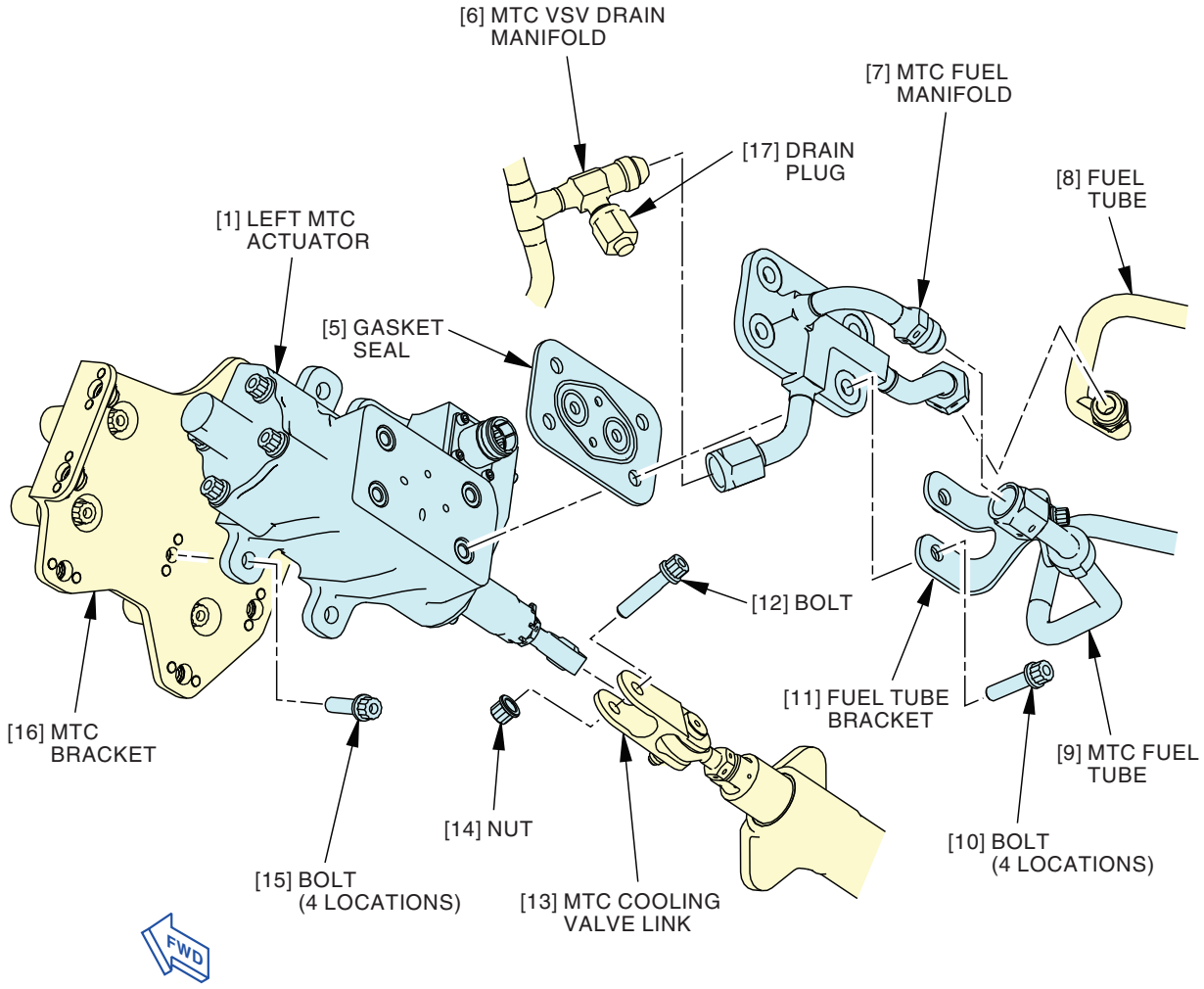
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Left MTC Actuator Installation
Figure 401/75-23-03-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Left MTC Actuator Installation
Figure 401/75-23-03-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-23-03-400-801-G00

3. Left Modulated Turbine Cooling Actuator Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the left Modulated Turbine Cooling (MTC) actuator.

(2) The left MTC actuator is at the 8:00 o'clock position on the engine compressor case.

(3) You must open the left thrust reverser to get access to the left MTC actuator.

(4) You must do the tests that are listed in the power plant test reference table after you install the left MTC actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Left MTC actuator	75-23-03-04-010	PTF ALL
5	Gasket seal	73-11-00-20-500	PTF ALL

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

G. Left Modulated Turbine Cooling Actuator Installation

SUBTASK 75-23-03-010-005-G00

(1) Remove the protective cover from the left MTC actuator [1] open port, the MTC fuel manifold [7], the fuel tube [8], the MTC fuel tube [9], and the MTC VSV drain manifold [6].

SUBTASK 75-23-03-410-001-G00

(2) Install the left MTC actuator [1] on the MTC bracket [16] as follows:

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(a) Put the left MTC actuator [1] in position on the MTC bracket [16].

NOTE: Make sure the left MTC actuator [1] rod end is aligned with the MTC cooling valve link [13]. You can turn the left MTC actuator [1] rod end if it is necessary to align it with the MTC cooling valve link [13].

(b) Apply the oil, D00599 [CP2442] on the threads of the four bolts [15].

(c) Install the four bolts [15] that attach the left MTC actuator [1] to the MTC bracket [16].

(d) Tighten the bolts [15] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

SUBTASK 75-23-03-410-002-G00

(3) Install the bolt [12] and the nut [14] that attaches the left MTC actuator [1] rod to the MTC cooling valve link [13].

(a) Apply the oil, D00599 [CP2442] on the threads of the bolt [12].

(b) Tighten the nut [14] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

SUBTASK 75-23-03-410-003-G00

(4) Install the MTC fuel manifold [7] on the left MTC actuator [1] as follows:

(a) Apply the engine oil, D00599 [CP2442] on the threads of the four bolts [10].

(b) Install a new gasket seal [5] on the left MTC actuator [1].

(c) Put the MTC fuel manifold [7] in position on the left MTC actuator [1].

(d) Install the four bolts [10] that attach the fuel tube bracket [11] and the MTC fuel manifold [7] to the left MTC actuator [1].

(e) Tighten the bolts [10] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

SUBTASK 75-23-03-410-004-G00

(5) Connect the MTC VSV drain manifold [6] to the MTC fuel manifold [7].

(a) Tighten the MTC VSV drain manifold [6] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

SUBTASK 75-23-03-410-005-G00

(6) Connect the MTC fuel tube [9] to the MTC fuel manifold [7].

(a) Tighten the MTC fuel tube [9] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

(b) Install the safety wire.

SUBTASK 75-23-03-410-006-G00

(7) Connect the fuel tube [8] to the MTC fuel manifold [7].

(a) Tighten the fuel tube [8] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

(b) Install the safety wire.

SUBTASK 75-23-03-410-007-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(CAUTION PRECEDES)

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(8) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [3] and the electrical connector [4] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical connector [3] and the electrical connector [4] from the left MTC actuator [1] and the left VSV actuator [2].

(b) Connect the electrical connector [3] to the left MTC actuator [1].

(c) Connect the electrical connector [4] to the left VSV actuator [2].

(d) Tighten the electrical connector [3] and the electrical connector [4].

SUBTASK 75-23-03-410-009-G00

(9) Install the segment shroud 4 (TASK 72-23-04-400-801-G00).

H. Put the Airplane Back to its Usual Condition

SUBTASK 75-23-03-860-003-G00

(1) Do these steps to put the fuel control valve and spar valve to the serviceable condition:

(a) Make sure the START LEVER switch is in the CUTOFF position.

(b) For the applicable engine, remove the safety tags and close these circuit breakers:

Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-23-03-410-010-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

I. Left MTC Actuator Installation Test

SUBTASK 75-23-03-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

PTF ALL

——— END OF TASK ———

TASK 75-23-03-000-802-G00

4. Right Modulated Turbine Cooling Actuator Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the right Modulated Turbine Cooling (MTC) actuator.

(2) The right MTC actuator is at the 2:30 o'clock position on the engine compressor case.

(3) You must open the right thrust reverser to get access to the right MTC actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

E. Prepare for the Removal

SUBTASK 75-23-03-010-006-G00

(1) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

SUBTASK 75-23-03-860-004-G00

(2) Do these steps to make sure that the fuel control valve and spar valve stay in the closed position:

(a) Make sure that the L (R) START LEVER switch is in the CUTOFF position.

1) Install a DO NOT OPERATE tag, STD-858, on the L (R) START LEVER switch.

(b) Make sure that the applicable SPAR VALVE CLOSED and ENG VALVE CLOSED lights, on the P5 forward overhead panel, are on dimly.

F. Right Modulated Turbine Cooling Actuator Removal

SUBTASK 75-23-03-010-007-G00

(1) Remove the segment shroud 2 (TASK 72-23-04-000-801-G00).

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SUBTASK 75-23-03-010-009-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR IS CLEAN BEFORE YOU DISCONNECT IT. DIRTY ELECTRICAL CONNECTORS CAN CAUSE INDICATION PROBLEMS.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO LOOSEN THE ELECTRICAL CONNECTOR. DO NOT USE PLIERS THAT HAVE METAL JAWS. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [41] from the right MTC actuator [32] (TASK 70-00-01-910-803-G00).

(a) Install the protective covers on the electrical receptacle on the right MTC actuator [32] and electrical connector [41].

SUBTASK 75-23-03-630-002-G00

(3) Drain the MTC fuel manifold [35] as follows:

(a) Cut and remove the safety wire or safety cable from the hydraulic fuel manifold [42] and MTC fuel manifold [35] B-nuts.

(b) Put a fuel resistant container (5 gal), STD-201, below the MTC fuel manifold [35].

(c) Disconnect the hydraulic fuel manifold [42] (2 locations) from the MTC fuel manifold [35].

1) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Disconnect the right MTC drain tube [39] from the MTC fuel manifold [35].

1) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

2) Remove the W200 electrical harness [40] and HJ5B electrical harness [36] from the clips on the right MTC drain tube [39].

(e) Move the fuel resistant container (5 gal), STD-201, away.

SUBTASK 75-23-03-010-010-G00

(4) Remove the right MTC actuator [32] as follows:

(a) Remove the four bolts [34] that attach the MTC fuel manifold [35] to the right MTC actuator [32].

(b) Remove the MTC fuel manifold [35] from the right MTC actuator [32].

(c) Remove and discard the gasket seal [33] from the right MTC actuator [32].

(d) Remove the bolt [43] and nut [45] that attach the right MTC actuator [32] rod to the MTC cooling valve link [44].

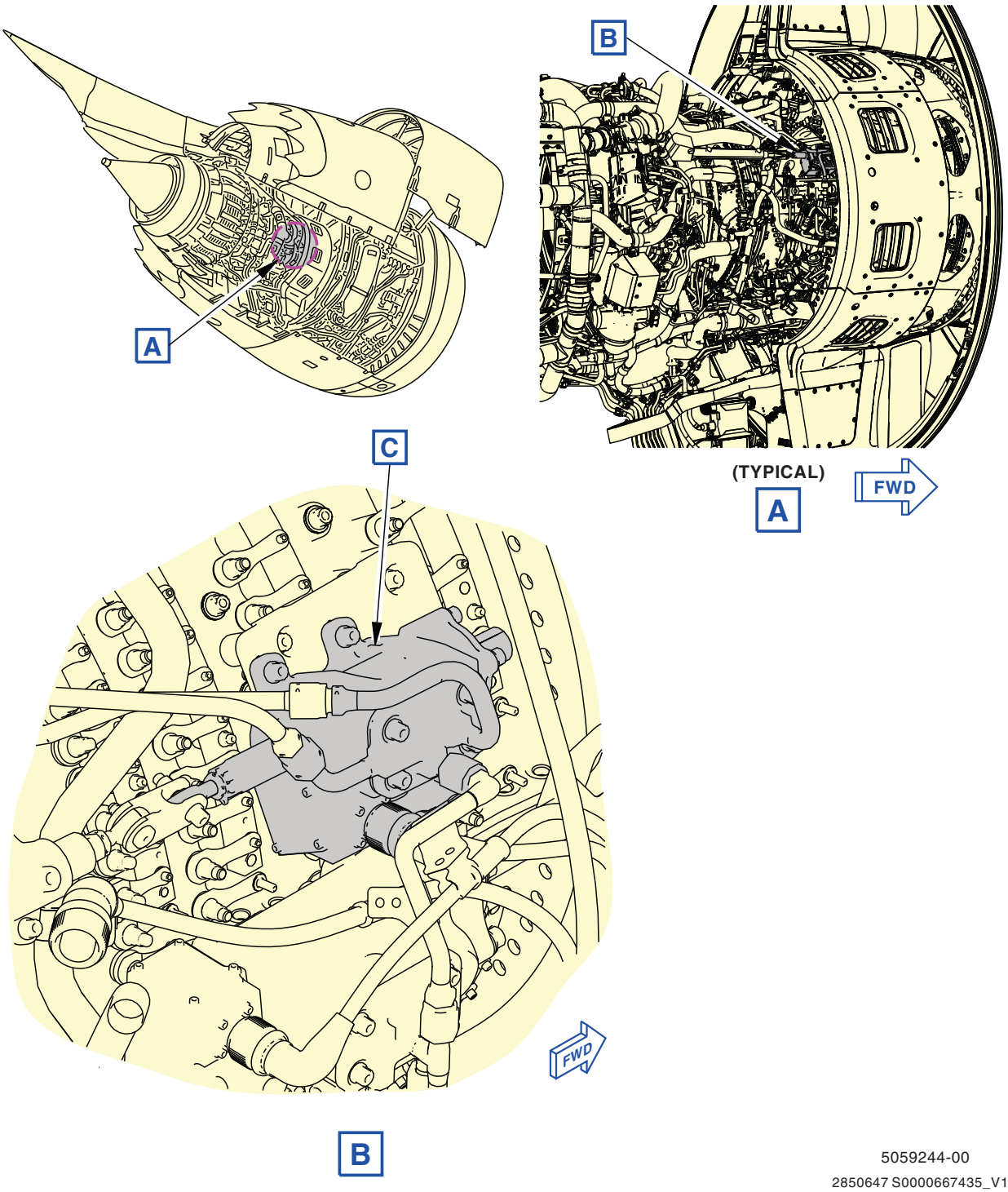
(e) Remove the four bolts [31] that attach the right MTC actuator [32] to the MTC actuator support bracket [46].

(f) Remove the right MTC actuator [32] from the engine.

(g) Install the protective covers on the right MTC actuator [32] open ports, MTC fuel manifold [35], hydraulic fuel manifold [42], and right MTC drain tube [39].

END OF TASK

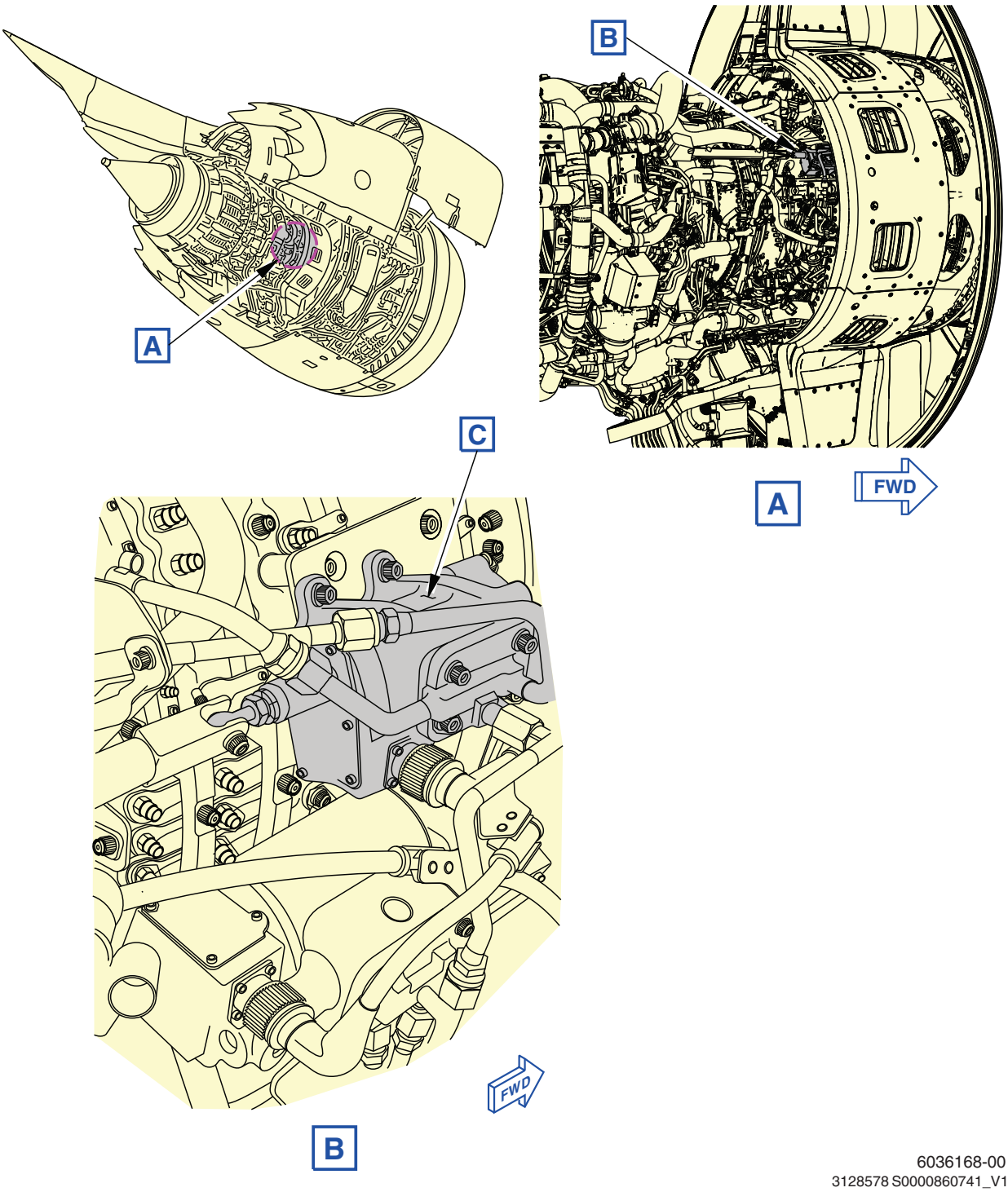
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2850647 S0000667435_V1

Right MTC Actuator Installation
Figure 402/75-23-03-990-802-G00 (Sheet 1 of 5)

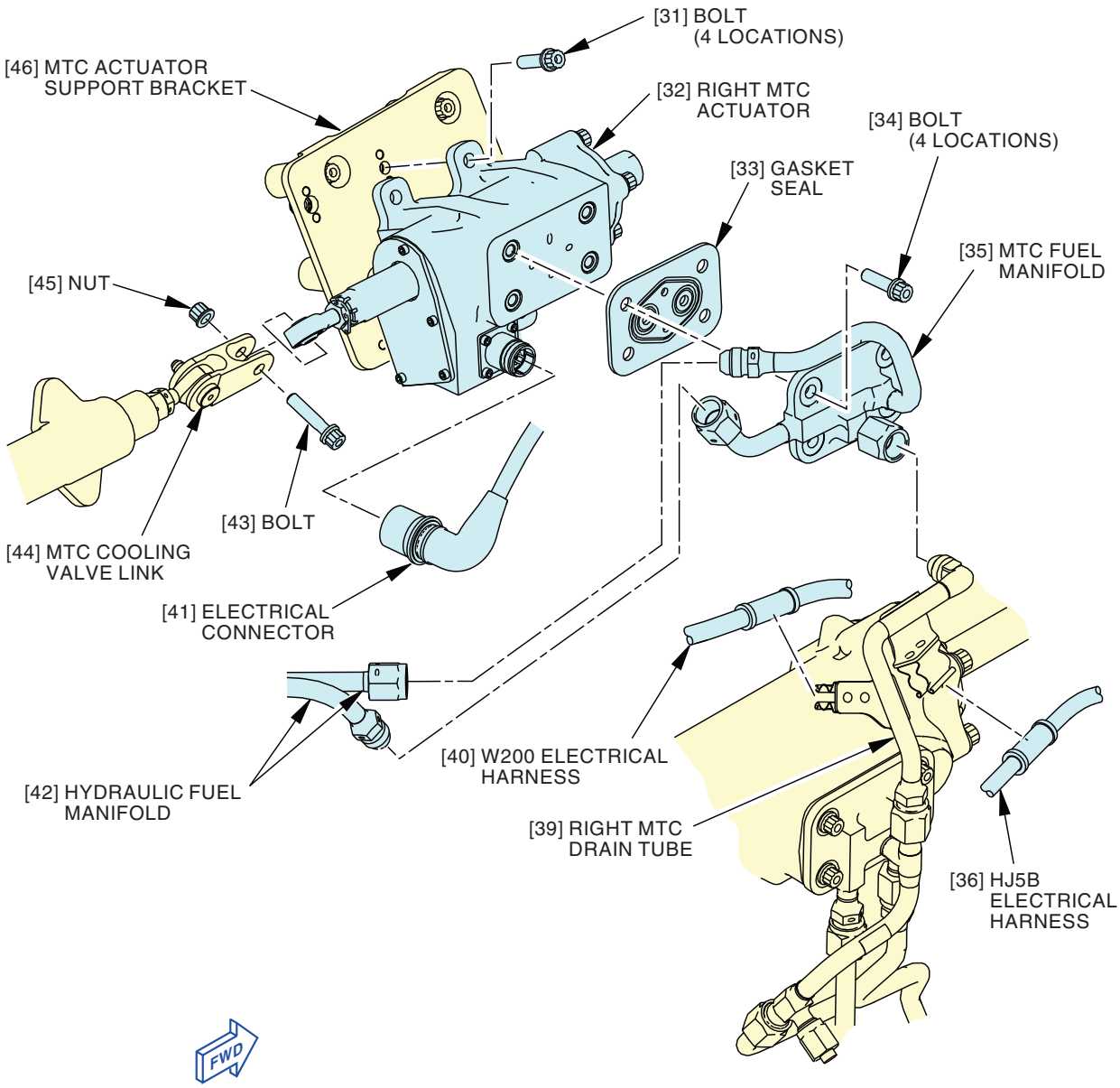
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6036168-00
3128578 S0000860741_V1

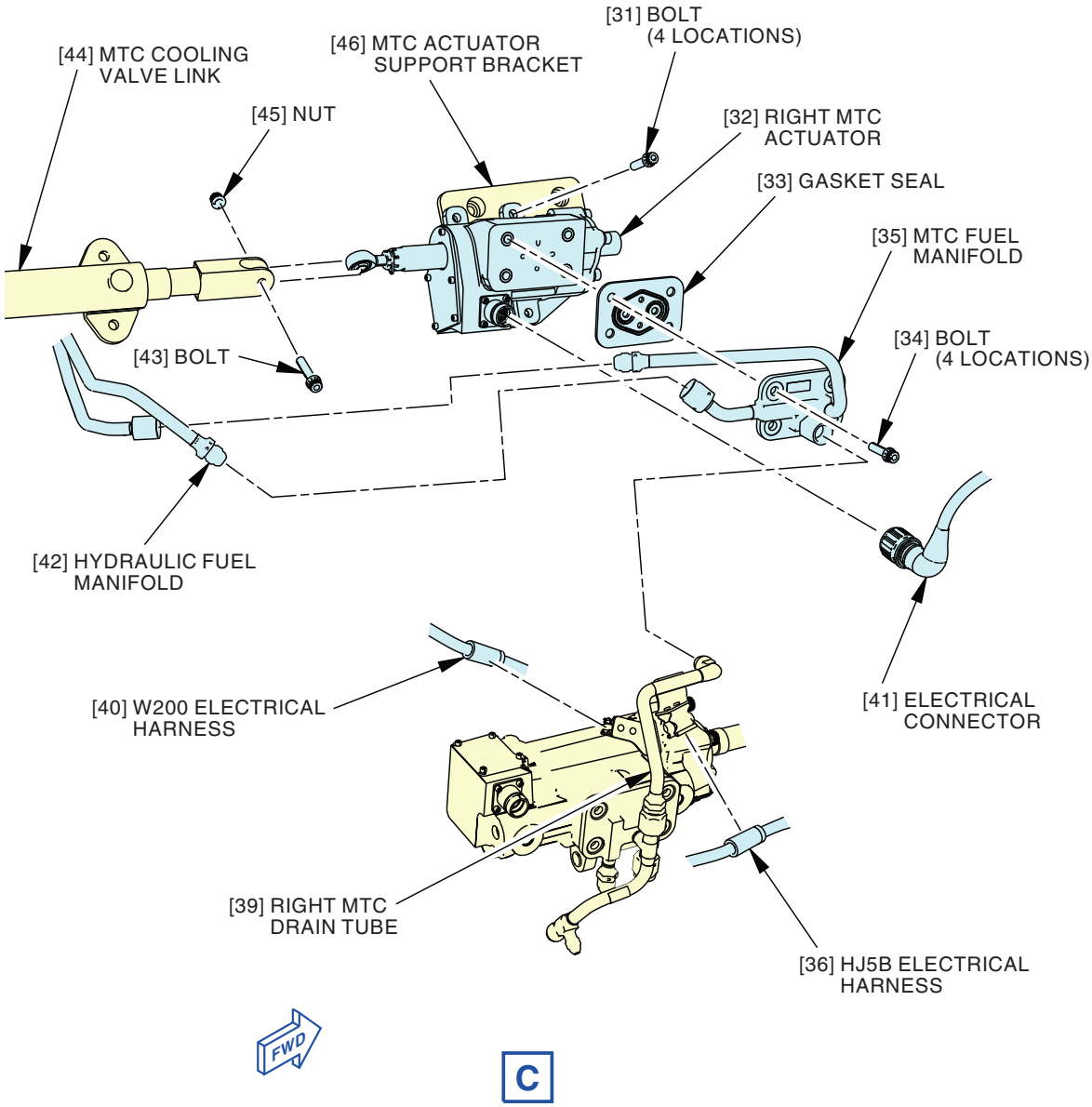
Right MTC Actuator Installation
Figure 402/75-23-03-990-802-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right MTC Actuator Installation
Figure 402/75-23-03-990-802-G00 (Sheet 3 of 5)

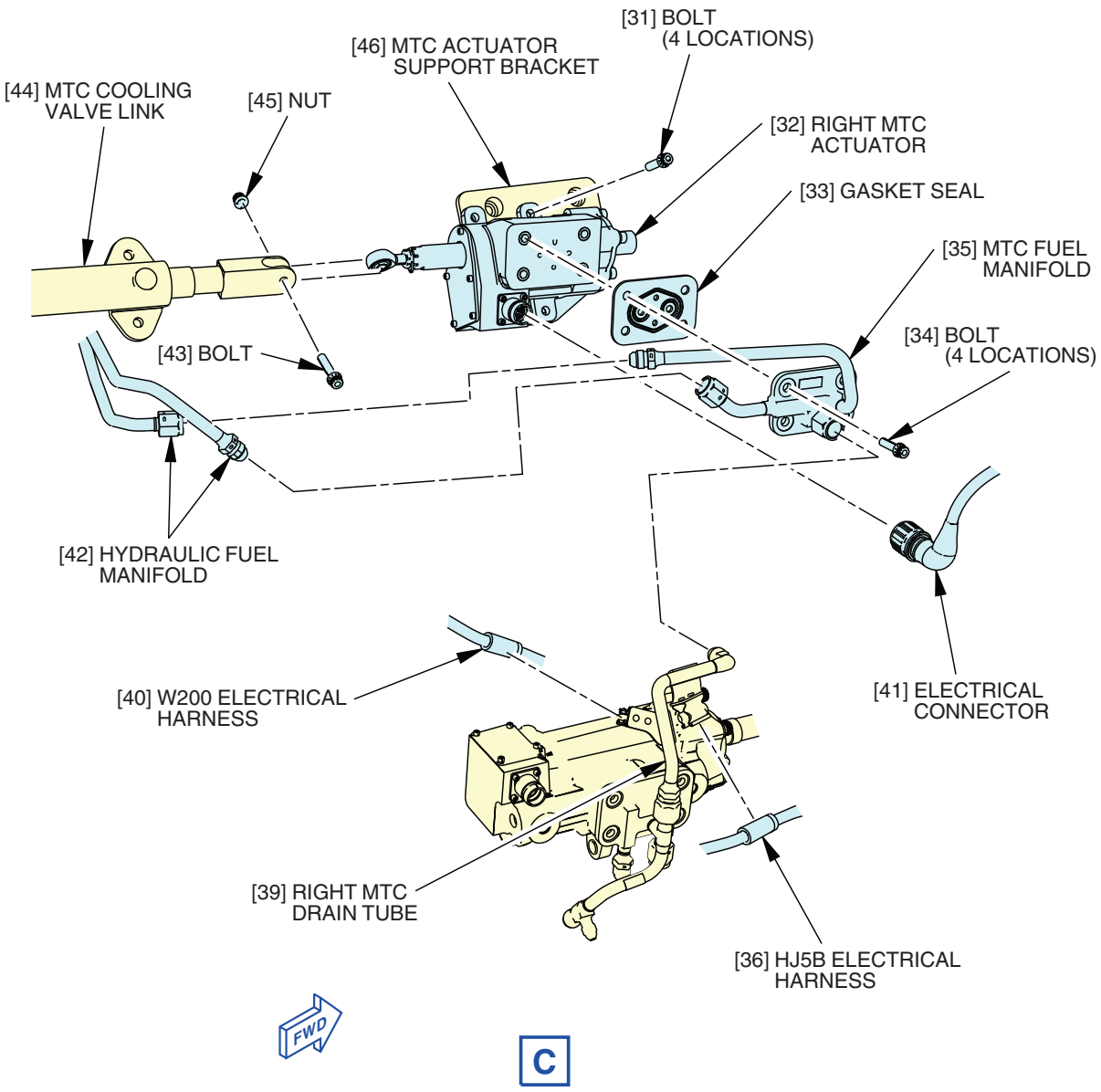
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2831354 S0000657224_V2

Right MTC Actuator Installation
Figure 402/75-23-03-990-802-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



6036166-00
3128588 S0000860742_V1

Right MTC Actuator Installation
Figure 402/75-23-03-990-802-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-23-03-400-802-G00

5. Right Modulated Turbine Cooling Actuator Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the right Modulated Turbine Cooling (MTC) actuator.

(2) The right MTC actuator is at the 2:30 o'clock position on the engine compressor case.

(3) You must open the right thrust reverser to get access to the right MTC actuator.

(4) You must do the tests that are listed in the power plant test reference table after you install the right MTC actuator.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
32	Right MTC actuator	75-23-03-03-010	PTF ALL
33	Gasket seal	73-11-00-20-540	PTF ALL

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

G. Right Modulated Turbine Cooling Actuator Installation

SUBTASK 75-23-03-010-011-G00

(1) Remove the protective cover from the MTC fuel manifold [35], right MTC actuator [32], hydraulic fuel manifold [42], and right MTC drain tube [39].

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SUBTASK 75-23-03-410-011-G00

(2) Install the right MTC actuator [32] on the MTC actuator support bracket [46] as follows:

(a) Put the right MTC actuator [32] in position on the MTC actuator support bracket [46].

1) Make sure that the right MTC actuator [32] rod end is aligned with the MTC cooling valve link [44].

a) If it is necessary, turn the right MTC actuator [32] rod end to align it with the MTC cooling valve link [44].

(b) Apply engine oil, D00599 [CP2442], on the threads of the four bolts [31].

(c) Install the four bolts [31] that attach the right MTC actuator [32] to the MTC actuator support bracket [46].

SUBTASK 75-23-03-410-012-G00

(3) Install the bolt [43] and nut [45] that attaches the right MTC actuator [32] rod to the MTC cooling valve link [44].

(a) Apply engine oil, D00599 [CP2442], on the threads of the bolt [43].

SUBTASK 75-23-03-410-013-G00

(4) Install the MTC fuel manifold [35] on the right MTC actuator [32] as follows:

(a) Apply engine oil, D00599 [CP2442], on the threads of the four bolts [34].

(b) Install a new gasket seal [33] on the right MTC actuator [32].

(c) Put the MTC fuel manifold [35] in position on the right MTC actuator [32].

(d) Install the four bolts [34] that attach the MTC fuel manifold [35] to the right MTC actuator [32].

SUBTASK 75-23-03-410-014-G00

(5) Connect the right MTC drain tube [39] to the MTC fuel manifold [35].

SUBTASK 75-23-03-410-015-G00

(6) Connect the hydraulic fuel manifold [42] to the MTC fuel manifold [35].

SUBTASK 75-23-03-420-001-G00

(7) Tighten the bolts [31] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

SUBTASK 75-23-03-420-002-G00

(8) Tighten the nut [45] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

SUBTASK 75-23-03-420-003-G00

(9) Tighten the bolts [34] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

SUBTASK 75-23-03-420-004-G00

(10) Tighten the MTC fuel manifold [35] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

SUBTASK 75-23-03-420-005-G00

(11) Tighten the hydraulic fuel manifold [42] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

SUBTASK 75-23-03-420-006-G00

(12) Tighten the MTC fuel manifold [35] B-nut to 262.0 in-lb (29.6 N·m) - 308.0 in-lb (34.8 N·m).

SUBTASK 75-23-03-420-007-G00

(13) Put the HJ5B electrical harness [36] and W200 electrical harness [40] in the clips on the right MTC drain tube [39].

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SUBTASK 75-23-03-420-008-G00

(14) Install safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006], on the hydraulic fuel manifold [42] and MTC fuel manifold [35] B-nuts.

SUBTASK 75-23-03-410-016-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(15) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [41] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle on the right MTC actuator [32] and electrical connector [41].

(b) Connect the electrical connector [41] to the right MTC actuator [32].

1) Tighten the electrical connector [41].

SUBTASK 75-23-03-410-018-G00

(16) Install the segment shroud 2 (TASK 72-23-04-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-23-03-860-005-G00

(1) Do these steps to put the fuel control valve and spar valve to the serviceable condition:

(a) Make sure the START LEVER switch is in the CUTOFF position.

(b) For the applicable engine, remove the safety tags and close these circuit breakers:

Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-23-03-410-019-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

I. Right MTC Actuator Installation Test

SUBTASK 75-23-03-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL VALVE (LPTACC) - REMOVAL/ INSTALLATION

1. General

A. This procedure has two tasks:

(1) Low Pressure Turbine Clearance Control (LPTACC) Valve Removal

(2) Low Pressure Turbine Clearance Control (LPTACC) Valve Installation.

TASK 75-24-01-000-801-G00

2. Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the Low Pressure Turbine Active Clearance Control Valve (LPTACC).

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-24-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-24-01-865-001-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-01-865-002-G00

(3) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-24-01-010-001-G00

(4) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. Low Pressure Turbine Active Clearance Control Valve Removal

SUBTASK 75-24-01-030-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

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(WARNING PRECEDES)

CAUTION MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use a teflon-jawed pliers, STD-664 to disconnect the electrical connector [10] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the electrical connector [10] from the LPTACC valve [17].

(b) Put the protective covers on the electrical connector [10] and electrical receptacles on the LPTACC valve [17].

SUBTASK 75-24-01-010-003-G00

(2) Remove the four bolts [9] that attach the ACC fuel manifold [6] to the LPTACC valve [17].

(a) Remove and discard the gasket seal [7].

(b) Put a protective cover on the ACC fuel manifold [6] and the LPTACC valve [17].

SUBTASK 75-24-01-020-001-G00

(3) Remove the LPTACC valve [17] as follows:

(a) Loosen the hose clamp [2] that attaches the LPT valve duct [3] to the LPTACC valve [17].

(b) Loosen the hose clamp [11] that attaches the ACC wye duct [12] to the LPTACC valve [17].

(c) Remove the bolt [26] and washer [27] that attach the LPTACC valve [17] to the ACC valve bracket [25].

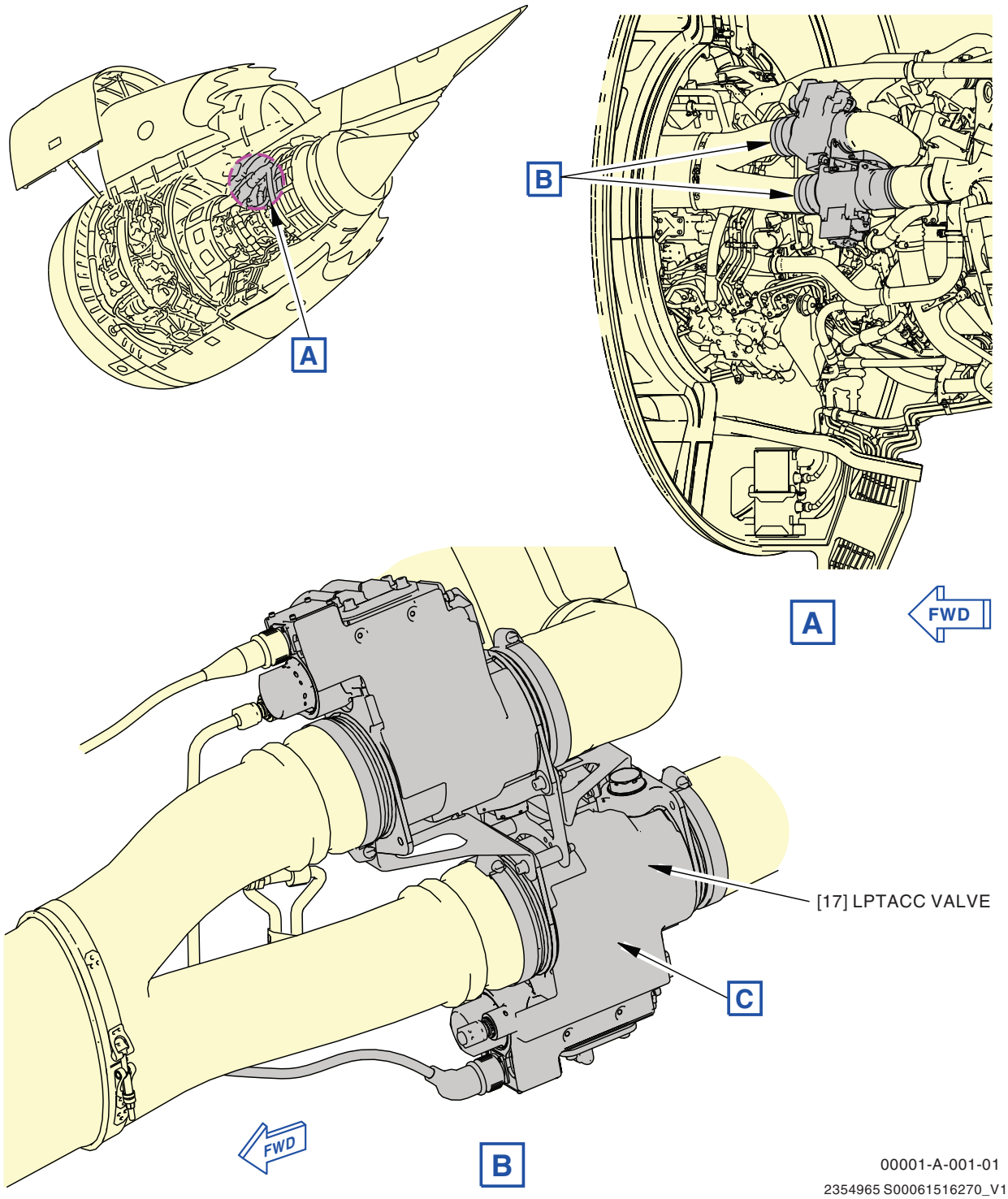
(d) Hold the LPTACC valve [17] and remove the two bolts [28] that attach the LPTACC valve [17] to the ACC valve bracket [29] and ACC valve bracket [30].

(e) Remove the LPTACC valve [17] from the engine.

(f) Put protective covers on the LPT valve duct [3], ACC wye duct [12], LPTACC valve [17] and HPTACC valve [18].

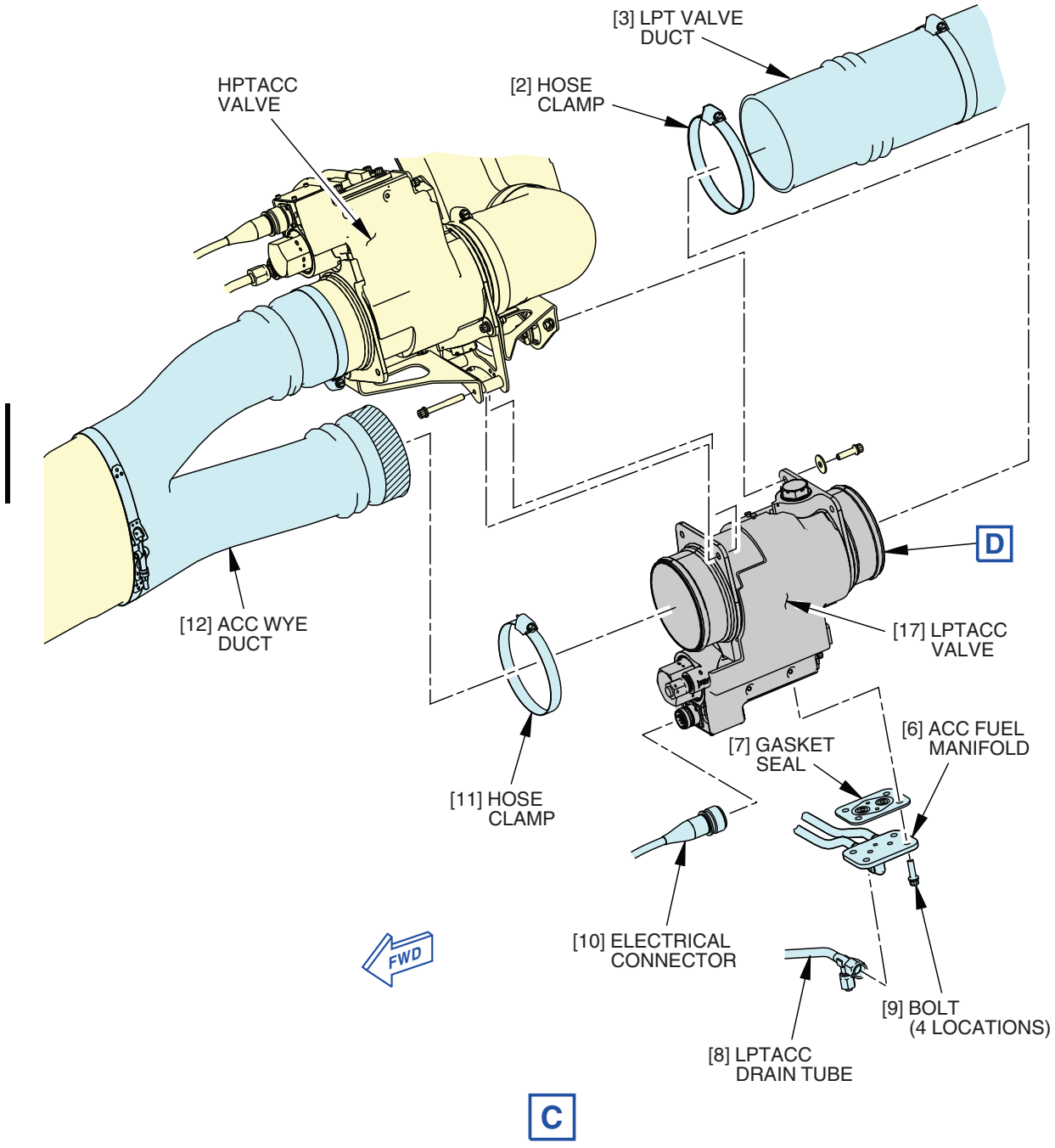
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPTACC Valve Installation
Figure 401/75-24-01-990-801-G00 (Sheet 1 of 3)

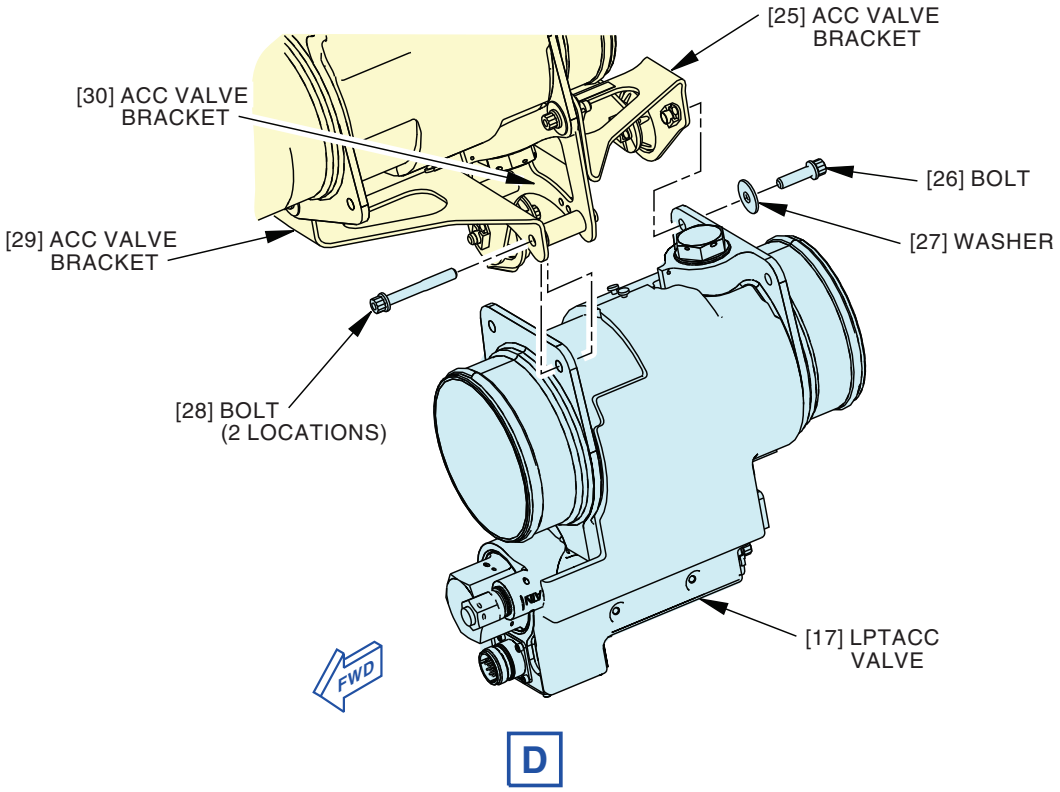
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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LPTACC Valve Installation
Figure 401/75-24-01-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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2386074 S0000547839_V3

LPTACC Valve Installation
Figure 401/75-24-01-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-24-01-400-801-G00

3. Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the Low Pressure Turbine Active Clearance Control (LPTACC).

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-06-210-801-G00	Split Control Unit Inspection (P/B 601)
73-21-09-400-801-G00	Data Entry Plug Installation (P/B 401)
73-21-09-700-801-G00	Data Entry Plug Configuration Check (P/B 601)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G51669 [CP2975]	Tape - Silicone Rubber, Glass Cloth Fabric	MIL-I-22444

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Gasket seal	73-11-00-20-270	PTF ALL
17	LPTACC valve	75-24-01-01-040	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation

SUBTASK 75-24-01-210-001-G00

(1) Check the LPTACC valve [17], Split Control Unit (SCU) part numbers, and Data Entry Plug (DEP) configuration to be used as follows:

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 73-0013

NOTE: It is not necessary to do the part number check.

PTF ALL

(a) Examine and record the part number of the installed SCU.

1) To get access to the SCU, do this task: Split Control Unit Inspection, TASK 73-11-06-210-801-G00.

(b) Examine and record the part number of the LPTACC valve [17] to be installed.

(c) Examine and record the hardware configuration data entry plug pin configuration from the DEP.

1) To get access to the DEP and do the inspection, do this task: Data Entry Plug Configuration Check, TASK 73-21-09-700-801-G00.

(d) Install the DEP again (TASK 73-21-09-400-801-G00).

CAUTION

THE USE OF ANY SCU AND LPTACC VALVE PART NUMBER AND DEP CONFIGURATION COMBINATION NOT LISTED BELOW IS NOT PERMITTED. THIS CAN LEAD TO INCORRECT COMPONENT AND ENGINE PERFORMANCE AND TOO MUCH ENGINE WEAR.

(e) Make sure that the LPTACC valve [17], SCU part numbers, and DEP configuration agree with the combinations given in Table 401.

Table 401/75-24-01-993-801-G00 SCU and LPTACC Valve Part Number and DEP Configuration Combination

Allowed Configurations	SCU	LPTACC Valve	Engine Configuration Plug
Pre LEAP-1B Service Bulletin 73-0013	8062-1151	8910-274	Make sure that the DEP hardware configuration is correct (TASK 73-21-09-700-801-G00).
Post LEAP-1B Service Bulletin 73-0013	8062-1180	8910-309	

(f) If the part numbers do not agree, replace the LPTACC valve [17] with a unit that agrees with the installed SCU part number and DEP configuration.

SUBTASK 75-24-01-420-001-G00

(2) Install the LPTACC valve [17] as follows:

(a) Remove the protective covers from the LPT valve duct [3], ACC wye duct [12], and LPTACC valve [17].

(b) Move the hose clamp [2] on the forward end of LPT valve duct [3] and put the head of the screw at approximately the 12 o'clock position.

(c) Use a back and forth motion to twist and push the LPT valve duct [3] forward end point on the LPTACC valve [17] past the LPTACC valve bead a minimum distance of the hose clamp width.

(d) Put the hose clamp [2] in position on the rubber joint between the raised LPTACC valve [17] bead and LPT valve duct [3].

(e) Put the LPTACC valve [17] in position on the LPT valve duct [3], ACC valve bracket [25], ACC valve bracket [29] and ACC valve bracket [30].

(f) Move the hose clamp [11] on the aft end of the ACC wye duct [12] and put it in position so that there is access to the head of the screw.

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(g) Use a back and forth motion to twist and push the ACC wye duct [12] aft end point on the LPTACC valve [17] past the LPTACC valve bead a minimum distance of the hose clamp width.

(h) Make sure that the ACC wye duct [12] has silicone rubber tape, G51669 [CP2975], in the aft side clamping area.

1) If the ACC wye duct [12] does not have silicone rubber tape, G51669 [CP2975], in the aft side clamping area, do this step:

a) Apply one row of silicone rubber tape, G51669 [CP2975], 0.020 in. (0.51 mm) thick and 1.00 in. (25.4 mm) wide to the ACC wye duct [12] end directly forward of the aft edge bead of the ACC wye duct [12].

(i) Put the hose clamp [11] in position on the rubber joint between the raised LPTACC valve [17] bead and ACC wye duct [12].

(j) Lubricate the threads of the bolt [26] and two bolts [28] with clean engine oil, D00599 [CP2442].

(k) Install the two bolts [28] that attach the LPTACC valve [17] to the ACC valve bracket [30].

NOTE: Do not tighten the bolts at this time.

(l) Install the bolt [26] and washer [27] that attaches the LPTACC valve [17] to the ACC valve bracket [25].

NOTE: Do not tighten the bolt at this time.

SUBTASK 75-24-01-420-002-G00

(3) Connect the ACC fuel manifold [6] to the LPTACC valve [17] as follows:

(a) Remove the protective covers from the ACC fuel manifold [6] and LPTACC valve [17].

(b) Install a new gasket seal [7] between the ACC fuel manifold [6] and LPTACC valve [17].

(c) Lubricate the threads of the four bolts [9] with clean engine oil, D00599 [CP2442].

(d) Install the four bolts [9] that attach the ACC fuel manifold [6] to the LPTACC valve [17].

(e) Tighten the four bolts [9] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(f) Tighten the two bolts [28] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(g) Tighten the bolt [26] to 106.0 in-lb (12.0 N·m) - 124.0 in-lb (14.0 N·m).

(h) Install the LPTACC valve [17] on the ACC wye duct [12].

(i) Tighten the hose clamp [2] and hose clamp [11] to 32.0 in-lb (3.6 N·m) - 38.0 in-lb (4.3 N·m).

SUBTASK 75-24-01-430-002-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(4) Use a teflon-jawed pliers, STD-664, to connect the electrical connector [10] as follows (TASK 70-00-01-910-803-G00):

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(a) Remove the protective covers from the electrical receptacle on the LPTACC valve [17] and electrical connector [10].

(b) Connect the electrical connector [10] to the LPTACC valve [17].

(c) Tighten the electrical connector [10].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 75-24-01-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-24-01-865-003-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-24-01-865-004-G00

(3) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-01-860-002-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

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J. Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation Test

SUBTASK 75-24-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL MANIFOLDS (LPTACC) - REMOVAL/ INSTALLATION

1. General

A. This procedure has the following tasks:

(1) A removal of the Low Pressure Turbine Active Clearance Control (LPTACC) upper manifold assembly

(2) An installation of the LPTACC upper manifold assembly

(3) A removal of the LPTACC lower manifold assembly

(4) An installation of the LPTACC lower manifold assembly.

TASK 75-24-02-000-801-G00

2. Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Assembly Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the Low Pressure Turbine Active Clearance Control (LPTACC) upper manifold assembly.

B. References

Reference	Title
26-11-01-000-802	Engine Fire Detector Assembly Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
75-33-01-000-801-G00	Transient Bleed Valve Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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F. Prepare for the Removal

SUBTASK 75-24-02-010-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-24-02-860-001-G00

(2) For the Engine 1, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-02-860-002-G00

(3) For the Engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-02-010-008-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

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1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 75-24-02-010-003-G00

(5) Remove the right core fire detector assembly (TASK 26-11-01-000-802).

SUBTASK 75-24-02-010-004-G00

(6) Remove the transient bleed valve duct (TASK 75-33-01-000-801-G00).

G. LPTACC Upper Manifold Assembly Removal

SUBTASK 75-24-02-020-001-G00

(1) Remove the swivel fitting [4] from the LPTACC upper manifold assembly [1].

(a) Remove the two bolts [5] that attach the swivel fitting [4] to the air duct.

SUBTASK 75-24-02-020-003-G00

(2) Remove the 12 nuts [8], 12 flat washers [9], 12 bolts [6], and 12 flat washers [7] that attach the LPTACC upper manifold assembly [1] to the Low Pressure Turbine (LPT) case at the 1:00, 2:00, 3:00, 9:00, 10:00, and 11:00 o'clock positions.

SUBTASK 75-24-02-020-004-G00

(3) Remove the four nuts [11], two retaining plates [10], four flat washers [13], and four bolts [12] that attach the LPTACC upper manifold assembly [1] to the bracket at the 9:00 and 11:00 o'clock positions.

SUBTASK 75-24-02-020-005-G00

(4) Remove the saddle clamp [17] from the bracket assembly [20] and air duct at the 10:00 o'clock position.

(a) Remove the two bolts [18] that attach the saddle clamp [17] to the bracket assembly [20].

SUBTASK 75-24-02-020-006-G00

(5) Remove the bracket assembly [20] from the LPTACC upper manifold assembly [1].

(a) While you hold the bracket assembly [20], remove the two nuts [16], two flat washers [15], two bolts [14], and bolt [19] that attach the LPTACC upper manifold assembly [1] to the bracket at the 10:00 o'clock position.

SUBTASK 75-24-02-020-007-G00

(6) Remove the two nuts [11], retaining plate [10], two flat washers [13], and two bolts [12] that attach the LPTACC upper manifold assembly [1] to the bracket at the 3:00 o'clock position.

SUBTASK 75-24-02-020-008-G00

(7) Remove the four nuts [11], two retaining plates [10], four flat washers [13], and four bolts [12] that attach the LPTACC upper manifold assembly [1] to the bracket at the 1:00 and 2:00 o'clock positions.

SUBTASK 75-24-02-020-009-G00

(8) While you hold the LPTACC upper manifold assembly [1], disconnect the LPTACC upper manifold assembly [1] from the LPT case at 12:00 o'clock position.

(a) Remove the two nuts [23], two flat washers [22], and two bolts [21] that attach the LPTACC upper manifold assembly [1] to the bracket.

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(b) Remove the two nuts [26], two flat washers [25], and two bolts [24] that attach the LPTACC upper manifold assembly [1] to the LPT case.

(c) Remove the two spring cups [27], two helical compression springs [28], and two flat washers [29] from the LPTACC upper manifold assembly [1].

SUBTASK 75-24-02-020-018-G00

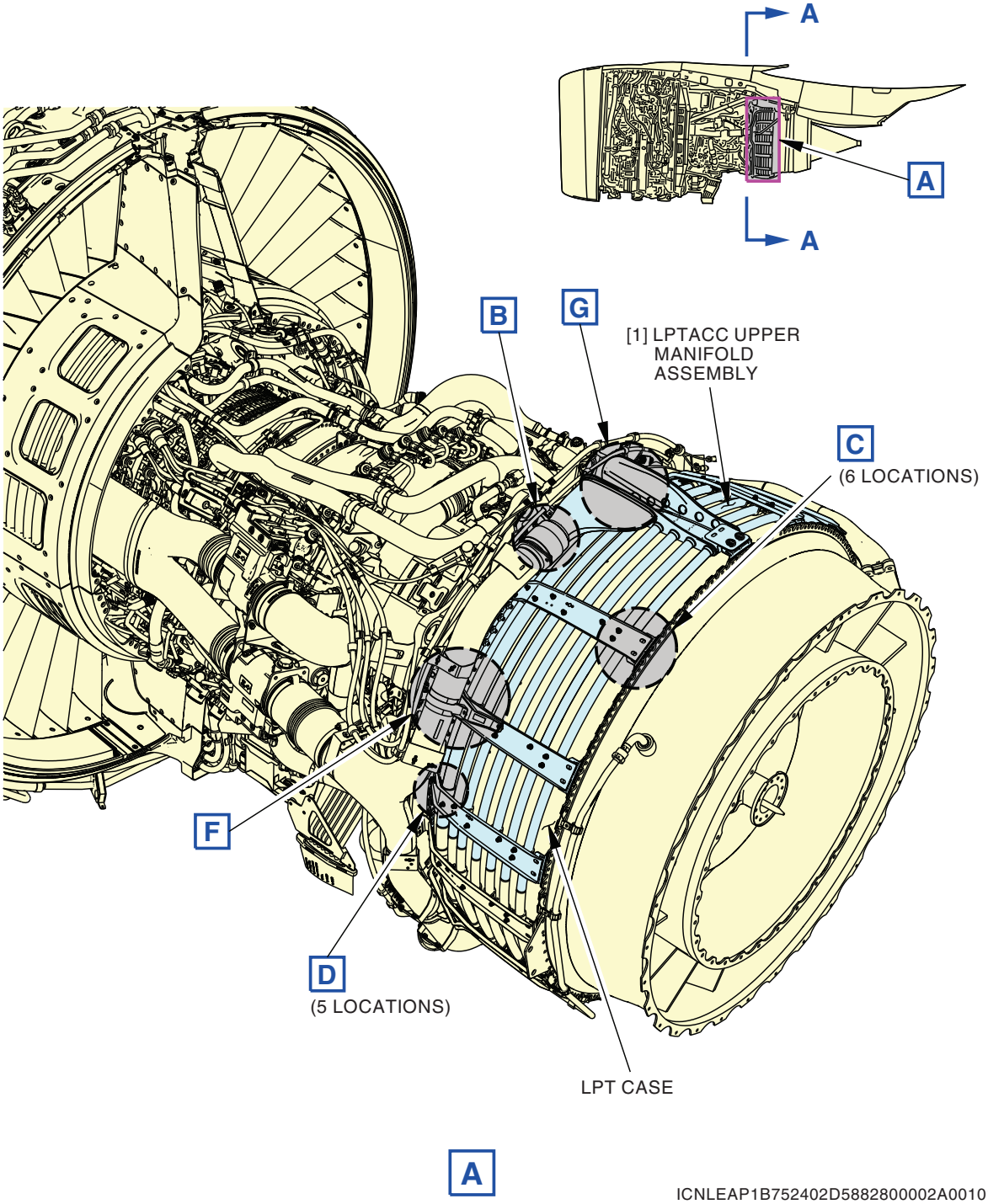
WARNING BE CAREFUL WHEN YOU LIFT THE LPTACC UPPER MANIFOLD ASSEMBLY. THE LPTACC UPPER MANIFOLD ASSEMBLY IS HEAVY. INJURIES TO PERSONNEL CAN OCCUR.

CAUTION MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC UPPER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(9) Carefully remove the LPTACC upper manifold assembly [1] from the LPT case.

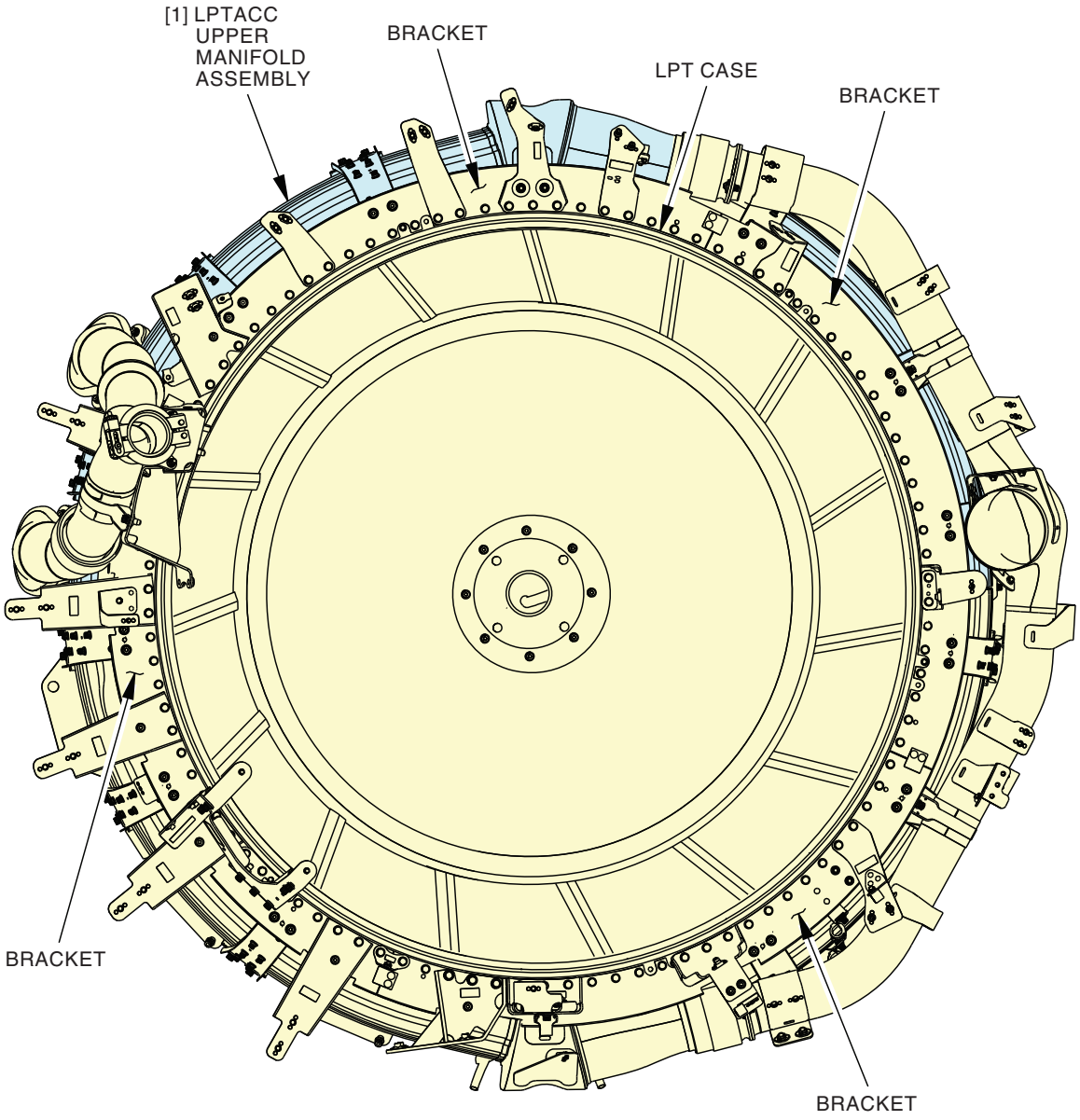
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPTACC Upper Manifold Assembly Installation
Figure 401/75-24-02-990-801-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE AFT DIRECTION)

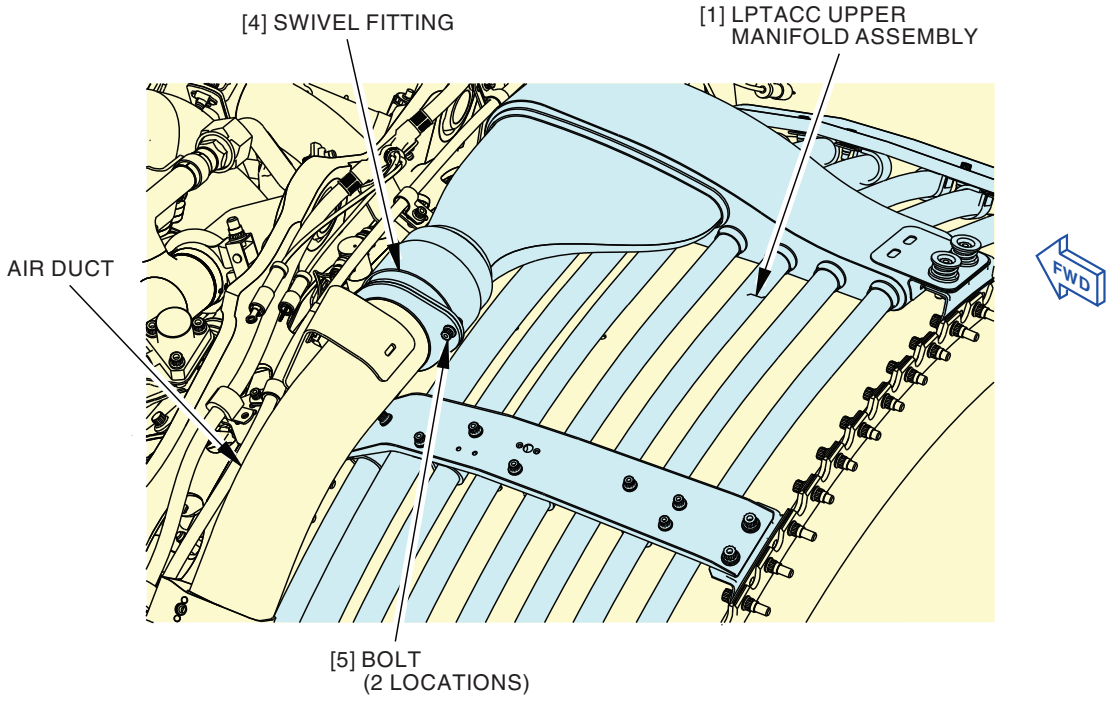
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LPTACC Upper Manifold Assembly Installation
Figure 401/75-24-02-990-801-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

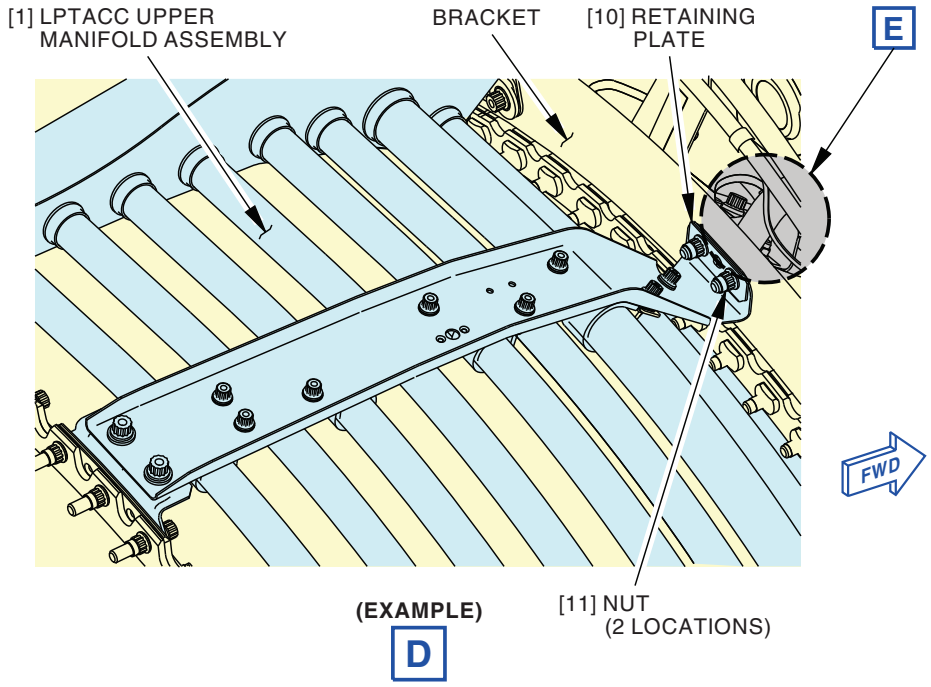


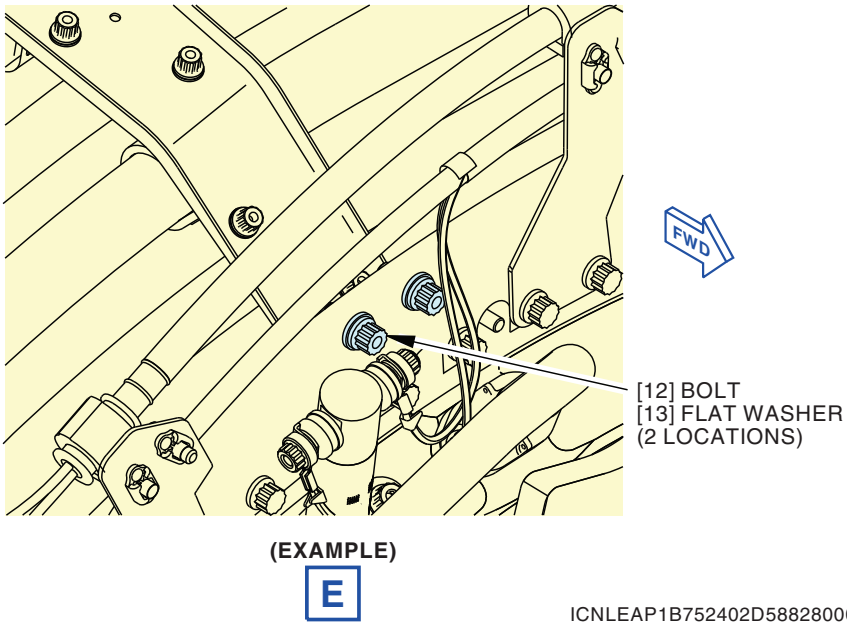
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LPTACC Upper Manifold Assembly Installation
Figure 401/75-24-02-990-801-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



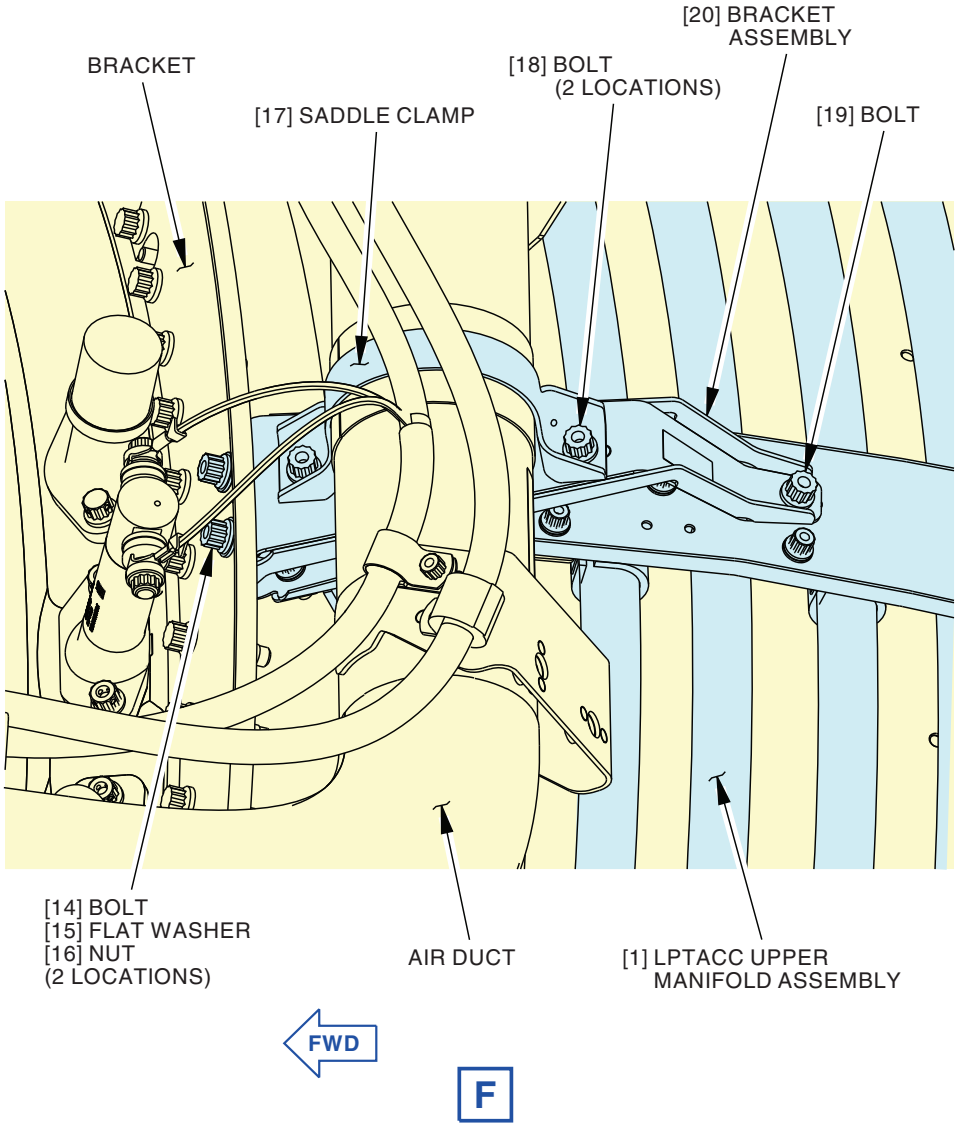


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LPTACC Upper Manifold Assembly Installation
Figure 401/75-24-02-990-801-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

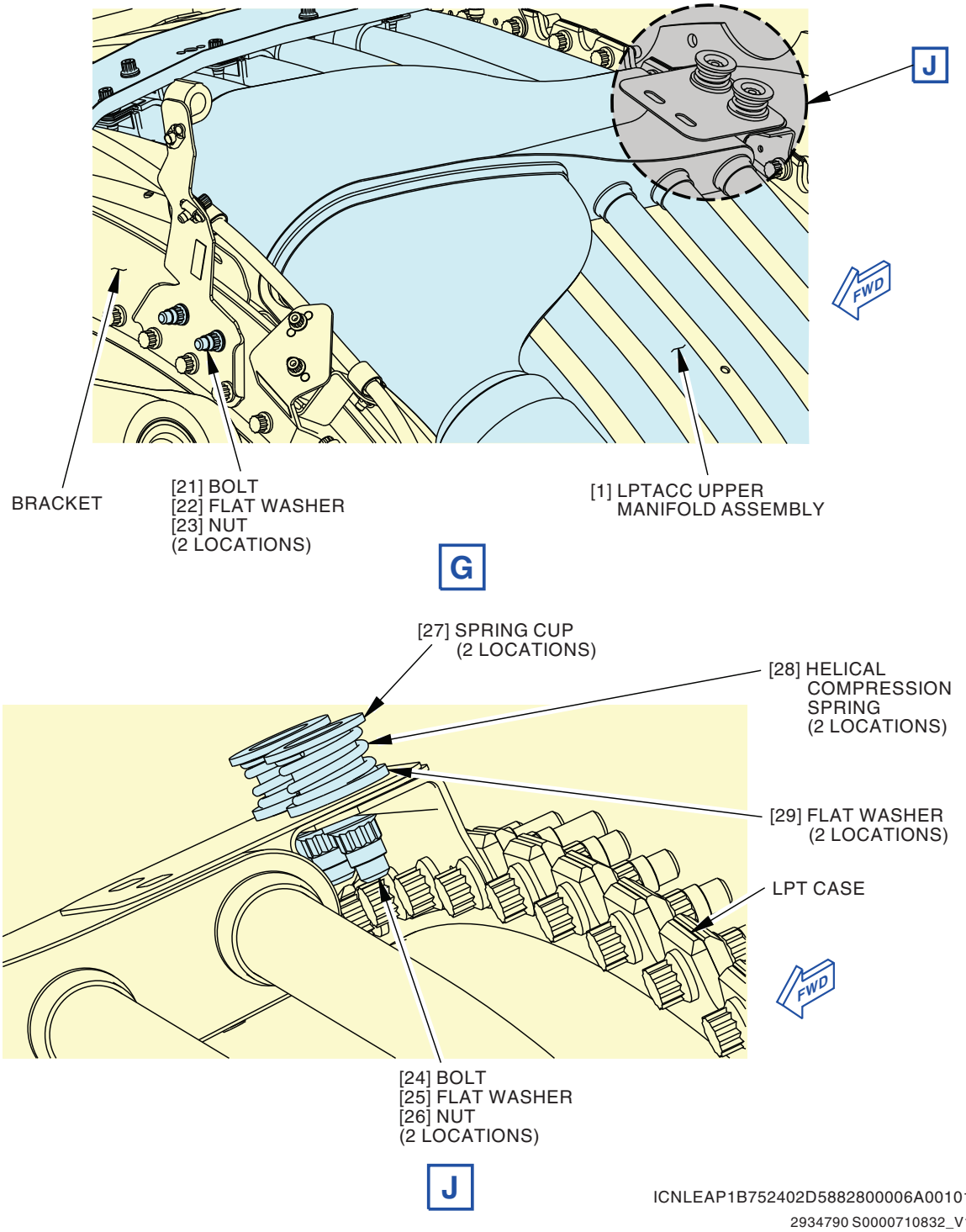


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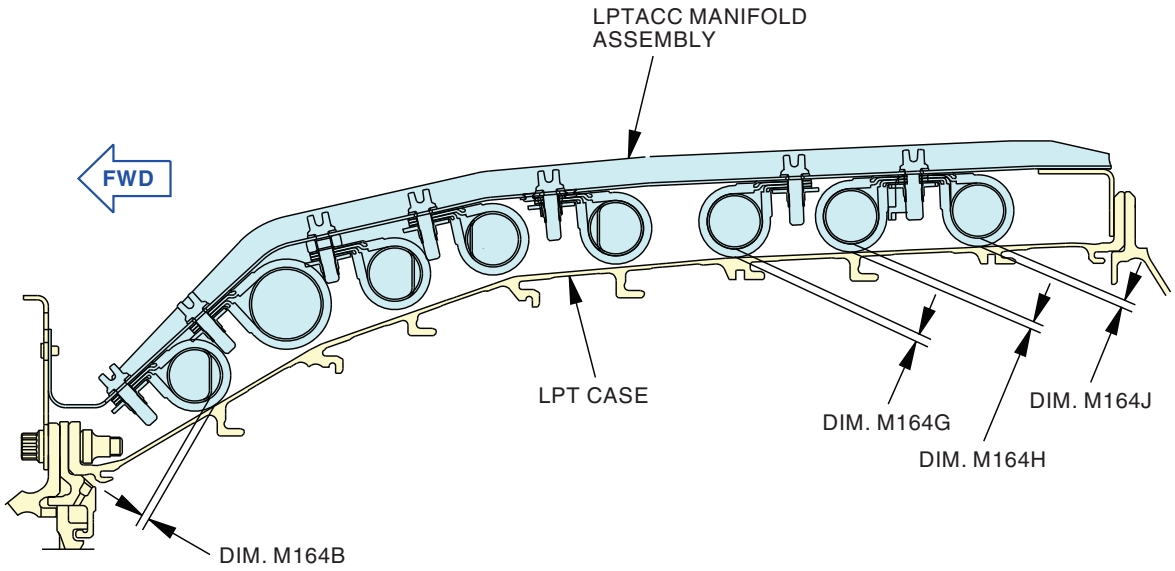
LPTACC Upper Manifold Assembly Installation
Figure 401/75-24-02-990-801-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

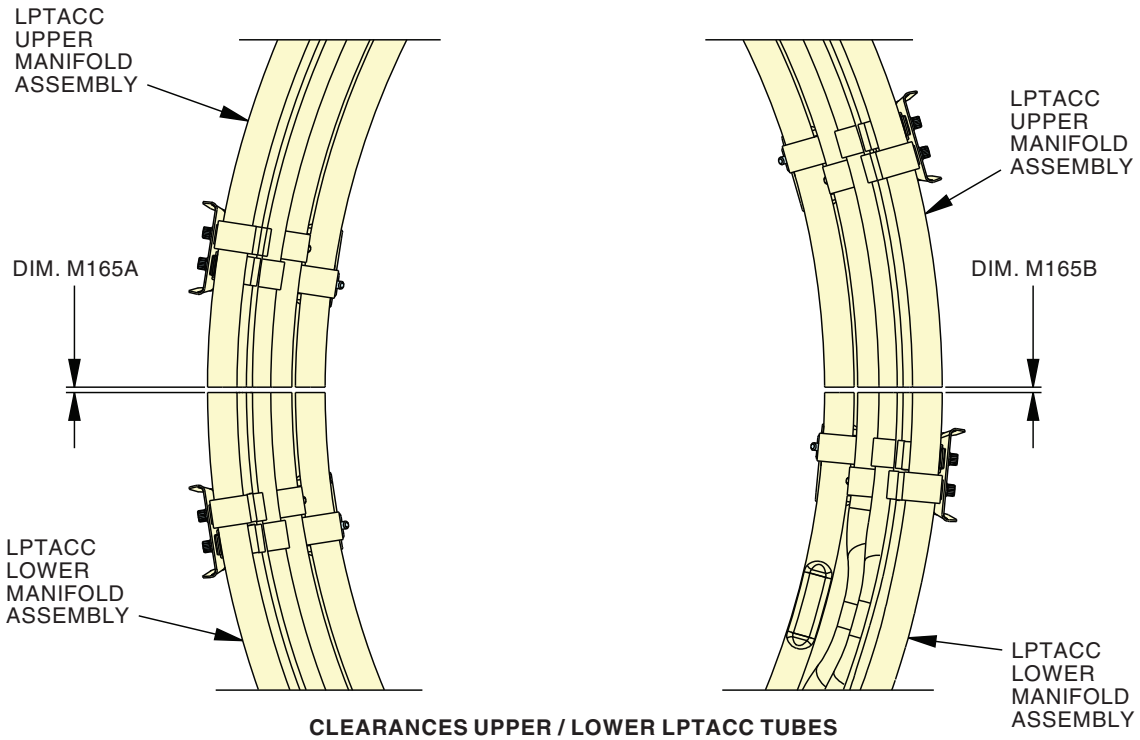


LPTACC Upper Manifold Assembly Installation
Figure 401/75-24-02-990-801-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CLEARANCES LPTACC TUBES / LPT CASE



CLEARANCES UPPER / LOWER LPTACC TUBES
(VIEW IN THE FORWARD DIRECTION)

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LPTACC Upper and Lower Manifold Assembly Adjustment
Figure 402/75-24-02-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-24-02-400-801-G00

3. Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Assembly Installation

(Figure 401, Figure 402)

A. General

(1) This task gives the instructions on how to install the Low Pressure Turbine Active Clearance Control (LPTACC) upper manifold assembly.

B. References

Reference	Title
26-11-01-400-802	Engine Fire Detector Assembly Installation (P/B 401)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
75-33-01-400-801-G00	Transient Bleed Valve Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. LPTACC Upper Manifold Assembly Installation

SUBTASK 75-24-02-420-001-G00

(1) Apply graphite mineral vaseline, D00601 [CP2101], on the thread of the bolts and pressure face of the nuts that follow:

(a) the 12 bolts [6]

(b) the two nuts [23]

(c) the two nuts [26]

(d) the two nuts [16]

(e) the 10 nuts [11]

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) the two bolts [18]

(g) the bolt [19].

SUBTASK 75-24-02-420-003-G00

WARNING BE CAREFUL WHEN YOU LIFT THE LPTACC UPPER MANIFOLD ASSEMBLY. THE LPTACC UPPER MANIFOLD ASSEMBLY IS HEAVY. INJURIES TO PERSONNEL CAN OCCUR.

CAUTION MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC UPPER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(2) Attach the LPTACC upper manifold assembly [1] to the Low Pressure Turbine (LPT) case at 12:00 o'clock position as follows:

(a) Put and hold the LPTACC upper manifold assembly [1] in position on the LPT case.

(b) Install the two bolts [21], two flat washers [22], and two nuts [23] that attach the LPTACC upper manifold assembly [1] to the bracket.

(c) Put the two helical compression springs [28], two spring cups [27], and two flat washers [29] in position on the LPTACC upper manifold assembly [1] and attach them to the LPT case with two bolts [24], two flat washers [25], and two nuts [26].

SUBTASK 75-24-02-420-004-G00

(3) Install the four bolts [12], four flat washers [13], two retaining plates [10], and four nuts [11] that attach the LPTACC upper manifold assembly [1] to the bracket at 1:00 and 2:00 o'clock positions.

SUBTASK 75-24-02-420-005-G00

(4) Install the two bolts [12], two flat washers [13], retaining plate [10], and two nuts [11] that attach the LPTACC upper manifold assembly [1] to the bracket at 3:00 o'clock position.

SUBTASK 75-24-02-420-006-G00

(5) Attach the bracket assembly [20] to the LPTACC upper manifold assembly [1].

(a) Put and hold the bracket assembly [20] on the LPTACC upper manifold assembly [1].

(b) Install the bolt [19] that attach the bracket assembly [20] to the LPTACC upper manifold assembly [1].

1) Tighten the bolt [19] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

SUBTASK 75-24-02-420-007-G00

(6) Install the saddle clamp [17] on the air duct.

(a) Install the two bolts [18] that attach the saddle clamp [17] to the bracket assembly [20].

1) Tighten the two bolts [18] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

SUBTASK 75-24-02-420-008-G00

(7) Install the two bolts [14], two flat washers [15], and two nuts [16] that attach the LPTACC upper manifold assembly [1] and the bracket assembly [20] to the bracket at 10:00 o'clock position.

SUBTASK 75-24-02-420-009-G00

(8) Install the four bolts [12], four flat washers [13], two retaining plates [10], and four nuts [11] that attach the LPTACC upper manifold assembly [1] to the bracket at 9:00 and 11:00 o'clock positions.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-24-02-420-010-G00

(9) Install the 12 bolts [6], 12 flat washers [7], 12 flat washers [9], and 12 nuts [8] that attach the LPTACC upper manifold assembly [1] to the LPT case at 1:00, 2:00, 3:00, 9:00, 10:00, and 11:00 o'clock positions.

SUBTASK 75-24-02-420-011-G00

(10) Adjust the LPTACC upper manifold assembly [1]:

(a) Measure and record the clearances M165A and M165B between the air cooling tube ends of the LPTACC lower manifold assembly [2] and the LPTACC upper manifold assembly [1].

1) Make sure that the clearances M165A and M165B between the air cooling tube ends of the two manifolds assemblies are not less than 0.04 in. (1.02 mm).

2) If the clearances are less than 0.04 in. (1.02 mm), adjust the positions of the LPTACC upper manifold assembly [1] until the clearances are in tolerance.

(b) Tighten the two nuts [26] to 34 in-lb (3.8 N·m) - 37 in-lb (4.2 N·m).

(c) Tighten the two nuts [23], 12 bolts [6], 10 nuts [11], and two nuts [16] to 59 in-lb (6.7 N·m) - 65 in-lb (7.3 N·m).

(d) Measure and record the clearances M164B, M164G, M164H, and M164J under all the 6 brackets.

1) Make sure that the clearances M164B, M164G, M164H, and M164J between the air cooling tubes of the manifold assemblies and the LPT case under all the 6 brackets are 0.111 in. (2.8 mm) - 0.236 in. (6 mm).

2) If the clearances M164B, M164G, M164H, and M164J are out of tolerances, stop the procedure and find the cause.

SUBTASK 75-24-02-420-012-G00

(11) Install the swivel fitting [4] on the LPTACC upper manifold assembly [1].

(a) Install the two bolts [5] that attach the swivel fitting [4] to the air duct.

1) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure face of the two bolts [5].

2) Tighten the two bolts [5] to 42 in-lb (4.7 N·m) - 46 in-lb (5.2 N·m).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-24-02-410-002-G00

(1) Install the transient bleed valve duct (TASK 75-33-01-400-801-G00).

SUBTASK 75-24-02-410-003-G00

(2) Install the right core fire detector assembly (TASK 26-11-01-400-802).

SUBTASK 75-24-02-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-24-02-860-003-G00

(4) For the Engine 1, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-02-860-004-G00

(5) For the Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-02-410-007-G00

(6) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Assembly Installation Test

SUBTASK 75-24-02-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-24-02-000-802-G00

4. Low Pressure Turbine Active Clearance Control (LPTACC) Lower Manifold Assembly Removal

(Figure 403)

A. General

(1) This task gives the instructions on how to remove the Low Pressure Turbine Active Clearance Control (LPTACC) lower manifold assembly.

B. References

Reference	Title
26-11-01-000-802	Engine Fire Detector Assembly Removal (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
75-33-01-000-801-G00	Transient Bleed Valve Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-24-02-010-009-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-24-02-860-007-G00

(2) For the Engine 1, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-02-860-008-G00

(3) For the Engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-02-010-012-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-24-02-010-002-G00

(5) Remove the right core fire detector assembly (TASK 26-11-01-000-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0175 (Continued)

SUBTASK 75-24-02-010-013-G00

(6) Remove the transient bleed valve duct (TASK 75-33-01-000-801-G00).

PTF ALL

G. LPTACC Lower Manifold Assembly Removal

SUBTASK 75-24-02-020-002-G00

(1) Remove the air tube [30] from the engine.

(a) Cut and remove the safety cable or safety wire installed between the B-nut of the air tube [30] and nipple of the exhaust air tube.

(b) Disconnect the B-nut between the air tube [30] and exhaust air tube.

(c) Disconnect the B-nut between the air tube [30] and air valve.

(d) Remove the bolt [33] that attached the hinged-loop clamp [34] to the bracket.

(e) Remove the bolt [35] that attach the hinged-loop clamp [36] to the bracket.

(f) Remove the three bolts [37] that attach the three hinged-loop clamps [38] to the three brackets.

SUBTASK 75-24-02-020-010-G00

(2) Remove the swivel fitting [40] from the LPTACC lower manifold assembly [2].

(a) Remove the two bolts [41] that attach the swivel fitting [40] to the air duct.

SUBTASK 75-24-02-020-011-G00

(3) Remove the 12 nuts [44], 12 flat washers [71], 12 bolts [42], and 12 flat washers [43] that attach the LPTACC lower manifold assembly [2] to the Low Pressure Turbine (LPT) case at 3:00, 4:00, 5:00, 7:00, 8:00, and 9:00 o'clock positions.

SUBTASK 75-24-02-020-012-G00

(4) Remove the four nuts [45], two retaining plates [46], four flat washers [48], and four bolts [47] that attach the LPTACC lower manifold assembly [2] to the bracket at the 7:00 and 09:00 o'clock positions.

SUBTASK 75-24-02-020-013-G00

(5) Remove the saddle clamp [52] from the air duct.

(a) Remove the two bolts [53] that attach the saddle clamp [52] to the bracket assembly [55].

SUBTASK 75-24-02-020-014-G00

(6) Remove the bracket assembly [55] from the LPTACC lower manifold assembly [2].

(a) While you hold the bracket assembly [55], remove the two nuts [51], two bolts [49], two flat washers [50], and bolt [54] that attach the LPTACC lower manifold assembly [2] to the bracket at the 08:00 o'clock position.

SUBTASK 75-24-02-020-015-G00

(7) Remove the two nuts [75], two flat washers [74], bracket assembly [76], and two bolts [73] that attach the LPTACC lower manifold assembly [2] to the bracket at the 4:00 o'clock position.

SUBTASK 75-24-02-020-016-G00

(8) Remove the four nuts [45], two retaining plates [46], four flat washers [48], and four bolts [47] that attach the LPTACC lower manifold assembly [2] to the bracket at the 3:00 and 5:00 o'clock positions.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-24-02-020-017-G00

(9) While you hold the LPTACC lower manifold assembly [2], disconnect the LPTACC lower manifold assembly [2] from the LPT case at 06:00 o'clock position.

(a) Remove the two nuts [58], two flat washers [57], and two bolts [56] that attach the LPTACC lower manifold assembly [2] to the bracket.

(b) Remove the two nuts [61], two flat washers [60], and two bolts [59] that attach the LPTACC lower manifold assembly [2] to the LPT case.

(c) Remove the two spring cups [62], two helical compression springs [63], and two flat washers [64] from the LPTACC lower manifold assembly [2].

SUBTASK 75-24-02-020-019-G00



WARNING

BE CAREFUL WHEN YOU LIFT THE LPTACC LOWER MANIFOLD ASSEMBLY. THE LPTACC LOWER MANIFOLD ASSEMBLY IS HEAVY. INJURIES TO PERSONNEL CAN OCCUR.



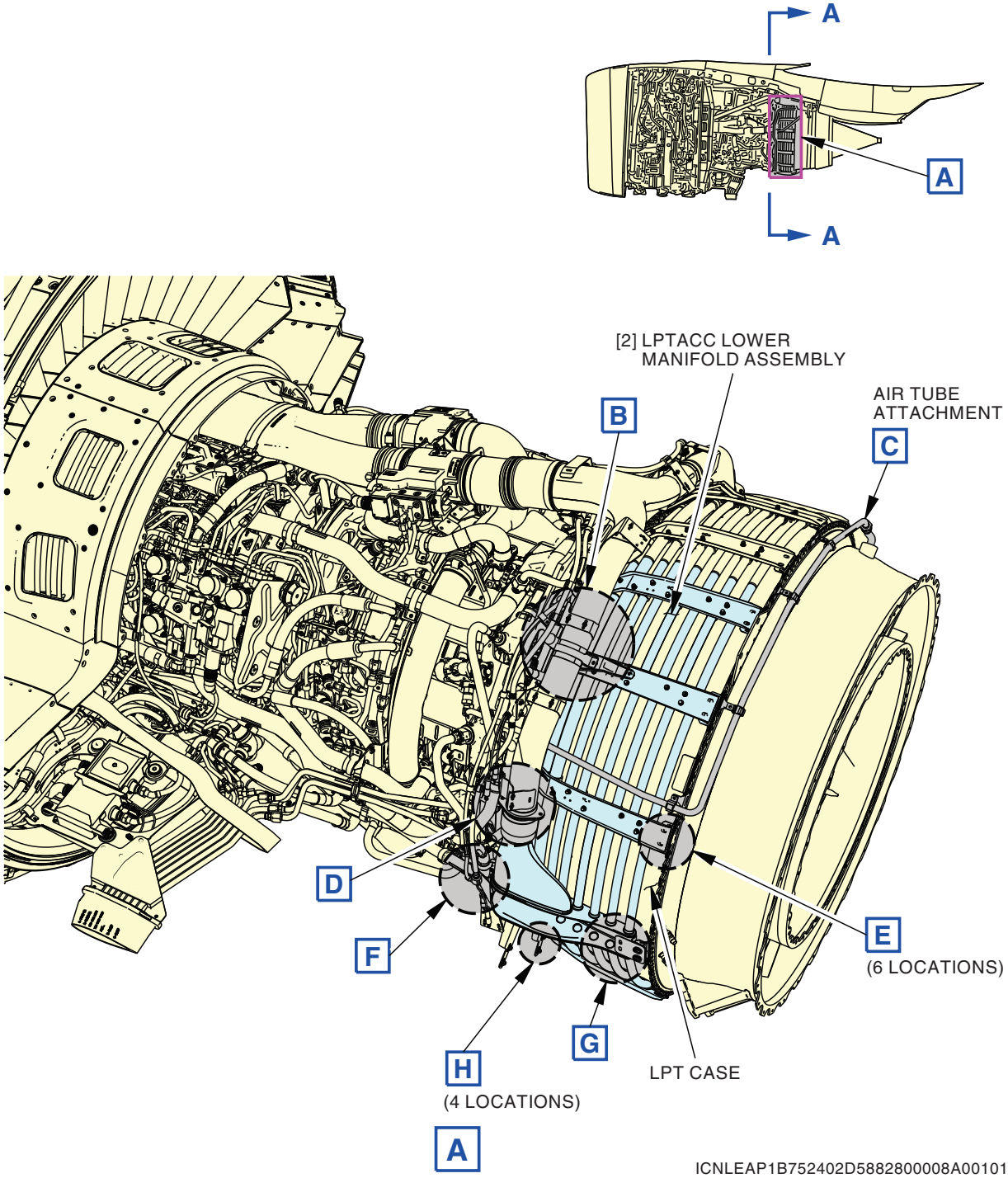
CAUTION

MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC LOWER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(10) Carefully remove the LPTACC lower manifold assembly [2] from the LPT case.

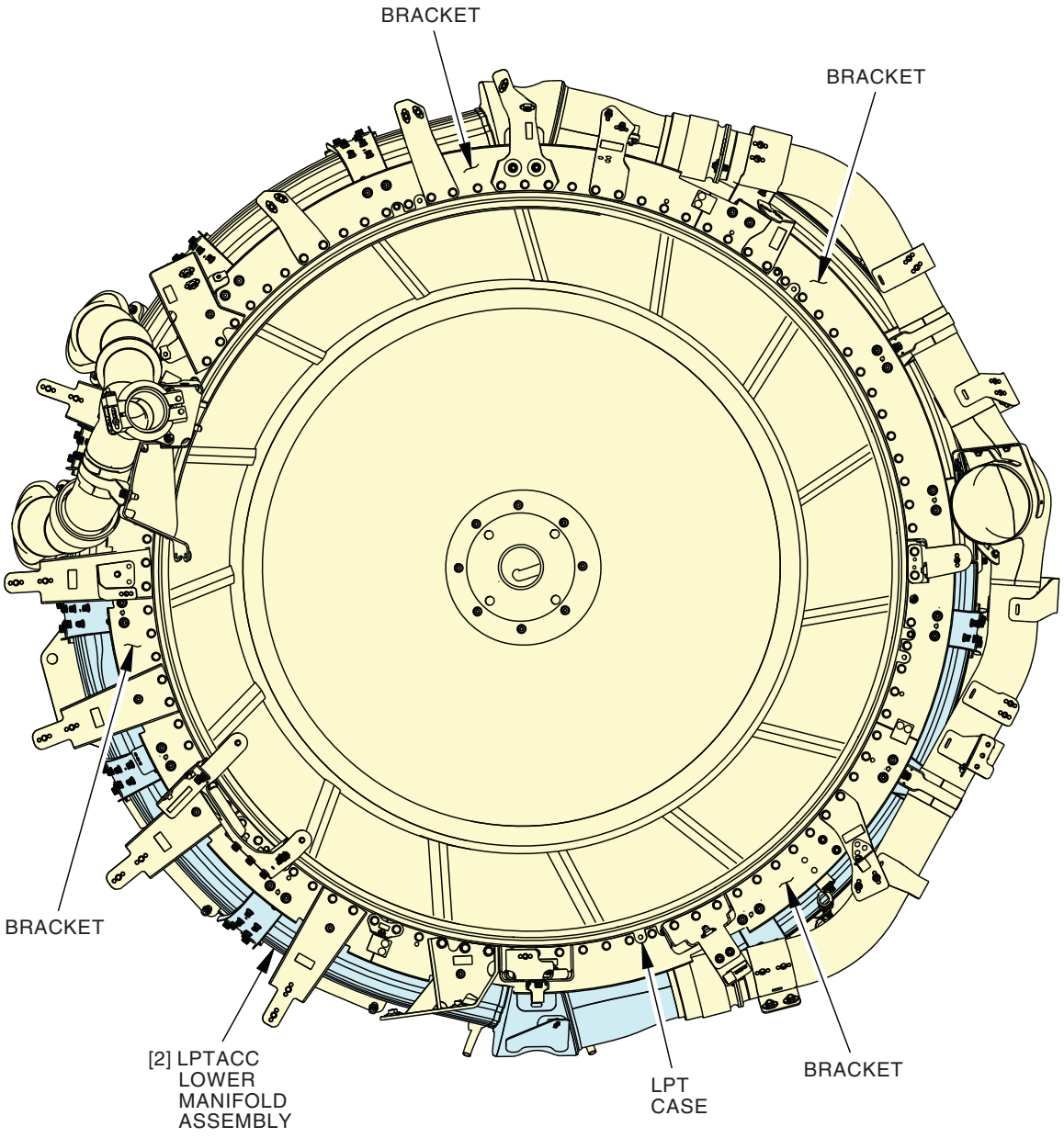
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPTACC Lower Manifold Assembly Installation
Figure 403/75-24-02-990-802-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE AFT DIRECTION)

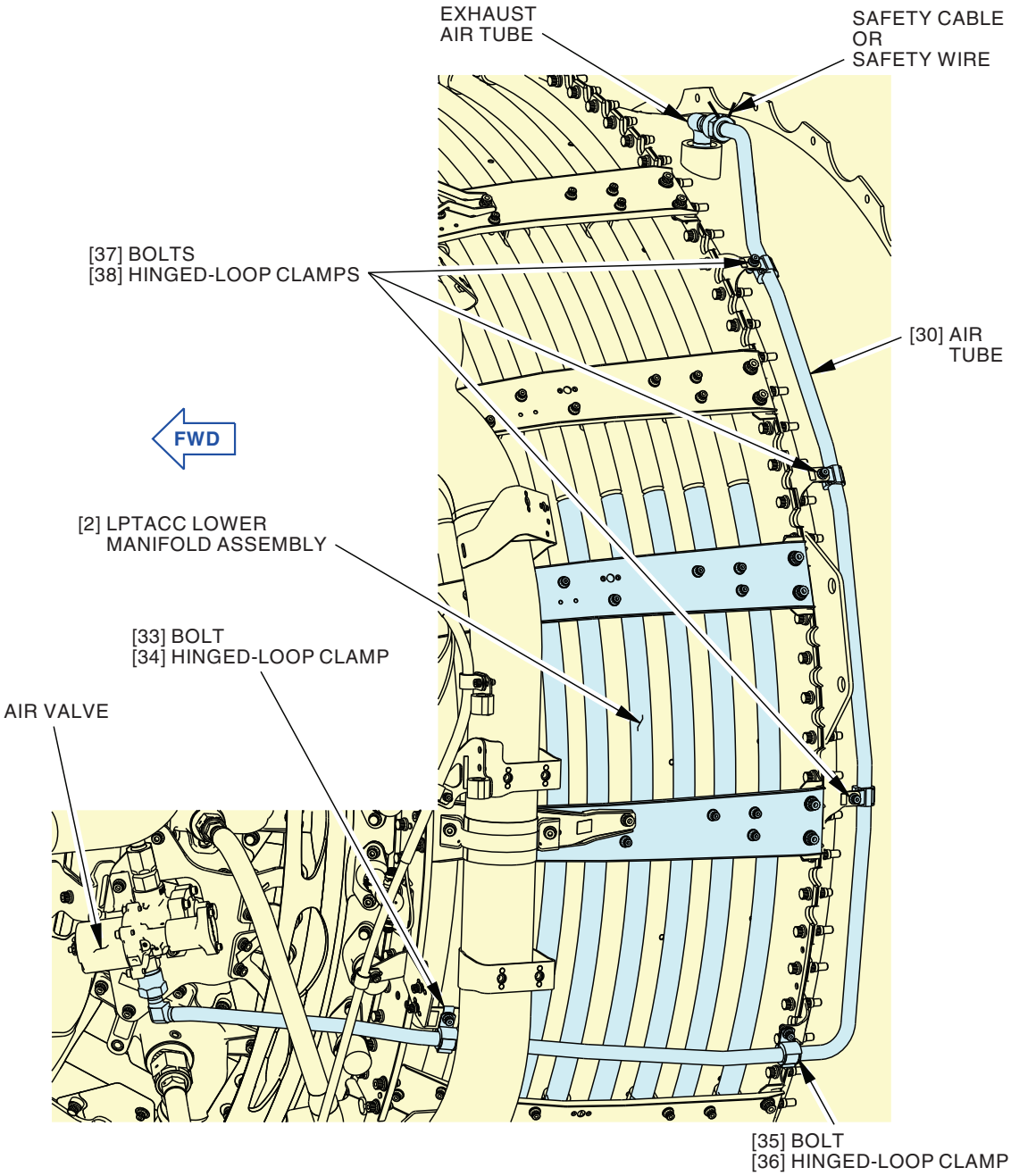
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LPTACC Lower Manifold Assembly Installation
Figure 403/75-24-02-990-802-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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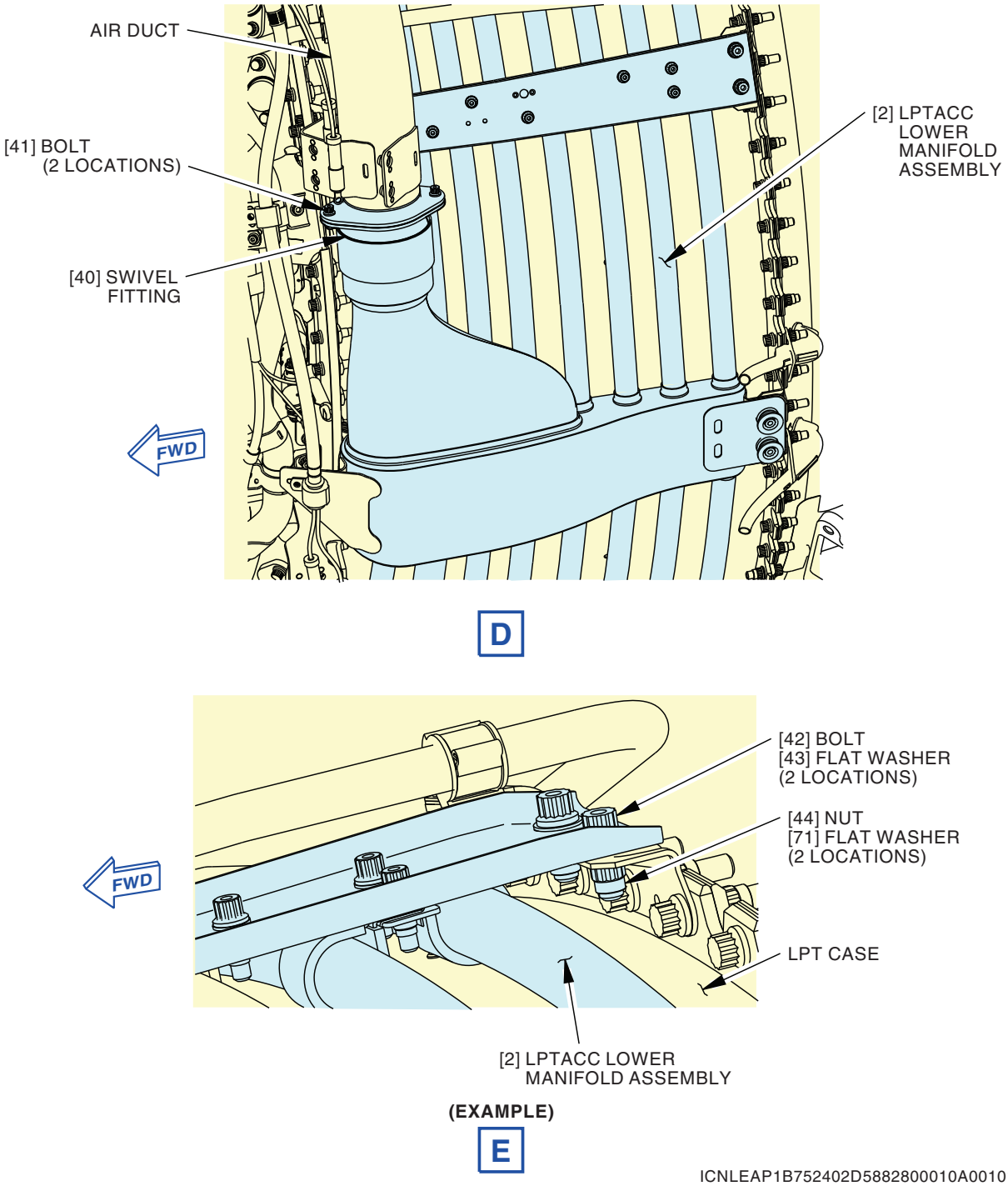
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LPTACC Lower Manifold Assembly Installation
Figure 403/75-24-02-990-802-G00 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

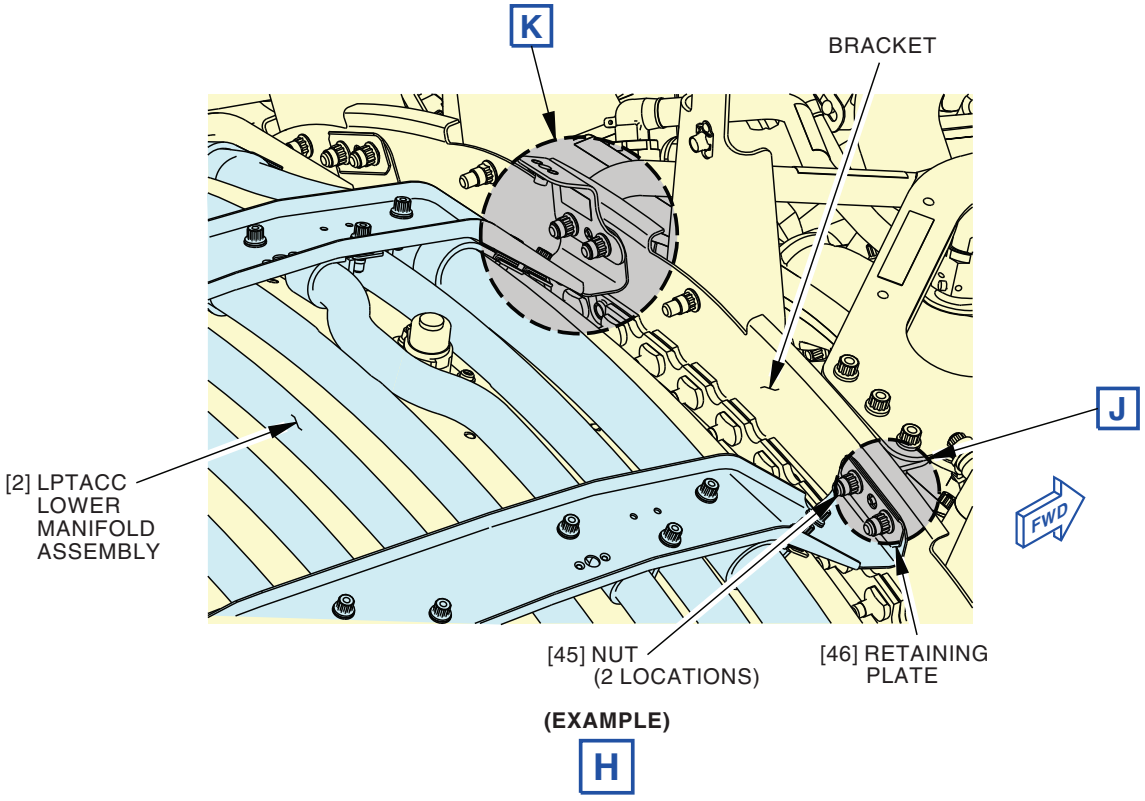


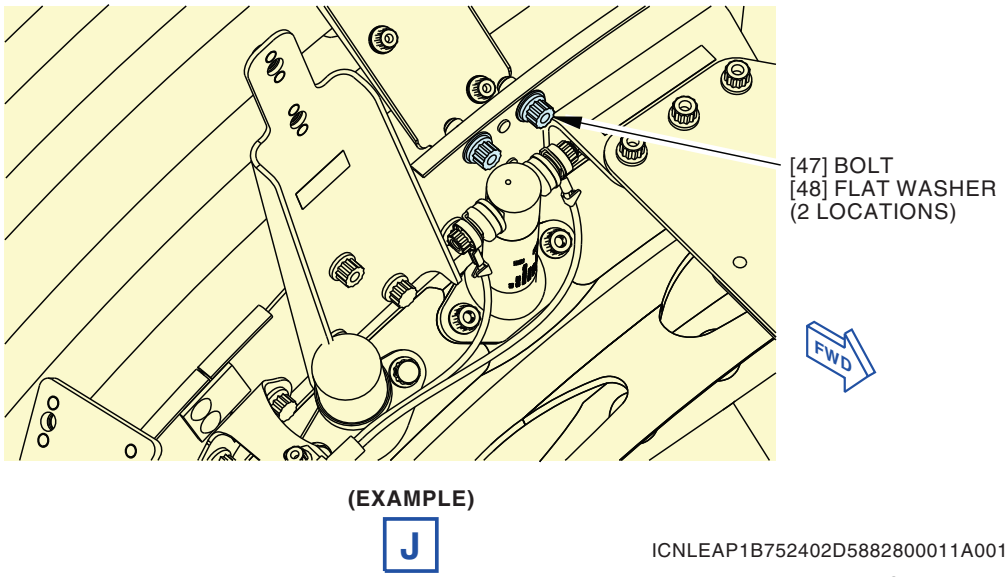
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LPTACC Lower Manifold Assembly Installation
Figure 403/75-24-02-990-802-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



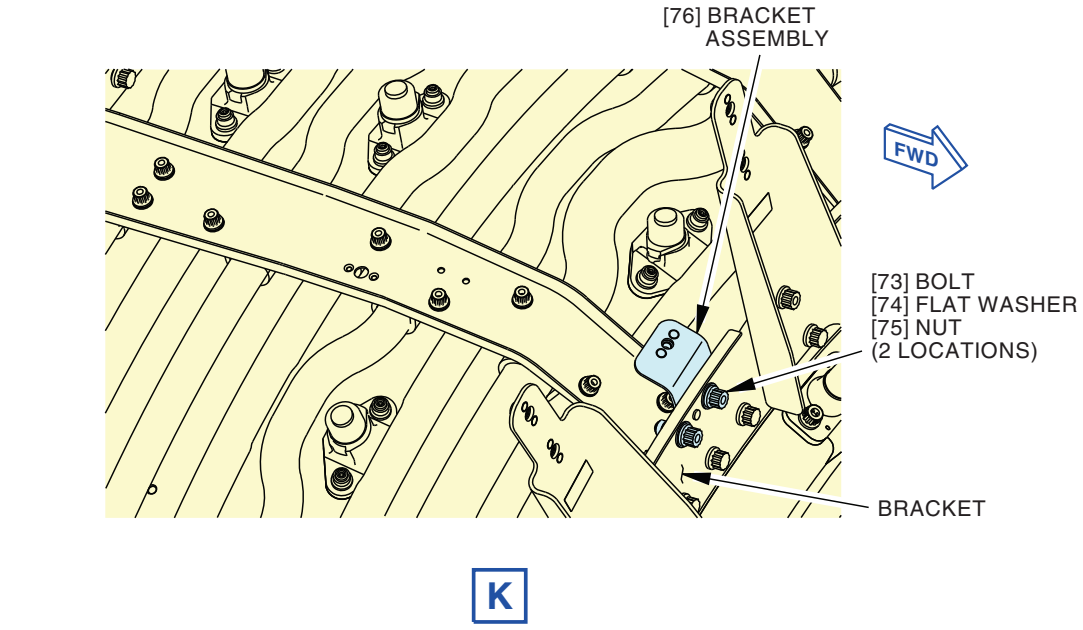


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LPTACC Lower Manifold Assembly Installation
Figure 403/75-24-02-990-802-G00 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

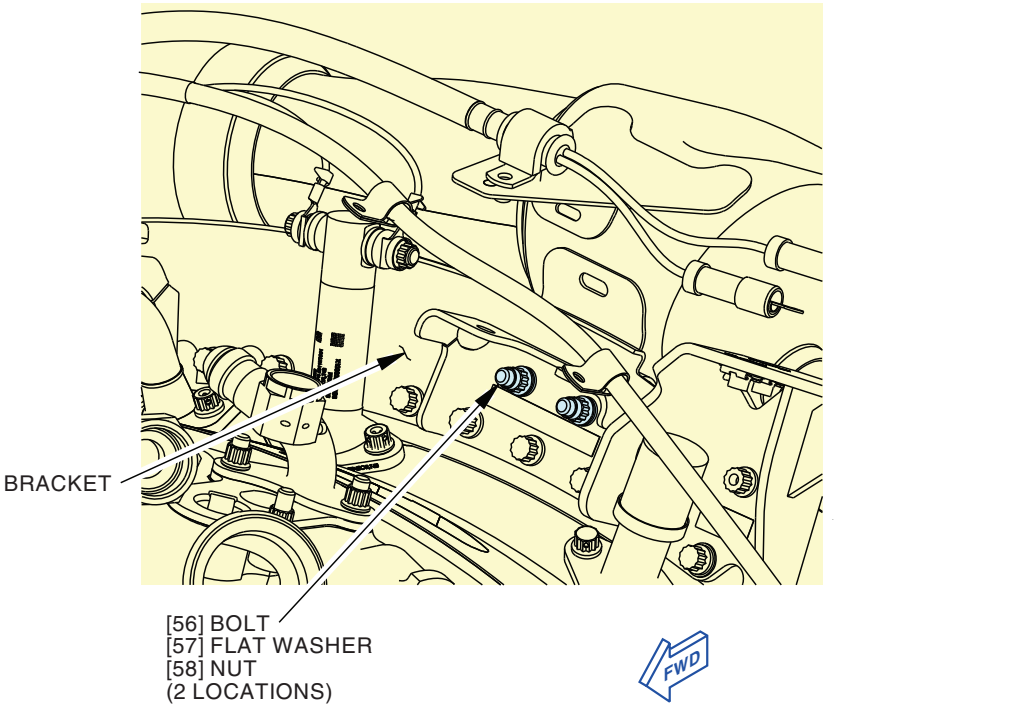


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LPTACC Lower Manifold Assembly Installation
Figure 403/75-24-02-990-802-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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LPTACC Lower Manifold Assembly Installation
Figure 403/75-24-02-990-802-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-24-02-400-802-G00

5. Low Pressure Turbine Active Clearance Control (LPTACC) Lower Manifold Assembly Installation

(Figure 403, Figure 402)

A. General

(1) This task gives the instructions on how to install the Low Pressure Turbine Active Clearance Control (LPTACC) lower manifold assembly.

B. References

Reference	Title
26-11-01-400-802	Engine Fire Detector Assembly Installation (P/B 401)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
75-33-01-400-801-G00	Transient Bleed Valve Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. LPTACC Lower Manifold Assembly Installation

SUBTASK 75-24-02-420-002-G00

(1) Apply graphite mineral vaseline, D00601 [CP2101], on the thread of the bolts and pressure face of the nuts that follow:

(a) the 12 bolts [42]

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) the two nuts [61]

(c) the two nuts [58]

(d) the eight nuts [45]

(e) the bolt [54]

(f) the two bolts [53]

(g) the two nuts [75]

(h) the two nuts [51].

SUBTASK 75-24-02-420-013-G00

WARNING BE CAREFUL WHEN YOU LIFT THE LPTACC LOWER MANIFOLD ASSEMBLY. THE LPTACC LOWER MANIFOLD ASSEMBLY IS HEAVY. INJURIES TO PERSONNEL CAN OCCUR.

CAUTION MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC LOWER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(2) Attach the LPTACC lower manifold assembly [2] to the Low Pressure Turbine (LPT) case at 06:00 o'clock position as follows:

(a) Put and hold the LPTACC lower manifold assembly [2] in position on the LPT case.

(b) Install the two bolts [56], two flat washers [57], and two nuts [58] that attach the LPTACC lower manifold assembly [2] to the bracket.

(c) Put the two helical compression springs [63], two spring cups [62], and two flat washers [64] in position on the LPTACC lower manifold assembly [2] and attach them to the LPT case with two bolts [59], two flat washers [60], and two nuts [61].

SUBTASK 75-24-02-420-014-G00

(3) Install the four bolts [47], four flat washers [48], two retaining plates [46], and four nuts [45] that attach the LPTACC lower manifold assembly [2] to the bracket at 3:00 and 5:00 o'clock positions.

SUBTASK 75-24-02-420-015-G00

(4) Install the two bolts [73], two flat washers [74], bracket assembly [76], and two nuts [75] that attach the LPTACC lower manifold assembly [2] to the bracket at 4:00 o'clock position.

SUBTASK 75-24-02-420-016-G00

(5) Attach the bracket assembly [55] to the LPTACC lower manifold assembly [2] and LPT case bracket.

(a) Put and hold the bracket assembly [55] on the LPTACC lower manifold assembly [2].

(b) Install the bolt [54] that attach the bracket assembly [55] to the LPTACC lower manifold assembly [2].

1) Tighten the bolt [54] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

(c) Install the two bolts [49], two flat washers [50], and two nuts [51] that attach the LPTACC lower manifold assembly [2] and bracket assembly [55] to the bracket at 8:00 o'clock position.

SUBTASK 75-24-02-420-017-G00

(6) Install the saddle clamp [52] on the air duct.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the two bolts [53] that attach the saddle clamp [52] to the bracket assembly [55].

1) Tighten the two bolts [53] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

SUBTASK 75-24-02-420-018-G00

(7) Install the four bolts [47], four flat washers [48], two retaining plates [46], and four nuts [45] that attach the LPTACC lower manifold assembly [2] to the bracket at the 7:00 and 09:00 o'clock positions.

SUBTASK 75-24-02-420-019-G00

(8) Install the 12 bolts [42], 12 flat washers [71], 12 flat washers [43], and 12 nuts [44] that attach the LPTACC lower manifold assembly [2] to the LPT case at 3:00, 4:00, 5:00, 7:00, 8:00, and 9:00 o'clock positions.

SUBTASK 75-24-02-820-001-G00

(9) Adjust the LPTACC lower manifold assembly [2]:

(a) Measure and record the clearances M165A and M165B between the air cooling tube ends of the LPTACC lower manifold assembly [2] and LPTACC upper manifold assembly [1].

1) Make sure that the clearances M165A and M165B between the air cooling tube ends of the two manifolds assemblies are not less than 0.04 in. (1 mm).

2) If the clearances are less than 0.04 in. (1 mm), adjust the positions of the LPTACC lower manifold assembly [2] until the clearances are in tolerance.

(b) Tighten the two nuts [58] to 34 in-lb (3.8 N·m) - 37 in-lb (4.2 N·m).

(c) Tighten the two nuts [61], 12 bolts [42], eight nuts [45], two nuts [75], and two nuts [51] to 59 in-lb (6.7 N·m) - 65 in-lb (7.3 N·m).

(d) Measure and record the clearances M164B, M164G, M164H, and M164J under all the 6 brackets.

1) Make sure that the clearances M164B, M164G, M164H, and M164J between the air cooling tubes of the manifold assemblies and LPT case under all the 6 brackets are 0.111 in. (2.8 mm) - 0.236 in. (6 mm).

2) If the clearances M164B, M164G, M164H, and M164J are out of tolerances, stop the procedure and find the cause.

SUBTASK 75-24-02-420-020-G00

(10) Install the swivel fitting [40] on the LPTACC lower manifold assembly [2].

(a) Install the two bolts [41] that attach the swivel fitting [40] to the air duct.

1) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure face of the two bolts [41].

2) Tighten the two bolts [41] to 42 in-lb (4.7 N·m) - 46 in-lb (5.2 N·m).

SUBTASK 75-24-02-420-021-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION

DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(11) Install the air tube [30] on the engine:

(a) Apply engine oil, D00599 [CP2442], on the threads of the nipples of the air valve and exhaust air tube and on the mating surface between the ends of the air tube [30] and its coupling nuts.

(b) Put the air tube [30] on the engine.

(c) Install the three bolts [37] that attach the three hinged-loop clamps [38] to the three brackets.

1) Apply graphite mineral vaseline, D00601 [CP2101], on the threads of the three bolts [37].

(d) Install the bolt [35] that attach the hinged-loop clamp [36] to the bracket.

1) Apply graphite mineral vaseline, D00601 [CP2101], on the thread of the bolt [35].

(e) Install the bolt [33] that attached the hinged-loop clamp [34] to the bracket.

1) Apply graphite mineral vaseline, D00601 [CP2101], on the thread of the bolt [33].

(f) Connect the B-nut between the air tube [30] and air valve.

1) Tighten the B-nut to 450 in-lb (51 N·m) - 550 in-lb (62 N·m).

(g) Connect the B-nut between the air tube [30] and exhaust air tube.

1) Tighten the B-nut to 450 in-lb (51 N·m) - 550 in-lb (62 N·m).

(h) Tighten the three bolts [37], bolt [35], and bolt [33] to 42 in-lb (4.7 N·m) - 46 in-lb (5.2 N·m).

(i) Safety the B-nuts of the air tube [30] to the nipple of the exhaust air tube with safety safety cable, G50065 [CP8006], or safety wire, G02345 [CP8001].

H. Put the Airplane Back to Its Usual Condition

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 75-24-02-410-001-G00

(1) Install the transient bleed valve duct (TASK 75-33-01-400-801-G00).

SUBTASK 75-24-02-410-008-G00

(2) Install the right core fire detector assembly (TASK 26-11-01-400-802).

PTF ALL

SUBTASK 75-24-02-410-009-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-24-02-860-005-G00

(4) For the Engine 1, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-02-860-006-G00

(5) For the Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-02-410-012-G00

(6) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Low Pressure Turbine Active Clearance Control (LPTACC) Lower Manifold Assembly Installation Test

SUBTASK 75-24-02-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOW PRESSURE TURBINE ACTIVE CLEARANCE CONTROL (LPTACC) MANIFOLD BRACKETS - REMOVAL/INSTALLATION

1. General

A. This procedure has the following tasks:

(1) A removal of the Low Pressure Active Clearance Control (LPTACC) upper manifold brackets

(2) An installation of the LPTACC upper manifold brackets

(3) A removal of the LPTACC lower manifold brackets

(4) An installation of the LPTACC lower manifold brackets.

TASK 75-24-03-000-801-G00

2. Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Brackets - Removal

(Figure 401)

A. General

(1) This task includes steps to remove the Low Pressure Turbine Active Clearance Control (LPTACC) upper manifold brackets.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-24-03-010-001-G00

(1) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

SUBTASK 75-24-03-010-019-G00

(2) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

(a) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 75-24-03-860-001-G00

(3) For the Engine 1, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-03-860-002-G00

(4) For the Engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-03-010-002-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. LPTACC Upper Manifold Brackets Removal

CAUTION

REPLACE THE ONE LPTACC MANIFOLD BRACKETS AT A TIME. IF YOU DO NOT OBEY INSTRUCTIONS, DAMAGE TO THE LPTACC MANIFOLD CAN OCCUR.

SUBTASK 75-24-03-000-001-G00

(1) Remove the LPTACC upper manifold bracket [1] or LPTACC upper manifold bracket [2] from the LPTACC upper manifold assembly as follows:

(a) Remove the six nuts [6], six washers [4], and six bolts [3] that attach the six clamps [5] to the LPTACC upper manifold assembly.

1) Keep the six clamps [5] installed in their positions on the LPTACC upper manifold assembly.

(b) Remove the nut [15], washer [13], and bolt [12] that attach the clamp [14] to the LPTACC upper manifold assembly.

1) Keep the clamp [14] installed in its position on the LPTACC upper manifold assembly.

(c) Remove the nut [11], bolt [7], washer [8], and spacer [9] that attach the clamp [10] to the LPTACC upper manifold assembly.

1) Keep the clamp [10] installed in its position on LPTACC upper manifold assembly.

SUBTASK 75-24-03-010-005-G00

(2) Remove the two nuts [19], two flat washers [18], and two bolts [16] that attach the aft end of the LPTACC upper manifold bracket [1] or LPTACC upper manifold bracket [2] to the Low Pressure Turbine (LPT) case bracket.

SUBTASK 75-24-03-010-006-G00

(3) Remove the two nut [21], retaining plate [20], two bolts [22], and two flat washer [23] that attach the forward end of the LPTACC upper manifold bracket [1] to the bracket.

SUBTASK 75-24-03-010-007-G00

CAUTION

MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC UPPER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(4) Remove the LPTACC upper manifold bracket [1] from the engine.

SUBTASK 75-24-03-000-002-G00

(5) Remove the two bolts [26] that attach the saddle clamp [27] to the bracket assembly [25] at the 10:00 o'clock position.

SUBTASK 75-24-03-000-006-G00

(6) Remove the saddle clamp [27] from the air duct and bracket assembly [25] at the 10:00 o'clock position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-24-03-000-007-G00

(7) Remove the two nuts [30], two bolts [28], and two flat washers [29] that attach the forward end of the LPTACC upper manifold bracket [2] and bracket assembly [25] to the bracket at the 10:00 o'clock position.

SUBTASK 75-24-03-010-008-G00

(8) While you hold the bracket assembly [25], remove the bolt [24] that attaches the bracket assembly [25] to the LPTACC upper manifold bracket [2].

SUBTASK 75-24-03-000-008-G00

(9) Remove the bracket assembly [25] from the LPTACC upper manifold bracket [2].

SUBTASK 75-24-03-010-009-G00

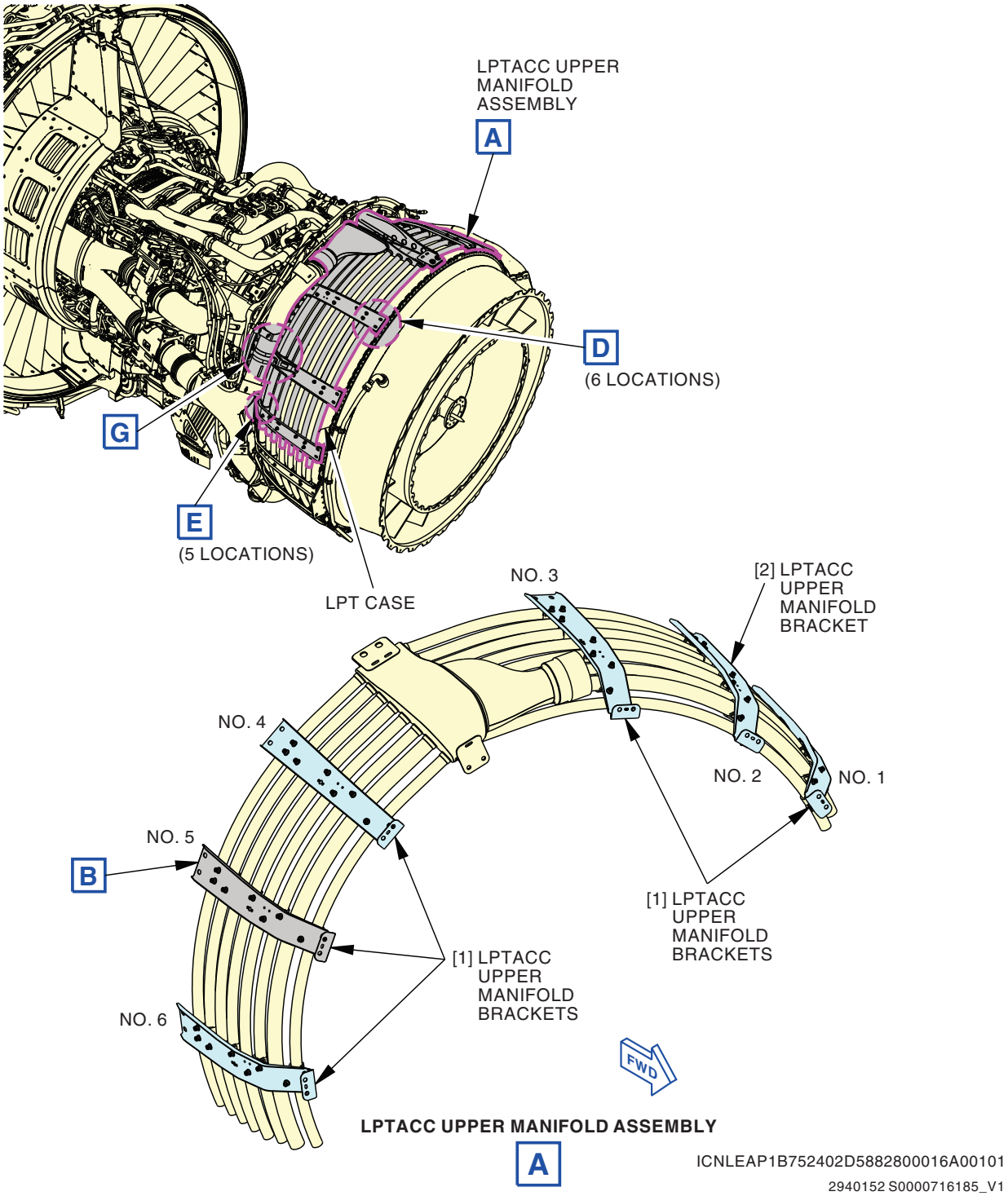
CAUTION

MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC UPPER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(10) Remove the LPTACC upper manifold bracket [2] from the engine.

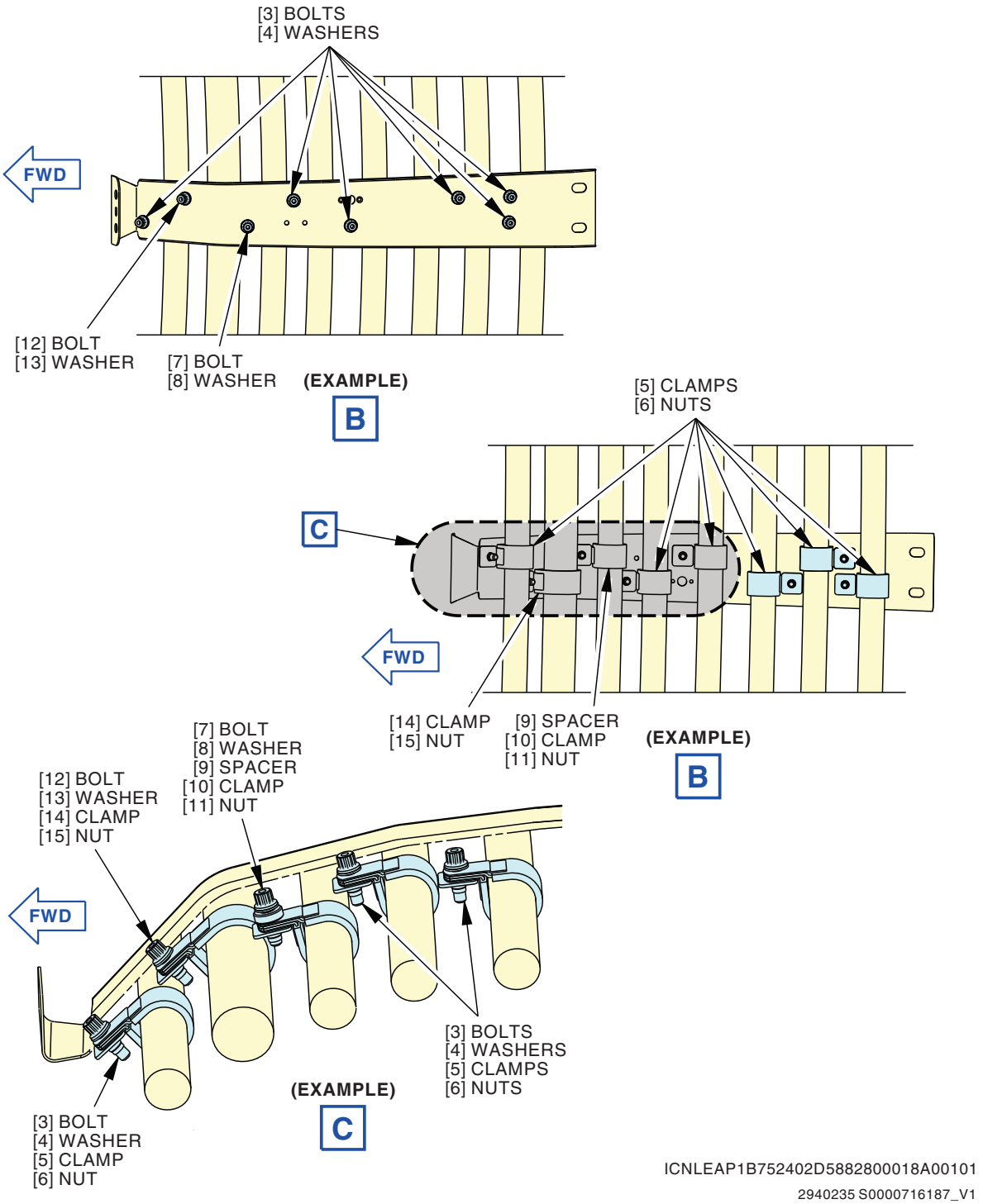
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPTACC Upper Manifold Brackets Installation
Figure 401/75-24-03-990-801-G00 (Sheet 1 of 4)

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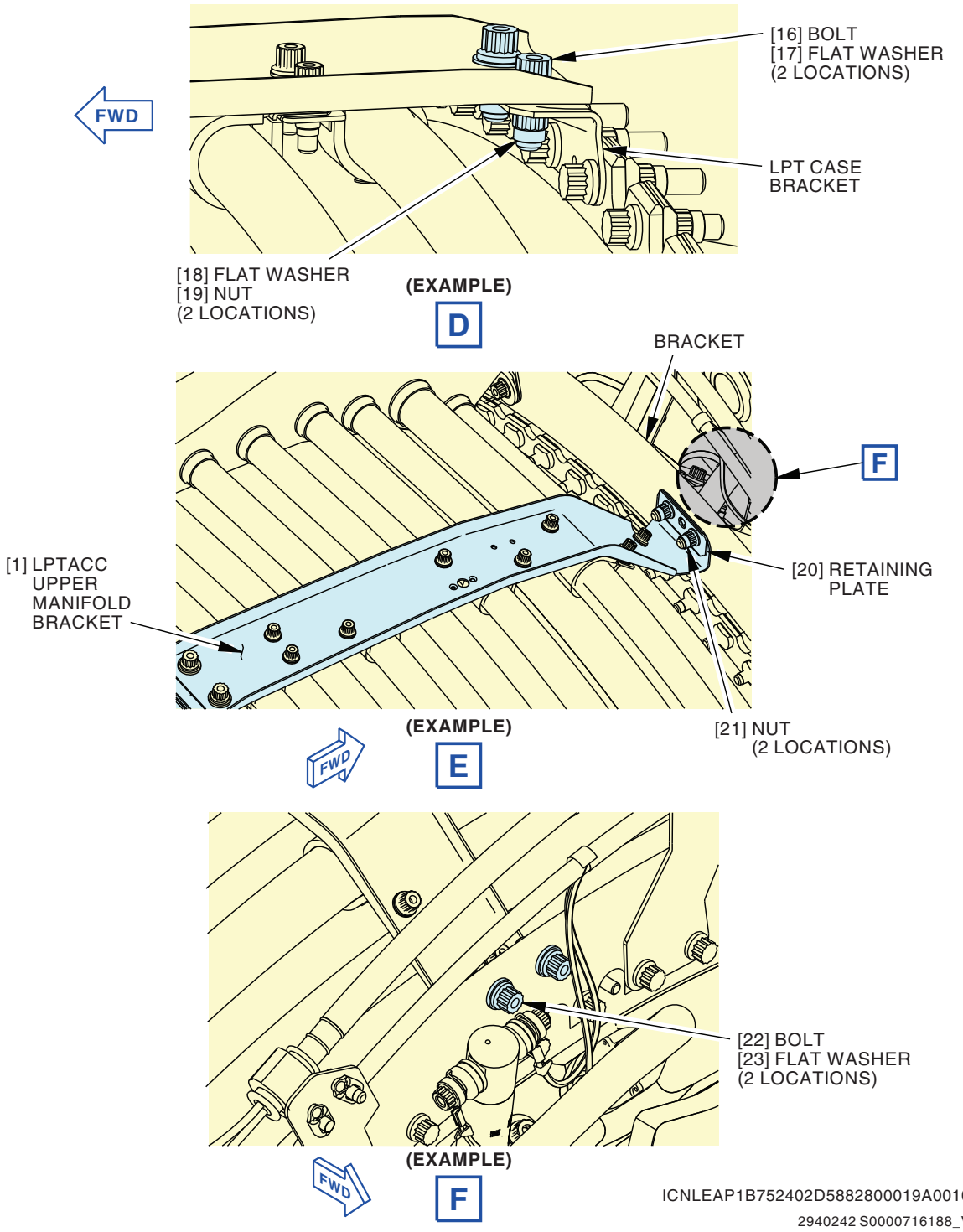


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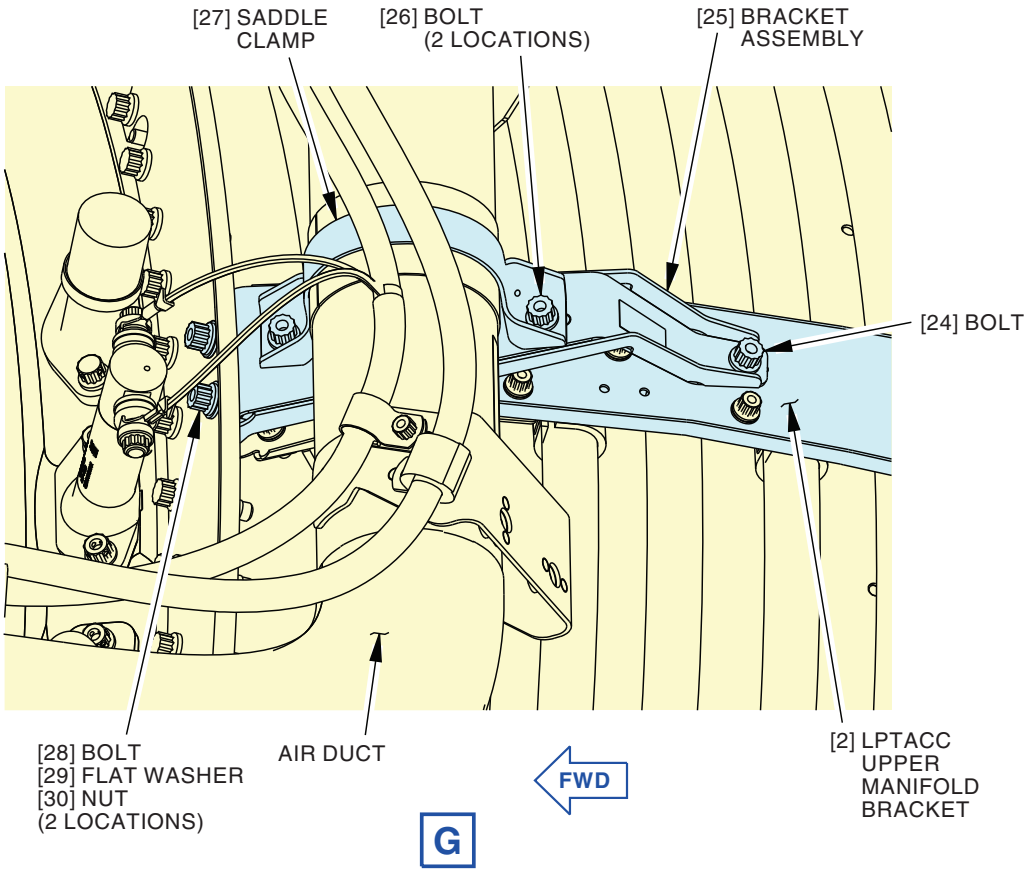
LPTACC Upper Manifold Brackets Installation
Figure 401/75-24-03-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPTACC Upper Manifold Brackets Installation
Figure 401/75-24-03-990-801-G00 (Sheet 3 of 4)

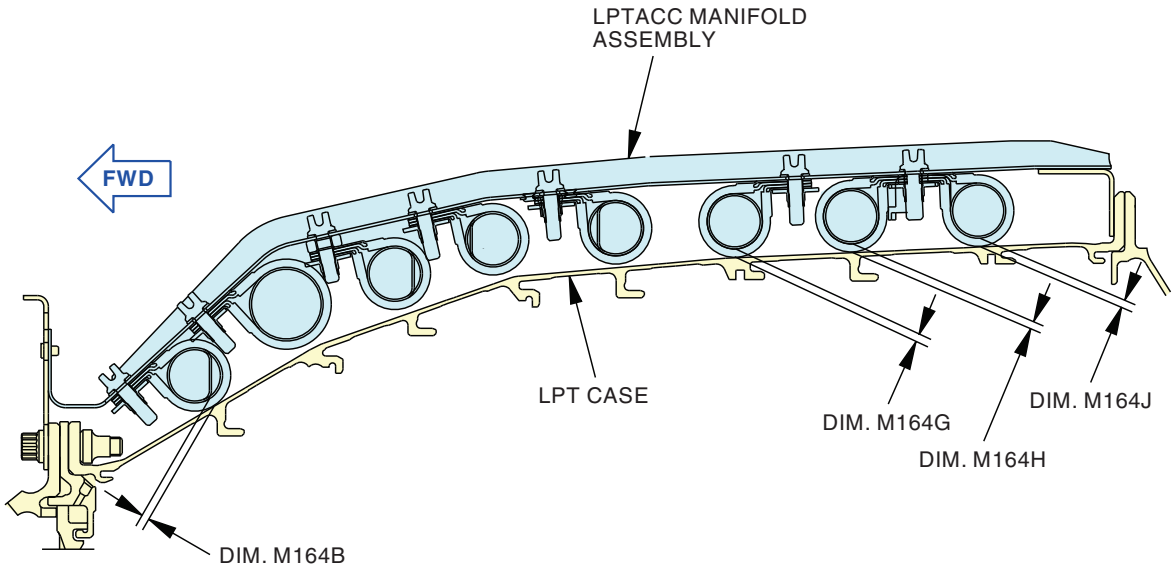
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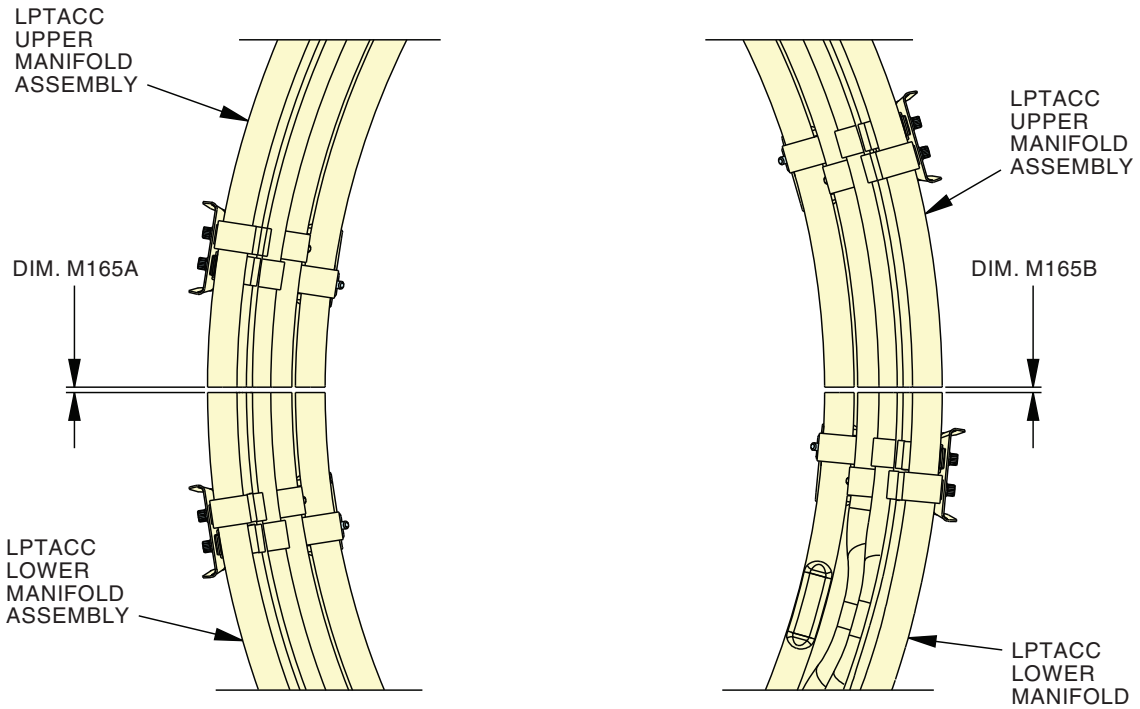
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LPTACC Upper Manifold Brackets Installation
Figure 401/75-24-03-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CLEARANCES LPTACC TUBES / LPT CASE



CLEARANCES UPPER / LOWER LPTACC TUBES
(VIEW IN THE FORWARD DIRECTION)

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LPTACC Upper Manifold Assembly Installation
Figure 402/75-24-03-990-806-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-24-03-400-801-G00

3. Low Pressure Turbine Active Clearance Control (LPTACC) Upper Manifold Brackets - Installation

(Figure 401, Figure 402)

A. General

(1) This task includes steps to install the Low Pressure Turbine Active Clearance Control (LPTACC) upper manifold brackets.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. LPTACC Upper Manifold Brackets Installation

CAUTION

REPLACE THE ONE LPTACC MANIFOLD BRACKETS AT A TIME. IF YOU DO NOT OBEY INSTRUCTIONS, DAMAGE TO THE LPTACC MANIFOLD CAN OCCUR.

SUBTASK 75-24-03-420-001-G00

(1) Apply graphite mineral vaseline, D00601 [CP2101], on the thread of the bolts and pressure face of the nuts that follow:

(a) the bolts [16]

(b) the nuts [21]

(c) the bolts [3]

(d) the bolt [12]

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) the bolt [7].

SUBTASK 75-24-03-420-002-G00

CAUTION

MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC UPPER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(2) Install the LPTACC upper manifold bracket [1] in its position on the bracket and Low Pressure Turbine (LPT) case bracket.

(a) Install the two bolts [22], two flat washers [23], retaining plate [20], and two nuts [21] that attach the LPTACC upper manifold bracket [1] to the bracket.

(b) Install the two bolts [16], two flat washers [17], two flat washers [18], and two nuts [19] that attach the LPTACC upper manifold bracket [1] to the LPT case bracket.

1) Tighten the two bolts [16] and two nuts [21] to 58.9 in-lb (6.7 N·m) - 65.1 in-lb (7.4 N·m).

SUBTASK 75-24-03-400-001-G00

(3) Install the LPTACC upper manifold bracket [1] or LPTACC upper manifold bracket [2] in its position on the clamp [10], clamp [14], and six clamps [5].

(a) Install the bolt [12], washer [13], and nut [15] that attach the clamp [14] to the LPTACC upper manifold bracket [2].

(b) Install the six bolts [3], six washers [4], and six nuts [6] that attach the six clamps [5] to the LPTACC upper manifold assembly.

(c) Install the bolt [7], washer [8], spacer [9], and nut [11] that attach the clamp [10] to the LPTACC upper manifold assembly.

SUBTASK 75-24-03-400-002-G00

(4) Install the LPTACC upper manifold bracket [2] in position on the bracket and on the LPT case bracket.

(a) Install the bolts [28], two flat washers [29], bracket assembly [25] and two nuts [30] that attach the LPTACC upper manifold bracket [2] to the bracket.

(b) Install the two bolts [16], two flat washers [17], two flat washers [18], and two nuts [19] that attach the LPTACC upper manifold bracket [2] to the LPT case bracket.

SUBTASK 75-24-03-420-003-G00

(5) Tighten the six bolts bolts [3], bolt [12], bolt [7], two bolts [16], and two nuts [30] to 58.9 in-lb (6.7 N·m) - 65.1 in-lb (7.4 N·m).

SUBTASK 75-24-03-420-004-G00

(6) Install the bolt [24] that attach the bracket assembly [25] to the LPTACC upper manifold bracket [2].

(a) Tighten the bolt [24] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

SUBTASK 75-24-03-420-005-G00

(7) Install the saddle clamp [27] on the air duct and on the bracket assembly [25].

(a) Install the two bolts [26] that attach the saddle clamp [27] to the bracket assembly [25].

1) Tighten the two bolts [26] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

SUBTASK 75-24-03-970-001-G00

(8) Measure and record the clearances M164B, M164G, M164H and M164J under the brackets.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the clearances M164B, M164G, M164H and M164J between the air cooling tubes of the manifold assemblies and LPT case under all 6 brackets are 0.111 in. (2.8 mm) - 0.236 in. (6 mm).

(b) If the clearances M164B, M164G, M164H and M164J are out of tolerances, stop the procedure and find the cause.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-24-03-410-001-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-24-03-410-002-G00

(2) For the Engine 1, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-03-410-003-G00

(3) For the Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-03-410-004-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 75-24-03-000-802-G00

4. Low Pressure Turbine Active Clearance Control (LPTACC) Lower Manifold Brackets - Removal

(Figure 403)

A. General

(1) This task includes steps to remove the Low Pressure Turbine Active Clearance Control (LPTACC) lower manifold brackets.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-24-03-010-010-G00

(1) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

(a) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

SUBTASK 75-24-03-010-020-G00

(2) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

(a) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 75-24-03-860-003-G00

(3) For the Engine 1, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-03-860-004-G00

(4) For the Engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-03-010-011-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING

DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. LPTACC Lower Manifold Brackets Removal

CAUTION REPLACE THE ONE LPTACC MANIFOLD BRACKETS AT A TIME. IF YOU DO NOT OBEY INSTRUCTIONS, DAMAGE TO THE LPTACC MANIFOLD CAN OCCUR.

SUBTASK 75-24-03-010-014-G00

(1) Remove the LPTACC lower manifold bracket [41] or LPTACC lower manifold bracket [42] from the LPTACC lower manifold assembly as follows:

(a) Remove the six nuts [46], six washers [44], and six bolts [43] that attach the six clamps [45] to the LPTACC lower manifold assembly.

1) Keep the six clamps [45] installed in their positions on the LPTACC lower manifold assembly.

(b) Remove the nut [50], washer [48], and bolt [47] that attach the clamp [49] to the LPTACC lower manifold assembly.

1) Keep the clamp [49] installed in its position on the LPTACC lower manifold assembly.

(c) Remove the nut [55], washer [52], bolt [51], and spacer [53] that attach the clamp [54] to the LPTACC lower manifold assembly.

1) Keep the clamp [54] installed in its position on LPTACC lower manifold assembly.

SUBTASK 75-24-03-010-015-G00

(2) Remove the two nuts [58], two flat washers [59], two flat washers [57], and two bolts [56] that attach the aft end of the LPTACC lower manifold bracket [41] or LPTACC lower manifold bracket [42] to the Low Pressure Turbine (LPT) case bracket.

SUBTASK 75-24-03-010-016-G00

(3) Remove the two nuts [63], retaining plate [62], two bolts [60], and two flat washers [61] that attach the forward end of the LPTACC lower manifold bracket [41] to the bracket.

SUBTASK 75-24-03-000-012-G00

(4) Remove the two bolts [64], two flat washers [65], bracket assembly [67], and two nuts [66] that attach the LPTACC lower manifold bracket [41] to the bracket at the 8:00 o'clock position.

SUBTASK 75-24-03-000-003-G00

CAUTION MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC UPPER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(5) Remove the LPTACC lower manifold bracket [41] from the engine.

SUBTASK 75-24-03-010-017-G00

(6) Remove the two bolts [70] that attach the saddle clamp [71] to the bracket assembly [69] at the 8:00 o'clock position.

SUBTASK 75-24-03-000-009-G00

(7) Remove the saddle clamp [71] from the air duct and bracket assembly [69] at the 8:00 o'clock position.

SUBTASK 75-24-03-000-010-G00

(8) Remove the two nuts [74], two bolts [72], and two flat washers [73] that attach the forward end of the LPTACC lower manifold bracket [42] and bracket assembly [69] to the bracket at the 8:00 o'clock position.

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SUBTASK 75-24-03-010-018-G00

(9) While you hold the bracket assembly [69], remove the bolt [68] that attach the bracket assembly [69] to the LPTACC lower manifold bracket [42].

SUBTASK 75-24-03-000-011-G00

(10) Remove the bracket assembly [69] from the LPTACC lower manifold bracket [42].

SUBTASK 75-24-03-000-004-G00

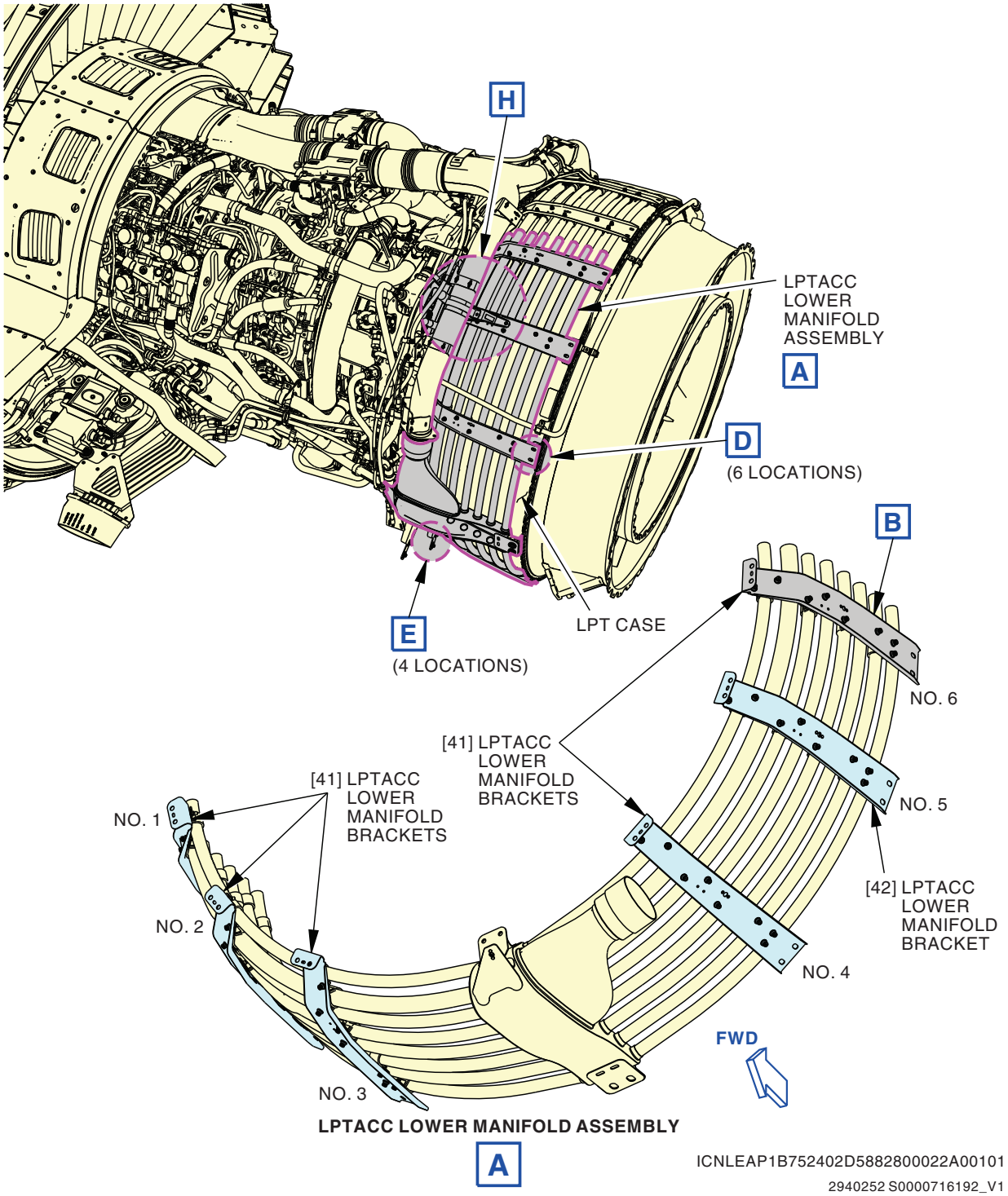
CAUTION

MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC UPPER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(11) Remove the LPTACC lower manifold bracket [42] from the engine.

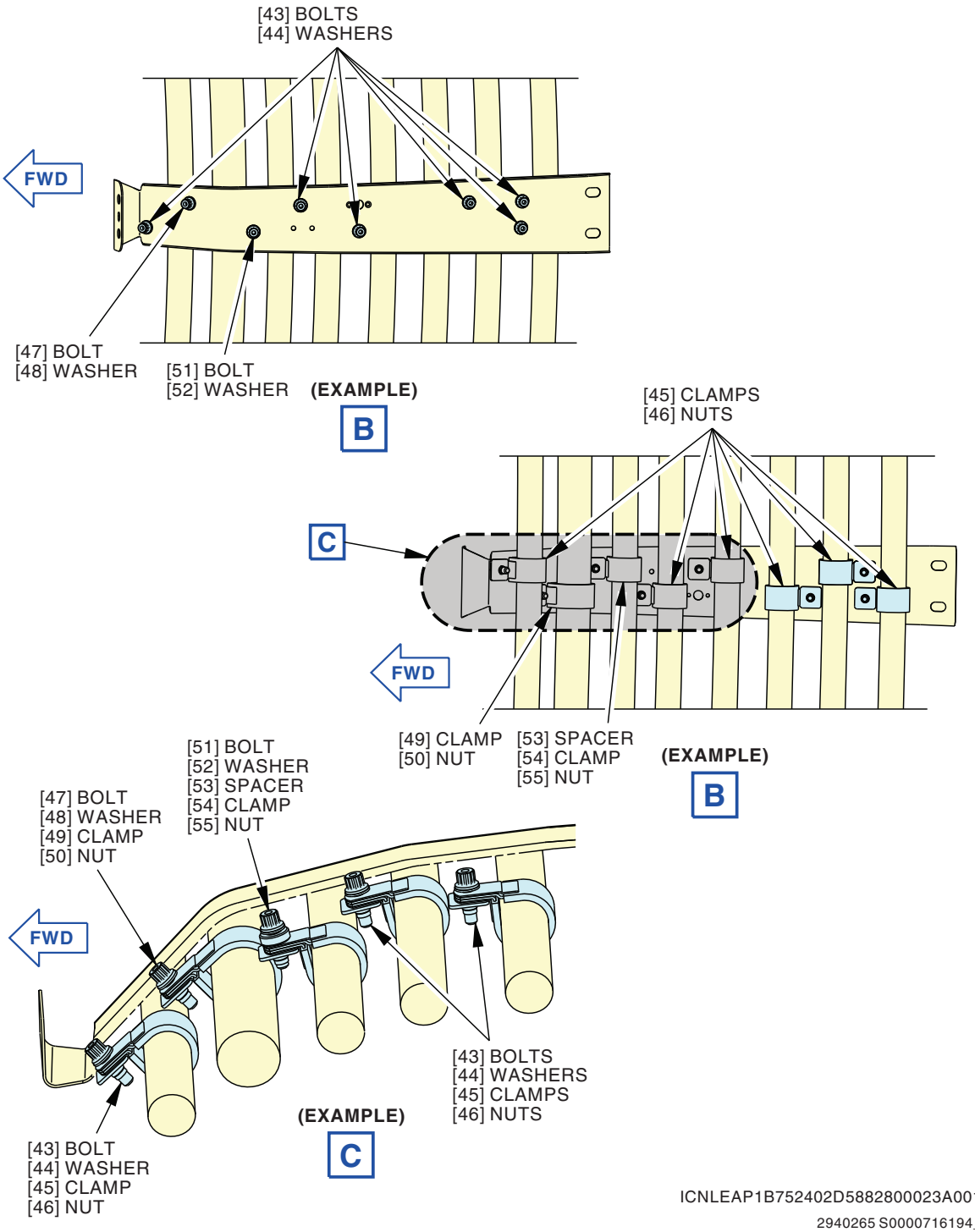
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPTACC Lower Manifold Brackets Installation
Figure 403/75-24-03-990-802-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

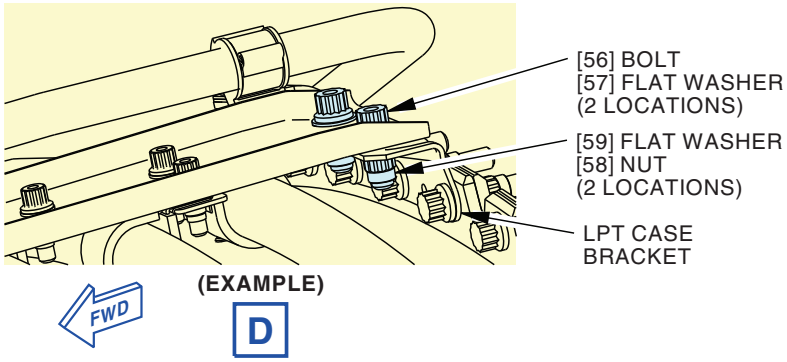


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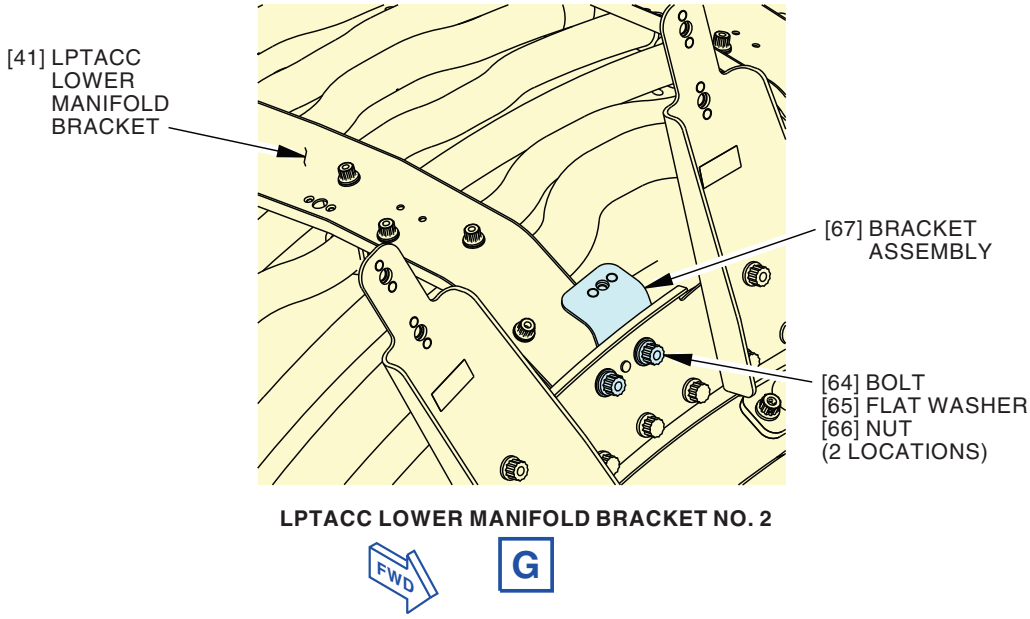
LPTACC Lower Manifold Brackets Installation
Figure 403/75-24-03-990-802-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPTACC Lower Manifold Brackets Installation
Figure 403/75-24-03-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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ICNLEAP1B752402D5882800026A00101

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LPTACC Lower Manifold Brackets Installation
Figure 403/75-24-03-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 75-24-03-400-802-G00

5. Low Pressure Turbine Active Clearance Control (LPTACC) Lower Manifold Brackets - Installation

(Figure 403, Figure 402)

A. General

(1) This task includes steps to install the Low Pressure Turbine Active Clearance Control (LPTACC) upper manifold brackets.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. LPTACC Upper Manifold Brackets Installation

CAUTION

REPLACE THE ONE LPTACC MANIFOLD BRACKETS AT A TIME. IF YOU DO NOT OBEY INSTRUCTIONS, DAMAGE TO THE LPTACC MANIFOLD CAN OCCUR.

SUBTASK 75-24-03-420-008-G00

(1) Apply graphite mineral vaseline, D00601 [CP2101], on the thread of the bolts and pressure face of the nuts that follow:

(a) the bolts [56]

(b) the nuts [63]

(c) the bolts [3]

(d) the bolt [12]

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) the bolt [7].

SUBTASK 75-24-03-420-009-G00

CAUTION

MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE LPTACC UPPER MANIFOLD ASSEMBLY. DAMAGE CAN OCCUR.

(2) Install the LPTACC lower manifold bracket [41] in its position on the bracket and Low Pressure Turbine (LPT) case bracket.

(a) Install the two bolts [64], two flat washers [65], bracket assembly [67], and two nuts [66] that attach the LPTACC lower manifold bracket [41] to the bracket at the 8:00 o'clock position.

(b) Install the two bolts [60], two flat washers [61], retaining plate [62], and twonuts [63] that attach the LPTACC lower manifold bracket [41] to the bracket.

(c) Install the two bolts [56], two flat washer [57], two flat washers [59], and two nuts [58] that attach LPTACC lower manifold bracket [41] to the LPT case bracket.

1) Tighten the two bolts [56], twonuts [63] and two nuts [66] to 58.9 in-lb (6.7 N·m) - 65.1 in-lb (7.4 N·m).

SUBTASK 75-24-03-420-010-G00

(3) Install the LPTACC lower manifold bracket [41] or LPTACC lower manifold bracket [42] in its position on the clamp clamp [49], clamp [54], and six clamps [45].

(a) Install the six bolts [43], the six washers [44], and six nuts [46] that attach the six clamp [45] to the LPTACC lower manifold assembly.

(b) Install the bolt [47], washer [48], and nut [50] that attach the clamp [49] to the LPTACC lower manifold assembly.

(c) Install the bolt [51], washer [52], spacer [53], andnut [55] that attach the clamp [54] to the LPTACC lower manifold assembly.

SUBTASK 75-24-03-400-003-G00

(4) Install the LPTACC lower manifold bracket [42] in its position on the bracket and on the LPT case bracket.

(a) Install two bolts [72], two flat washers [73], bracket assembly [69] and two nuts [74] that attach theLPTACC lower manifold bracket [42] to the bracket.

(b) Install the two bolts [56], two flat washer [57], two flat washers [59], and two nuts [58] that attach LPTACC lower manifold bracket [42] to the LPT case bracket.

SUBTASK 75-24-03-400-004-G00

(5) Tighten the six bolts [43], bolt [47], bolt [51], two bolts [56], and two nuts [74] to 58.9 in-lb (6.7 N·m) - 65.1 in-lb (7.4 N·m).

SUBTASK 75-24-03-420-011-G00

(6) Install the bolt [68] that attaches the bracket assembly [69] to the LPTACC lower manifold bracket [41].

(a) Tighten the bolt [68] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

SUBTASK 75-24-03-420-012-G00

(7) Install the saddle clamp [71] on the air duct and on the bracket assembly [69].

(a) Install the two bolts [70] that attach the saddle clamp [71] to the bracket assembly [69].

1) Tighten the two bolts [70] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

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SUBTASK 75-24-03-970-002-G00

(8) Measure and record the clearances M164B, M164G, M164H and M164J under the brackets

(a) Make sure that the clearances M164B, M164G, M164H and M164J between the air cooling tubes of the manifold assemblies and LPT case under all 6 brackets are 0.111 in. (2.8 mm) - 0.236 in. (6 mm).

(b) If the clearances M164B, M164G, M164H and M164J are out of tolerances, stop the procedure and find the cause.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 75-24-03-420-015-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 75-24-03-000-005-G00

(2) For the Engine 1 remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 75-24-03-020-001-G00

(3) For the Engine 2 remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-24-03-840-001-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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HIGH PRESSURE TURBINE ACTIVE CLEARANCE CONTROL VALVE (HPTACC) - REMOVAL/ INSTALLATION

1. General

A. This procedure has two tasks:

(1) High Pressure Turbine Clearance Control (HPTACC) valve removal

(2) High Pressure Turbine Clearance Control (HPTACC) valve installation.

TASK 75-24-04-000-801-G00

2. High Pressure Turbine Active Clearance Control (HPTACC) Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the High Pressure Turbine Active Clearance Control (HPTACC) valve.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 75-24-04-860-001-G00

(1) For the left engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 75-24-04-860-005-G00

(2) For the right engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 75-24-04-865-001-G00

(3) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 75-24-04-010-001-G00

(4) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. High Pressure Turbine Active Clearance Control Valve Removal

SUBTASK 75-24-04-030-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

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(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use a teflon-jawed pliers, STD-664 to disconnect the electrical connector [15] from the HPTACC valve [1] as follows:

(a) Disconnect the electrical connector [15] from the HPTACC valve [1] (TASK 70-00-01-910-803-G00).

(b) Install the protective covers on the electrical receptacle on the HPTACC valve [1], and the electrical connector [15].

SUBTASK 75-24-04-030-002-G00

(2) Disconnect the ACC drain tube [14] from the HPTACC valve [1].

SUBTASK 75-24-04-020-001-G00

(3) Put the protective covers on the ACC drain tube [14] and the HPTACC valve [1] fitting.

SUBTASK 75-24-04-030-003-G00

(4) Remove the four bolts [17] that attach the hydraulics fuel manifold [16] to the HPTACC valve [1].

SUBTASK 75-24-04-860-002-G00

(5) Put the protective covers on the hydraulics fuel manifold [16] and the HPTACC valve [1].

SUBTASK 75-24-04-010-002-G00

(6) Remove and discard the gasket seal [5].

SUBTASK 75-24-04-020-002-G00

(7) Remove the HPTACC valve [1] as follows:

(a) Remove the hose clamp [6] that attaches the HPT valve elbow duct [4] to the HPTACC valve [1].

(b) Loosen the hose clamp [3] that attaches the HPT valve elbow duct [4] to the HPT ACC air manifold [2].

(c) Rotate the HPT valve elbow duct [4].

(d) Remove the hose clamp [13] that attaches the ACC wye duct [9] to the HPTACC valve [1].

(e) Put protective covers on these components:

1) The HPT valve elbow duct [4]

2) The ACC wye duct [9]

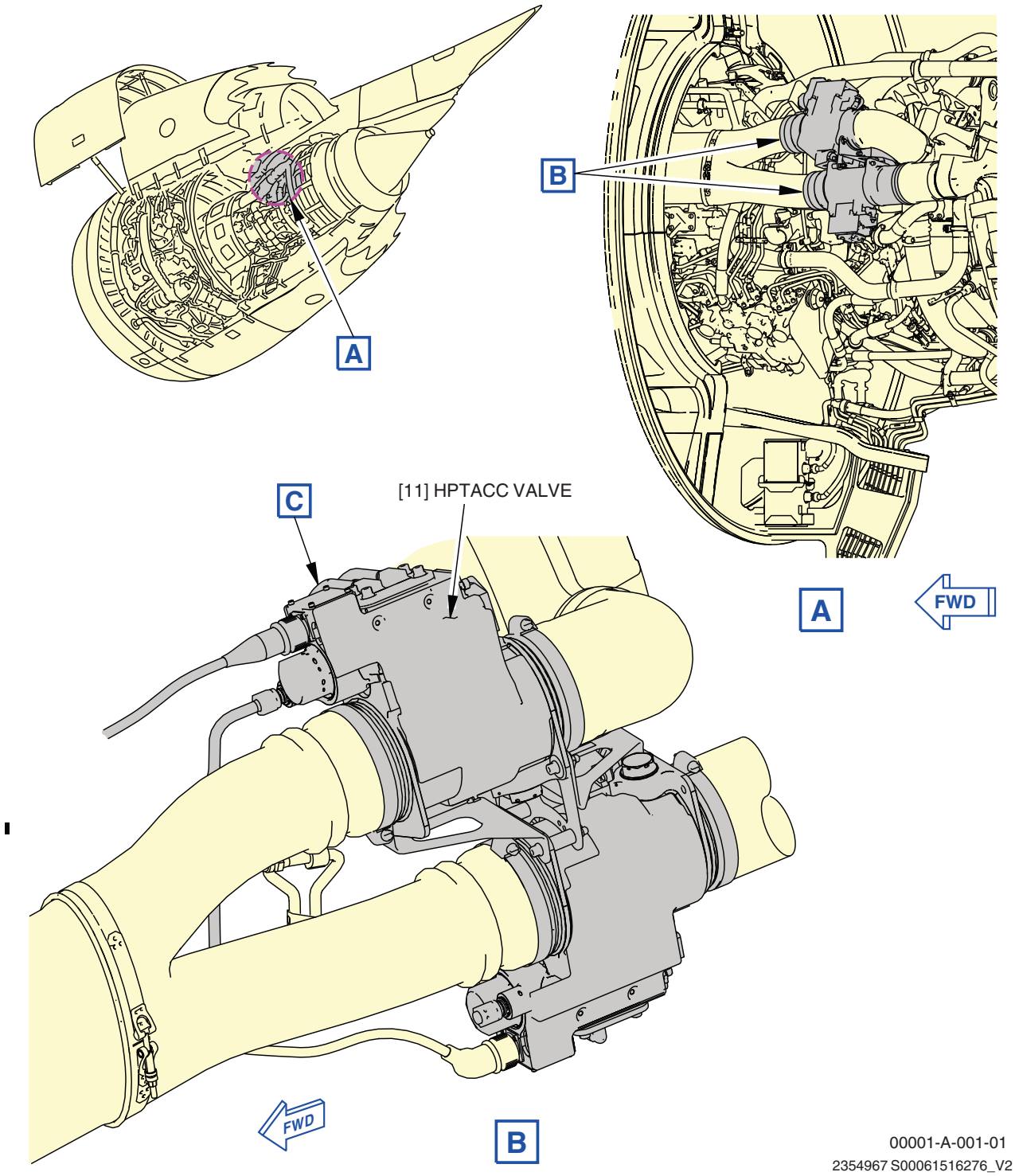
3) The HPTACC valve [1].

(f) Remove the two bolts [18] and the washers [19] that attach the HPTACC valve [1] to the ACC valve bracket [20].

(g) Remove the bolt [23] and the washer [22] that attach the HPTACC valve [1] to the ACC valve bracket [21].

END OF TASK

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HPTACC Valve Installation
Figure 401/75-24-04-990-801-G00 (Sheet 1 of 3)

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